

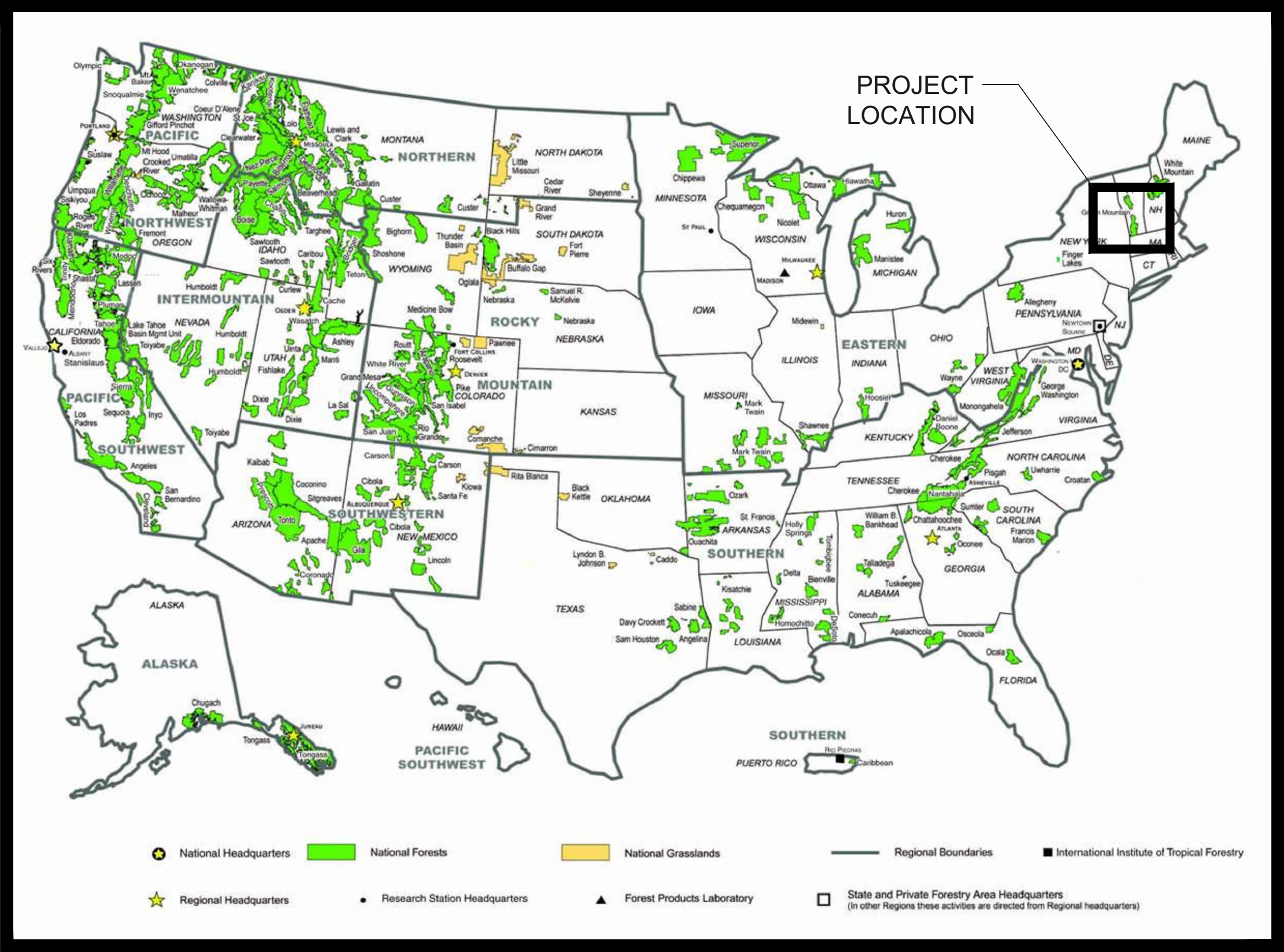


United States Department of Agriculture
Forest Service

(R09) EASTERN REGION
GREEN MOUNTAIN NATIONAL FOREST
BENNINGTON COUNTY
VERMONT

FR 278-0.24 BRIDGE SUPERSTRUCTURE
REPLACEMENT OVER BOLLES BROOK
STRUCTURE NUMBER: 092002000001068

PROJECT
LOCATION



FOREST LOCATION



VICINITY MAP

TRAVEL DIRECTIONS:

FROM VT 9 IN WOODFORD, VT, TURN LEFT ONTO HARBOUR ROAD. TRAVEL NORTH ALONG HARBOUR ROAD FOR APPROXIMATELY 1.7 MILES TO THE END OF THE PAVED ROADWAY. CONTINUE ON FOREST ROAD 278 FOR 0.24 MILES TO REACH THE BRIDGE AT MILE POST 0.24. 42°55'18"N, 73°6'21"W.

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STANDARD SHEETS

FIGURE 3	WOOD BRIDGE RAILING DESIGN DETAILS	
FIGURE 4	APPROACH GUARDRAIL TRANSITION DESIGN DETAILS	

VTRANS STANDARDS

STANDARD	DATE
G-1D	STEEL BEAM GUARDRAIL END TERMINALS
	3/10/2017

RECOMMENDED BY:

FOREST ENGINEER

DATE

DISTRICT RANGER

DATE

FOREST SUPERVISOR

DATE

REGIONAL STRUCTURAL ENGINEER

DATE

APPROVED:

REGIONAL DIRECTOR OF ENGINEERING

DATE



5/31/24 16:11 BGRAVES I:\629065 QMNF FR278 BRIDGE SUPERSTRUCTURE REPLACEMENT\DRAWINGS\CAD\629065_02 GENERAL NOTES.DWG;

DESIGN LOADS, MATERIALS AND SPECIFICATIONS:

(1)	DESIGN LOADING:	HL-93
(2)	DESIGN METHOD:	LOAD AND RESISTANCE FACTOR DESIGN (LRFD)
(3)	SPECIFICATIONS:	AASHTO LRFD BRIDGE DESIGN SPECIFICATIONS, 9TH ED., 2020 APA- THE ENGINEERED WOOD ASSOCIATION GLULAM PRODUCT GUIDE, JUNE 2023 U.S. DEPARTMENT OF TRANSPORTATION FEDERAL HIGHWAY ADMINISTRATION STANDARD SPECIFICATIONS FOR THE CONSTRUCTION OF ROADS AND BRIDGES ON FEDERAL HIGHWAY PROJECTS, FP-14 and FSSS NATIONAL DESIGN SPECIFICATION (NDS) FOR WOOD CONSTRUCTION, 2024

GENERAL NOTES:

- ALL MATERIALS AND CONSTRUCTION SHALL CONFORM TO THE U.S. DEPARTMENT OF TRANSPORTATION FEDERAL HIGHWAY ADMINISTRATION STANDARD SPECIFICATIONS FOR THE CONSTRUCTION OF ROADS AND BRIDGES ON FEDERAL HIGHWAY PROJECTS, FP-14 (2014) AND FOREST SERVICE SUPPLEMENTAL SPECIFICATIONS (FSSS).
- ALL DIMENSIONS ARE HORIZONTAL AND VERTICAL, AND ARE GIVEN AT 68 DEGREES FAHRENHEIT.
- DRAWINGS ARE ON ASSUMED DATUM. NORTHING, EASTING, AND ELEVATION ARE NOT TIED TO VT STATE PLANE. NORTH WAS DERIVED FROM A COMPASS BEARING TAKEN ON 5-24-23
- ALL DIMENSIONS AND FIELD CONDITIONS SHALL BE VERIFIED BY THE CONTRACTOR PRIOR TO COMMENCEMENT OF THE WORK.
- THE CONTRACTOR SHALL REVIEW AND UNDERSTAND ALL APPLICABLE ENVIRONMENTAL PERMITS AND ENSURE THAT ALL CONSTRUCTION CONDITIONS ARE MET.
- THE CONTRACTOR SHALL BE RESPONSIBLE FOR CONSTRUCTION SAFETY, AND MEANS AND METHODS TO PERFORM AND COMPLETE THE WORK. THE CONSTRUCTION CONTRACTOR SHALL PREPARE OR OBTAIN THE APPROPRIATE SAFETY SYSTEMS WHICH MAY BE REQUIRED BY THE U.S. OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA) AND ALL APPLICABLE REGULATIONS. THESE PLANS AND PROFESSIONAL SEALS DO NOT EXTEND TO OR INCLUDE SYSTEMS PERTAINING TO THE SAFETY OF THE CONSTRUCTION CONTRACTOR OR THEIR EMPLOYEES, AGENTS, OR REPRESENTATIVES IN THE PERFORMANCE OF THE WORK.
- THE CONTRACTOR SHALL SUBMIT LITERATURE (MANUFACTURERS LITERATURE, CUT SHEETS, APPLICATION PROCEDURES, ETC.) FOR ALL PRODUCTS PROPOSED FOR USE ON THE PROJECT, FOR APPROVAL BY THE CONTRACTING OFFICER'S REPRESENTATIVE (C.O.R).
- THE CONTRACTOR SHALL COMPACT THE ROADWAY SUBGRADE AFTER EXISTING MATERIAL IS REMOVED AND PRIOR TO THE PLACEMENT OF ANY ADDITIONAL OR NEW MATERIAL. SUBGRADE SHALL MEET 95% COMPACTION.
- THE CONTRACTOR SHALL TAKE ALL NECESSARY MEASURES TO ENSURE THAT DEBRIS DOES NOT FALL ON ANY WATERWAY BELOW THE EXISTING STRUCTURE. ALL COSTS, INCLUDING ERECTION, MAINTENANCE AND REMOVAL OF TEMPORARY STRUCTURES OR OTHER SUCH APPROVED METHODS, SHALL BE SUBSIDIARY TO THE APPROPRIATE ITEMS OF WORK BEING PERFORMED.
- ALL WASTE MATERIAL INCLUDING UNUSED EXCAVATED MATERIAL SHALL BECOME THE PROPERTY OF THE CONTRACTOR AND SHALL BE DISPOSED OF OFF U.S. GOVERNMENT (NATIONAL FOREST) LAND IN AN APPROVED LANDFILL. THERE ARE NO GOVERNMENT DISPOSAL SITES OR PITS AVAILABLE FOR THIS PROJECT.
- ALL SOIL MOVING EQUIPMENT SHALL BE THOROUGHLY CLEANED TO MAKE IT FREE OF SOIL, NON-NATIVE INVASIVE SPECIES OR OTHER DEBRIS THAT COULD CONTAIN OR HOLD SEEDS PRIOR TO BEING DELIVERED TO THE PROJECT SITE. EQUIPMENT SHALL BE CONSIDERED FREE OF NON-NATIVE OR INVASIVE SPECIES AND OTHER SUCH DEBRIS WHEN A VISUAL INSPECTION BY THE C.O.R., COMPLETED PRIOR TO THE EQUIPMENT BEING MOVED TO THE SITE, DOES NOT DISCLOSE SUCH MATERIAL PRESENT.
- PRIOR TO REMOVAL OF EQUIPMENT FROM THE PROJECT SITE, ALL SOIL MOVING EQUIPMENT SHALL BE THOROUGHLY CLEANED TO MAKE IT FREE OF SOIL, VEGETATION, OR OTHER DEBRIS.
- FR 278 SHALL BE CLOSED WITHIN THE PROJECT LIMITS AS SHOWN ON SHEET C-06 FOR THE DURATION OF CONSTRUCTION. THE CONTRACTOR SHALL PROVIDE SIGNS AND CLOSE GATE AS APPROVED BY THE C.O.R..
- ALL EXPOSED RIPRAP AND AGGREGATE SURFACE COURSE SHALL BE BROWN, DARK GREY OR EARTH TONE IN COLOR. THE C.O.R. SHALL APPROVE COLOR PRIOR TO PURCHASE OR PLACEMENT BY THE CONTRACTOR.
- THE CONTRACTOR SHALL PERFORM ALL IN-STREAM WORK BETWEEN JULY 1ST AND OCTOBER 1ST.
- THE WORK COVERED BY THESE PLANS INCLUDES THE REMOVAL OF THE EXISTING STEEL BEAMS, TIMBER DECK, AND RAILING SYSTEM. INSTALLATION OF THE NEW GLU-LAM DECK, STEEL BEAMS, STEEL DIAPHRAGMS, AND TIMBER WEARING SURFACE, CONCRETE ABUTMENT REPAIRS, RAILING SYSTEM, APPROACH ROADWAY WORK INCLUDING GUARDRAIL STEEL BEAMS AND PLACEMENT OF EROSION CONTROL MEASURES.
- THE CONTRACTOR SHALL NOTIFY DIG SAFE (DIAL 888-344-7233 OR 811 OR SUBMIT TICKET AT DIGSAFE.COM) A MINIMUM OF 3 DAYS PRIOR TO THE START OF WORK. IF RELOCATION OF UTILITIES IS NECESSARY, THE CONTRACTOR IS RESPONSIBLE FOR COORDINATING WORK WITH THE UTILITY COMPANY.
- THE CONTRACTOR WILL BE RESPONSIBLE FOR ALL CONSTRUCTION STAKING. CONSTRUCTION STAKING SHALL BE IN ACCORDANCE WITH FP-14 REQUIREMENTS. ALL COSTS ASSOCIATED WITH THIS SHALL BE PAID FOR UNDER ITEM 15201, CONSTRUCTION AND SURVEY.
- WATER LEVEL IS SUBJECT TO CHANGE. THE CONTRACTOR IS RESPONSIBLE FOR DETERMINING THE EXPECTED WATER LEVELS DURING CONSTRUCTION.
- LOCATE STAGING AREAS AWAY FROM SENSITIVE AREAS INCLUDING WETLANDS AND STREAM BUFFERS IN COMPLIANCE WITH REGULATIONS AND PERMIT REQUIREMENTS
- TO LIMIT INADVERTENT DAMAGE TO AND UNAUTHORIZED REMOVAL OF TREES ADJACENT TO THE WORK AREA, CONTRACTOR SHALL COORDINATE WITH THE C.O.R. AND OBTAIN APPROVAL PRIOR TO CUTTING OR TRIMMING TREES WITHIN AND ADJACENT TO THE WORK AREA.

- ALL MATERIALS USED ON THIS PROJECT SHALL CONFORM TO THE TECHNICAL SPECIFICATIONS. FOR ANY DISCREPANCY BETWEEN THE PLANS AND MATERIAL SPECIFICATIONS, THE TECHNICAL SPECIFICATIONS SHALL TAKE PRECEDENCE OVER NOTES CONTAINED WITHIN THESE PLANS.
- VERIFY ALL INDICATED CONDITIONS AND DIMENSIONS IN THE FIELD BEFORE COMMENCING ANY FABRICATION, ORDERING OF MATERIAL, OR PERFORMING ANY WORK. NOTIFY THE C.O.R. OF ANY CONDITIONS OR DIMENSIONS THAT WOULD PREVENT OR HAMPER THE PERFORMANCE OF THE WORK IN ACCORDANCE WITH THE CONTRACT DRAWINGS AND SPECIFICATIONS.
- SUBMIT SHOP DRAWINGS FOR BRIDGE DECK PANELS, STEEL STRINGERS, AND BRIDGE RAILINGS. ALL SHOP DRAWINGS TO BE STAMPED BY VERMONT P.E.

BRIDGE REMOVAL NOTES:

- THE CONTRACTOR SHALL SUBMIT, FOR DOCUMENTATION, A DETAILED OUTLINE OR PLAN OF THE PROPOSED METHOD OF REMOVAL OF THE EXISTING BRIDGE COMPONENTS PRIOR TO THE COMMENCEMENT OF ANY WORK.
 - THE EXISTING BRIDGE SUPERSTRUCTURE SHALL BE REMOVED IN ITS ENTIRETY INCLUDING THE TIMBER DECK, TIMBER RAILING, STEEL GIRDERS, STEEL DIAPHRAGMS, AND BEARINGS AND ALL ASSOCIATED HARDWARE, AND SHALL BE PAID FOR UNDER ITEM 20303 REMOVAL OF BRIDGE SUPERSTRUCTURE. DAMAGE TO THE ABUTMENT BRIDGE SEAT INCLUDING AT EXISTING BEARING PAD ANCHOR BOLTS SHALL BE SEALED AND GROUTED. THIS WORK WILL BE INCIDENTAL TO ITEM 20303. ALL MATERIALS REMOVED SHALL BE THE PROPERTY OF THE CONTRACTOR AND SHALL BE DISPOSED OF APPROPRIATELY.
- GRADING:
- SEED AND MULCH ALL DISTURBED AREAS. TURF ESTABLISHMENT AND SEEDING SHALL BE IN ACCORDANCE WITH FHWA FP14 SECTION 625 TURF ESTABLISHMENT, DRY METHOD AND SHALL BE PAID FOR UNDER ITEM 62502 TURF ESTABLISHMENT. MULCH SHALL BE ROLLED EROSION CONTROL PRODUCT, TYPE 1A, WHICH SHALL BE PAID FOR UNDER 62901. SEED SHALL BE APPLIED AT A RATE OF 75 LBS/ACRE.
 - THE CONTRACTOR SHALL FURNISH AGGREGATE SURFACE COURSE TO THE GRADING SPECIFIED IN THESE PLANS
 - COMPACTION TESTING IS REQUIRED ON BACKFILL AREAS. REFER TO APPLICABLE DETAILS AND SPECIFICATIONS.

TIMBER MATERIALS AND TREATMENT:

- LUMBER FOR GLULAM PRODUCTION TO BE VISUALLY OR MECHANICALLY GRADED IN CONFORMANCE WITH THE CURRENT EDITION OF "AASHTO LRFD BRIDGE DESIGN SPECIFICATION" OR "NATIONAL DESIGN SPECIFICATIONS FOR WOOD CONSTRUCTION."
- GLULAM MEMBERS TO BE FINISHED TO INDUSTRIAL APPEARANCE GRADE PER AITC 110-2001.
- ALL GLULAM MATERIAL SHALL CONFORM TO THE REQUIREMENTS OF ANSI/AITC A190.1 AND BE STAMPED WITH AITC QUALIFY MARK OR AN APA-EWS TRADEMARK. PLACE STAMPS ON SURFACES THAT WILL NOT BE EXPOSED TO VIEW IN THE COMPLETED STRUCTURE. SUBMIT CERTIFICATES OF CONFORMANCE BY AN ACCREDITED INSPECTION AND TESTING AGENCY INDICATING THE GLULAM MEMBERS CONFORM TO THE REQUIREMENTS OF ASTM A190.1.
- GLULAM MEMBERS SHALL BE MANUFACTURED FROM SOUTHERN PINE LAMINATING LUMBER.
- THE CONTRACTOR SHALL HAVE THE OPTION TO SUBMIT AN ALTERNATIVE DECK PANEL DESIGN UTILIZING DOWEL LAMINATED PANELS. ALTERNATIVE DESIGN SHALL INCORPORATE MATERIEL DETAILS AND CALCULATIONS STAMPED BY A STRUCTURAL ENGINEER LICENSED IN THE STATE OF VERMONT SUBMITTED TO COR FOR APPROVAL. IF CONTRACTOR ELECTS TO SUBMIT ALTERNATIVE DECK PANEL, PAYMENT FOR ALTERNATIVE DECK, DESIGN, REVIEW, AND APPROVAL SHALL BE PAID FOR UNDER ITEM 55701A.
- ADHESIVES USED IN THE GLULAM MANUFACTURING PROCESS SHALL CONFORM TO THE AITC 405 FOR WET USE ADHESIVES.
- GLULAM MEMBERS SHALL BE AS FOLLOWS AND CONFORM TO AASHTO TABLE 8.4.1.2.3-2:
 - DECK PANELS: COMBINATION 48, GRADE N2D14
 - CURBS AND SPACER BLOCKS: COMBINATION 47, GRADE N2M14
 - RAIL AND RAIL POSTS: COMBINATION 48, GRADE N2D14
- SIZES OF GLULAM MEMBERS SHALL BE AS SHOWN ON THE DRAWINGS.
- TO THE EXTENT POSSIBLE, SHOP-FABRICATE ALL GLULAM CONNECTIONS INCLUDING CUTTING TO THE LENGTH AND DRILLING BOLT HOLES. END GAPS BETWEEN MEMBERS SHALL BE UNIFORM AND NOT EXCEED $\frac{3}{8}$ ". HOLES SHALL BE DRILLED PERPENDICULAR TO THE FACE OF THE MEMBER. DRILL WIDE MEMBERS FROM BOTH SIDES AS REQUIRED TO PREVENT MISALIGNMENT OF HOLES.
- ALL GRADING, COMBINATIONS, AND SIZE REFER TO SOUTHERN YELLOW PINE UNLESS NOTED OTHERWISE ON THE PLANS. THE DESIGN VALUES FOLLOW LRFD SECTION 8.4.1. THESE ARE THE MINIMUM REQUIREMENTS. OTHER SPECIES MAY BE USED WITH APPROVAL FROM THE C.O.R. AND AT NO ADDITIONAL COST TO THE GOVERNMENT.
- TIMBER PRESERVATION TREATMENT SHALL BE IN ACCORDANCE WITH CURRENT STATE AND/ OR AASHTO SPECIFICATIONS. ALL TIMBER SHALL BE TREATED WITH COPPER NAPHTHENATE HYDROCARBON SOLVENT A, PER AWP, UC4C (MINIMUM).
- TO EXTENT PRACTICAL, ALL BRIDGE COMPONENTS TO BE FABRICATED (HOLES DRILLED, CUTS MADE, ETC.) PRIOR TO PRESERVATIVE TREATMENTS PROCESS. FIELD FABRICATION TO BE TREATED IN ACCORDANCE WITH AWP, STANDARD M4.
- ABUTMENT BACKWALL TO BE NO.1 GRADE VISUALLY GRADED SOUTHERN YELLOW PINE, P.T.
- TIMBER WEARING SURFACE TO BE ROUGH SAWN HEMLOCK. PLANK LENGTH SHALL BE A MINIMUM OF 8 FEET, EXCEPT FOR END PANELS, WHICH SHALL BE 4 FEET IN LENGTH.
- ALL TIMBER TO BE CUT EXACT LENGTH, DRESSED TO SIZE REQUIRED AND ALL PRACTICAL FRAMING TO BE DONE PRIOR TO TREATMENT.
- ALL DECK PLANKS SHALL BE PREDRILLED PRIOR TO TREATMENT.
- ALL PLANK FOR DECK PANELS SHALL BE PRECISION END TRIMMED TO LENGTH WITH 1/4" UNDERLENGTH & NO OVERLENGTH TOLERANCE PERMITTED.
- DECK PANELS WILL BE DELIVERED TO JOBSITE AFTER BEING FULLY ASSEMBLED AT FABRICATION PLANT.
- ALL TIMBER HARDWARE TO MEET ASTM A307-97 GALVANIZED TO A153. ALL HIGH STRENGTH HARDWARE TO MEET ASTM A325 OR A449 GALVANIZED TO A153.

STRUCTURAL STEEL:

- STRUCTURAL STEEL BEAMS, DIAPHRAGMS, STIFFENERS AND CONNECTION PLATES WILL BE PAID FOR UNDER ITEM 55502, "STRUCTURAL STEEL, FURNISHED, FABRICATED, AND ERECTED" AND SHALL CONFORM TO AASHTO 270M/M270 GRADE 50 UNLESS NOTED OTHERWISE.
- ALL BOLTED CONNECTIONS SHALL BE MADE WITH ASTM F3125 GRADE A325 TYPE 3 7/8" DIAMETER HIGH STRENGTH BOLTS IN STANDARD 15/16" HOLES UNLESS OTHERWISE NOTED.
- ANY CONNECTIONS THAT ARE NOT DETAILED ON THE PLANS SHALL BE DETAILED BY THE FABRICATOR AND SUBMITTED TO THE C.O.R. FOR APPROVAL. ALL ASSOCIATED COSTS SHALL BE INCIDENTAL TO THE CONTRACT.
- BEARING STIFFENERS AND GIRDER ENDS SHALL BE PLUMB UNDER FULL DEAD LOAD DEFLECTION.
- UNLESS NOTED OTHERWISE, ALL BOLTED CONNECTIONS SHALL BE SLIP CRITICAL WITH CLASS B FAYING SURFACES.
- THE STRUCTURAL STEEL SHALL BE PAINTED WITH A SHOP APPLIED PAINT SYSTEM IN ACCORDANCE WITH SECTION 563 OF THE SPECIFICATIONS. ANY AREAS OF DAMAGE DURING THE SHIPPING OR PLACEMENT OF THE STEEL SHALL BE REPAIRED IN ACCORDANCE WITH THIS SECTION. THE COLOR SHALL CONFORM TO FEDERAL STANDARD NO. 595, COLOR CHIP NUMBER 20059 AND SHALL BE APPROVED BY THE C.O.R.

CONCRETE:

- CONCRETE FOR ANY NECESSARY SUBSTRUCTURE REPAIRS SHALL BE PAID FOR UNDER ITEM 55221, "REPAIR CONCRETE" AND SHALL BE IN ACCORDANCE WITH SECTION 552. ALL NECESSARY SUBSTRUCTURE REPAIRS SHALL BE APPROVED BY THE C.O.R. IN ADVANCE OF THE START OF ANY ABUTMENT REPAIR WORK.
- CONCRETE STRENGTH SHALL BE $f'c = 4,500$ PSI MINIMUM.
- REINFORCING STEEL SHALL BE BLACK AND BE GRADE 60.



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STAMPS, LOGOS, AND SEALS



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PROJECT NAME

FR 278-0.24
BRIDGE
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REPLACEMENT

GREEN MOUNTAIN
NATIONAL FOREST

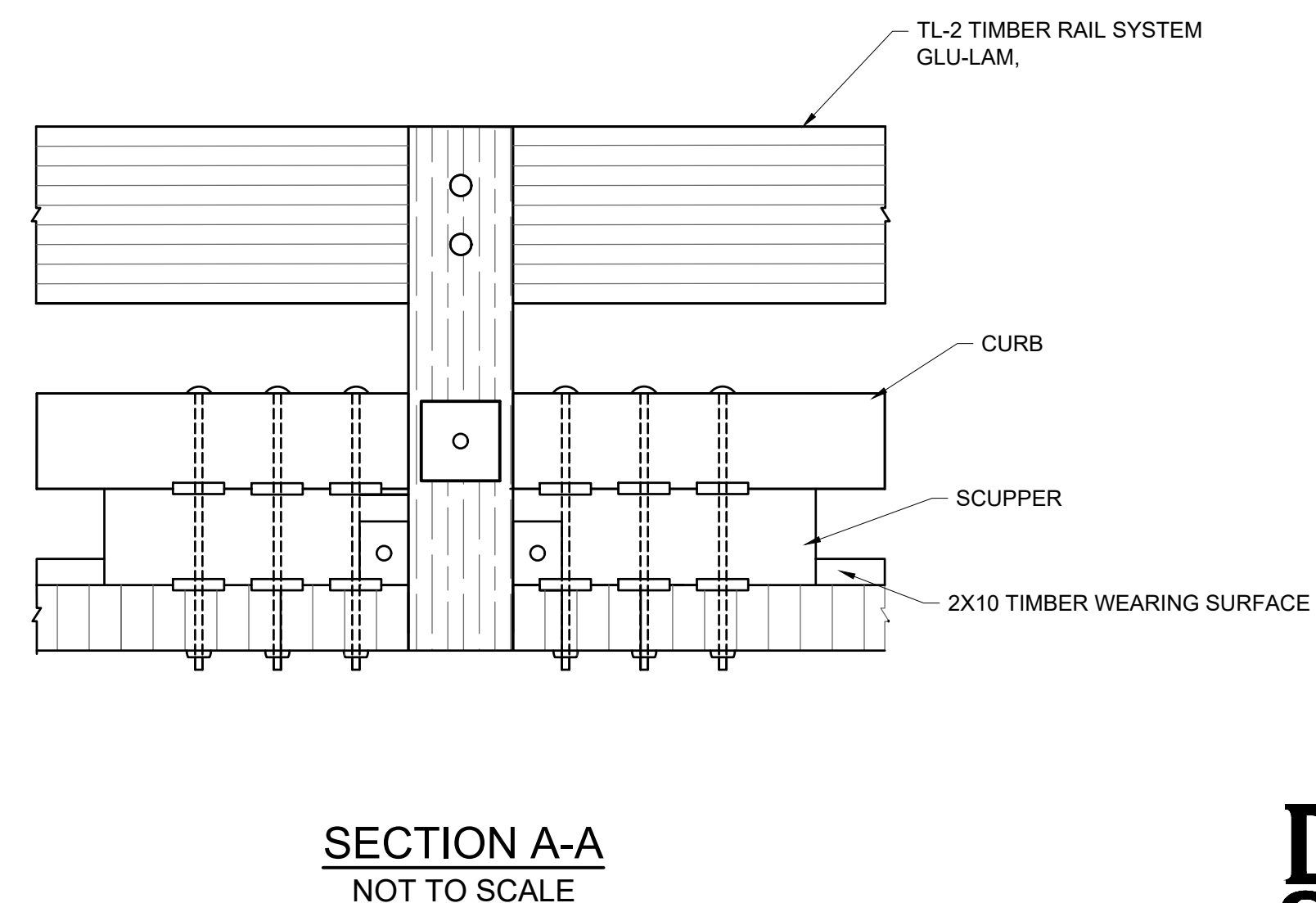
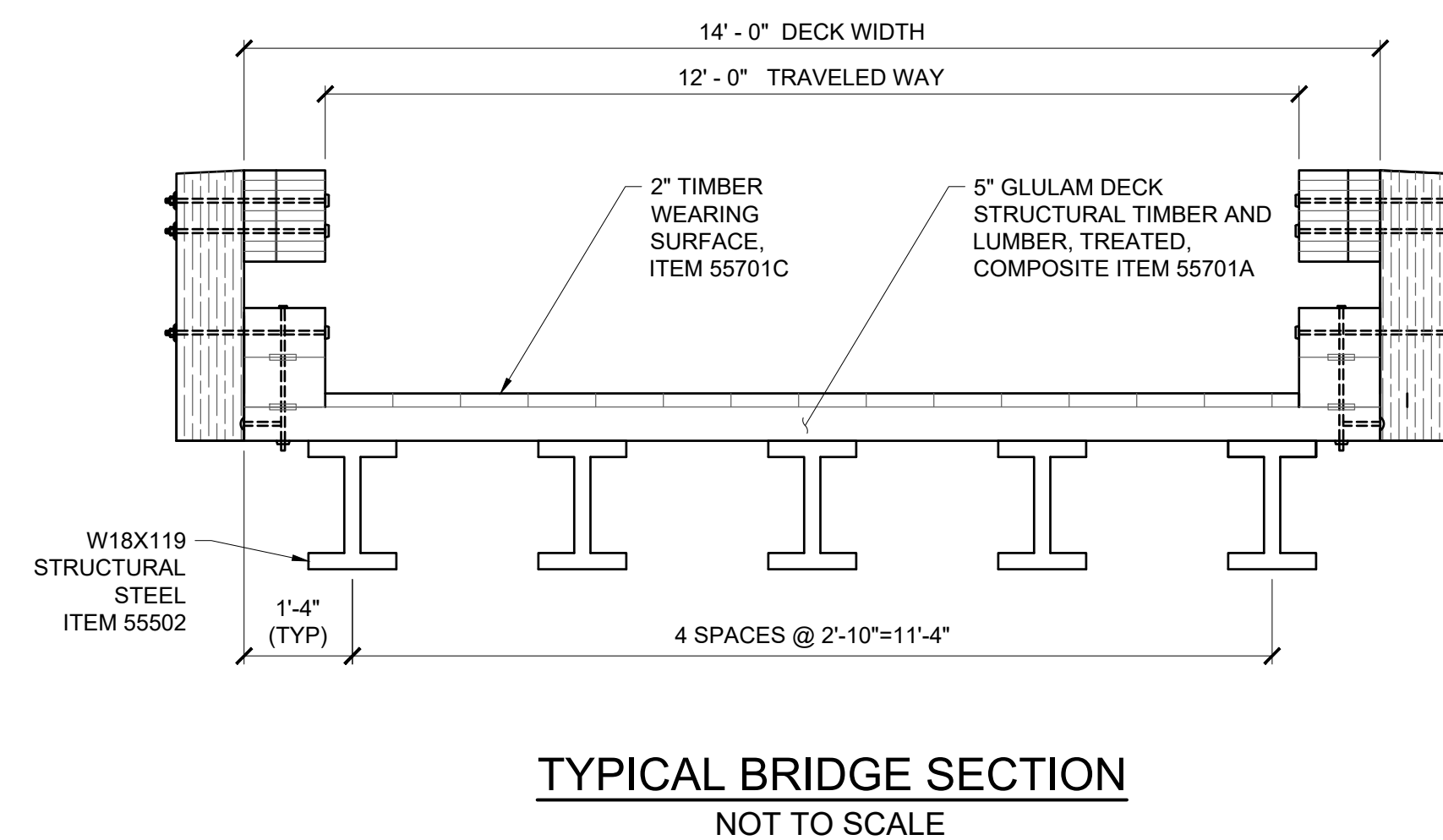
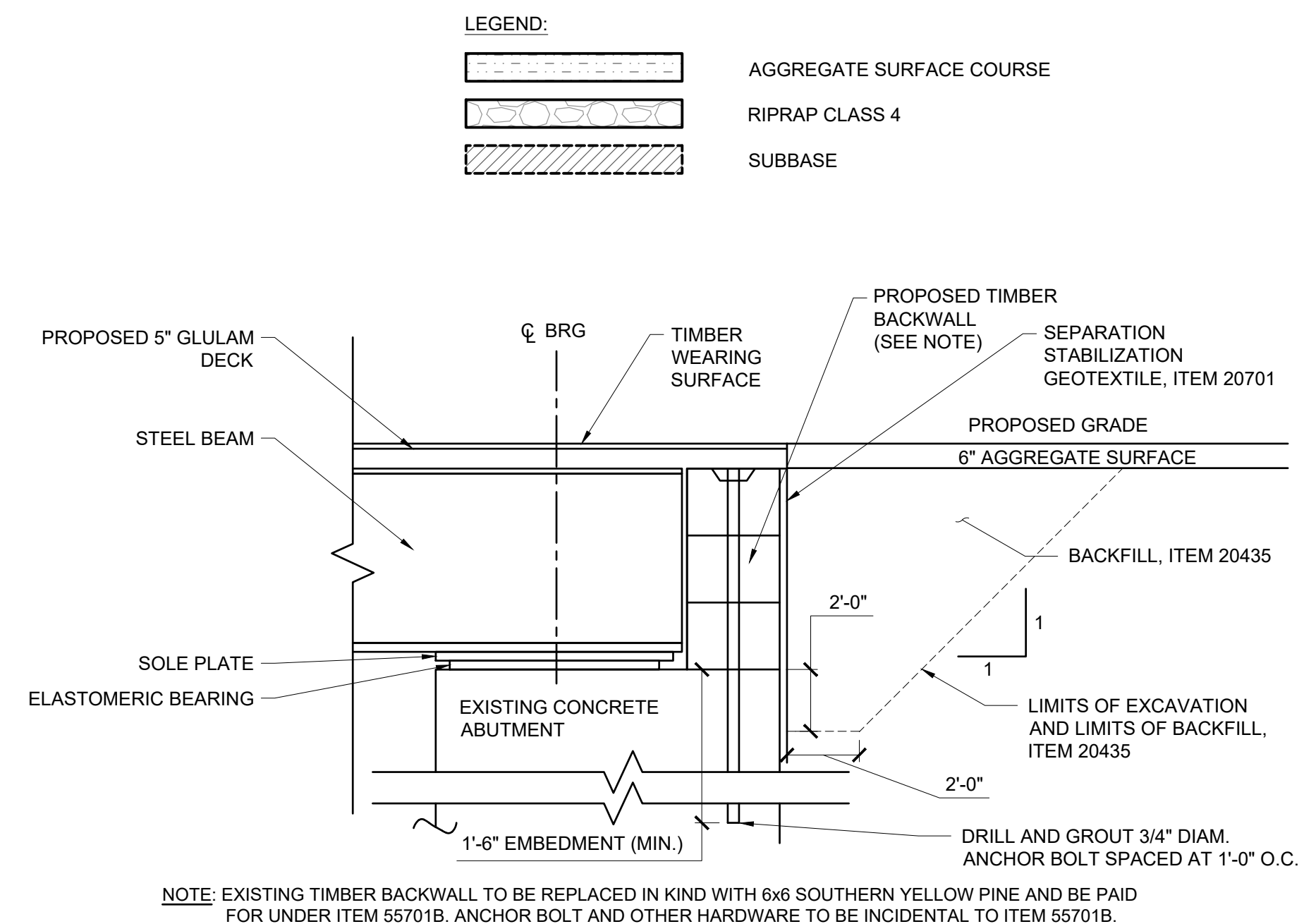
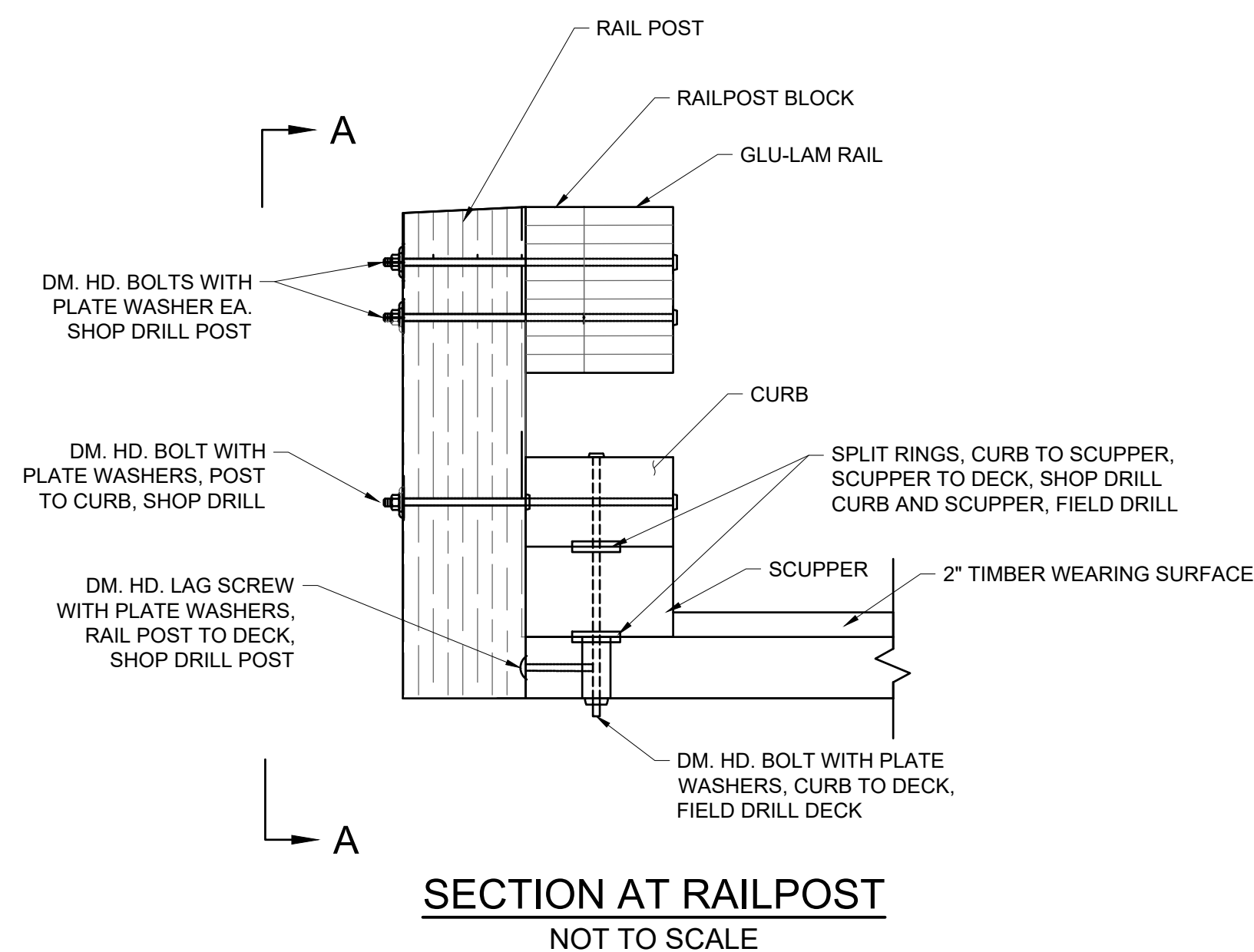
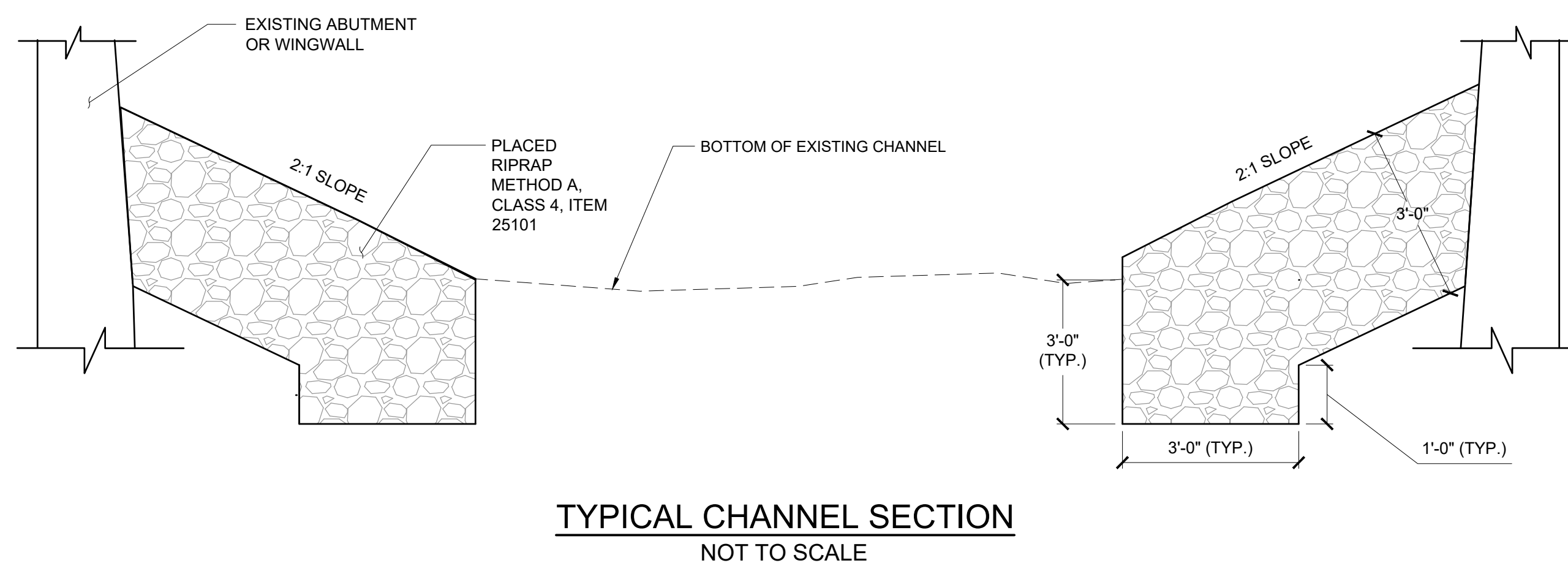
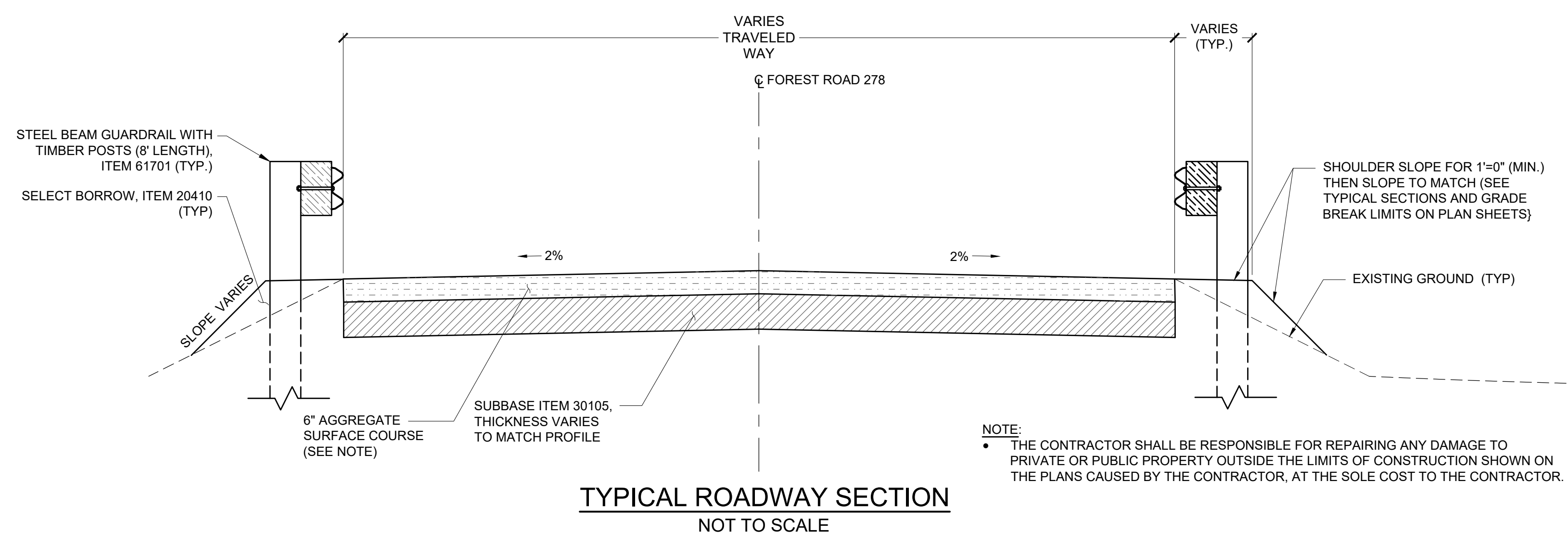
MANCHESTER RANGER DISTRICT

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GENERAL NOTES

DATE 7/12/2024	ARCHIVE NO.
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DRAWN B. GRAVES	C-02
CHECKED D. CONGER	
PROJECT NO. 629065	SHEET 02 OF 18

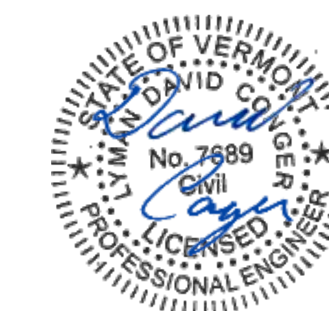




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MANCHESTER RANGER DISTRICT

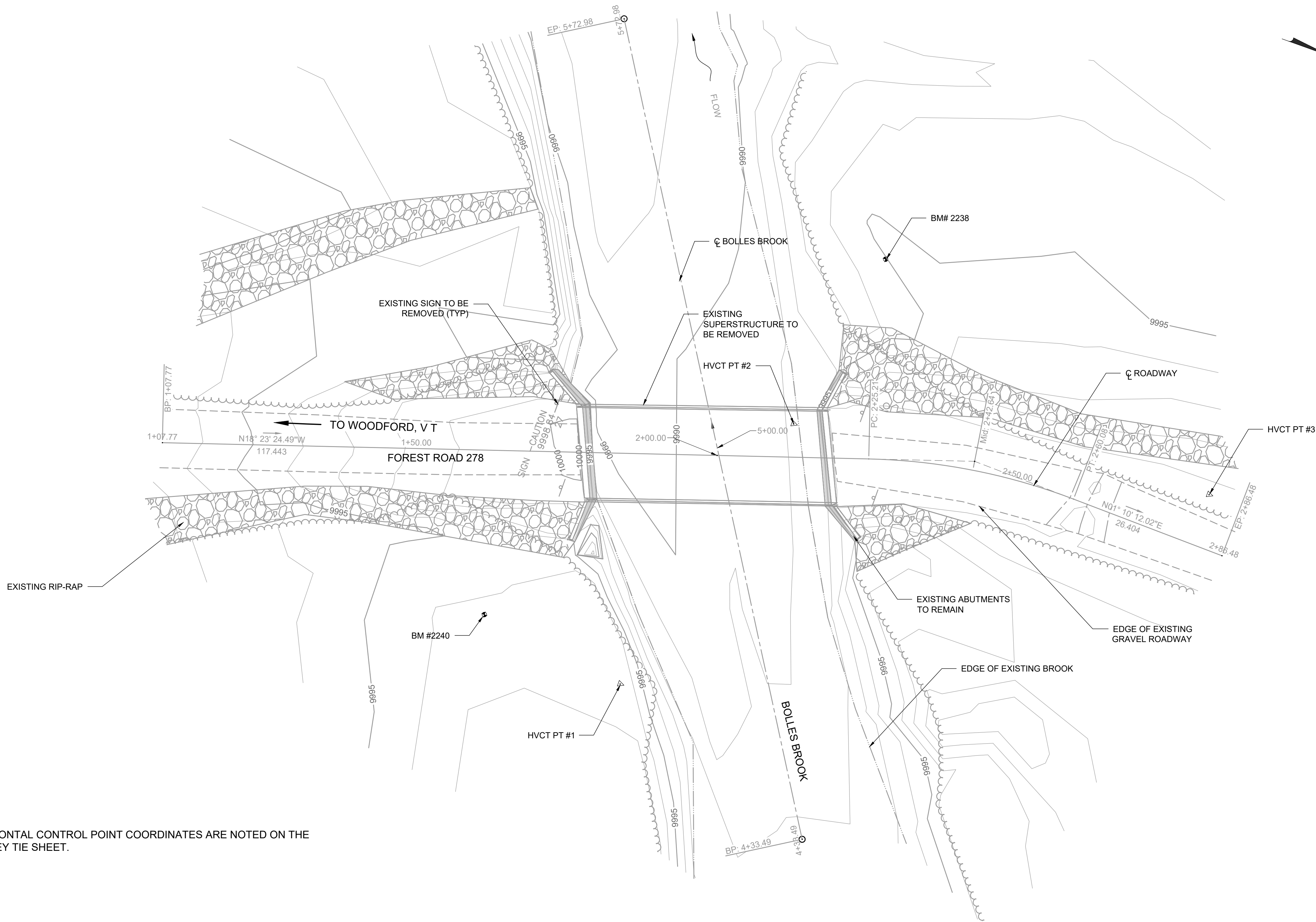
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TYPICAL SECTIONS AND DETAILS

DATE 7/17/2024	ARCHIVE NO.
DESIGNER A. HUMMERS	DRAWING SHEET NO. C-03
DRAWN B. GRAVES	
CHECKED D. CONGER	
PROJECT NO. 629065	SHEET 03 OF 18

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& King** inc.

7/12/24 12:39 BGRAVES I:\629065 GMINF FR278 BRIDGE SUPERSTRUCTURE REPLACEMENT\DRAWINGS\CAD\629065_03 EXISTING CONDITIONS PLANDWG



NOTES:

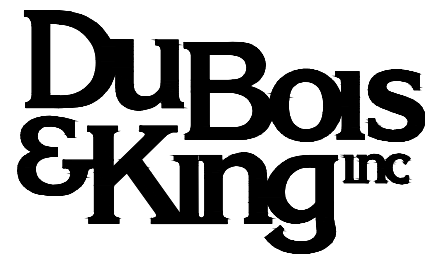
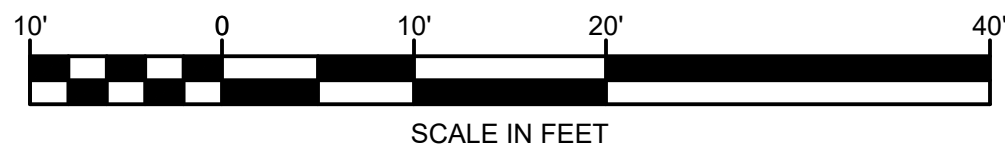
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LEGEND:



EXISTING RIP-RAP

EXISTING CONDITIONS PLAN



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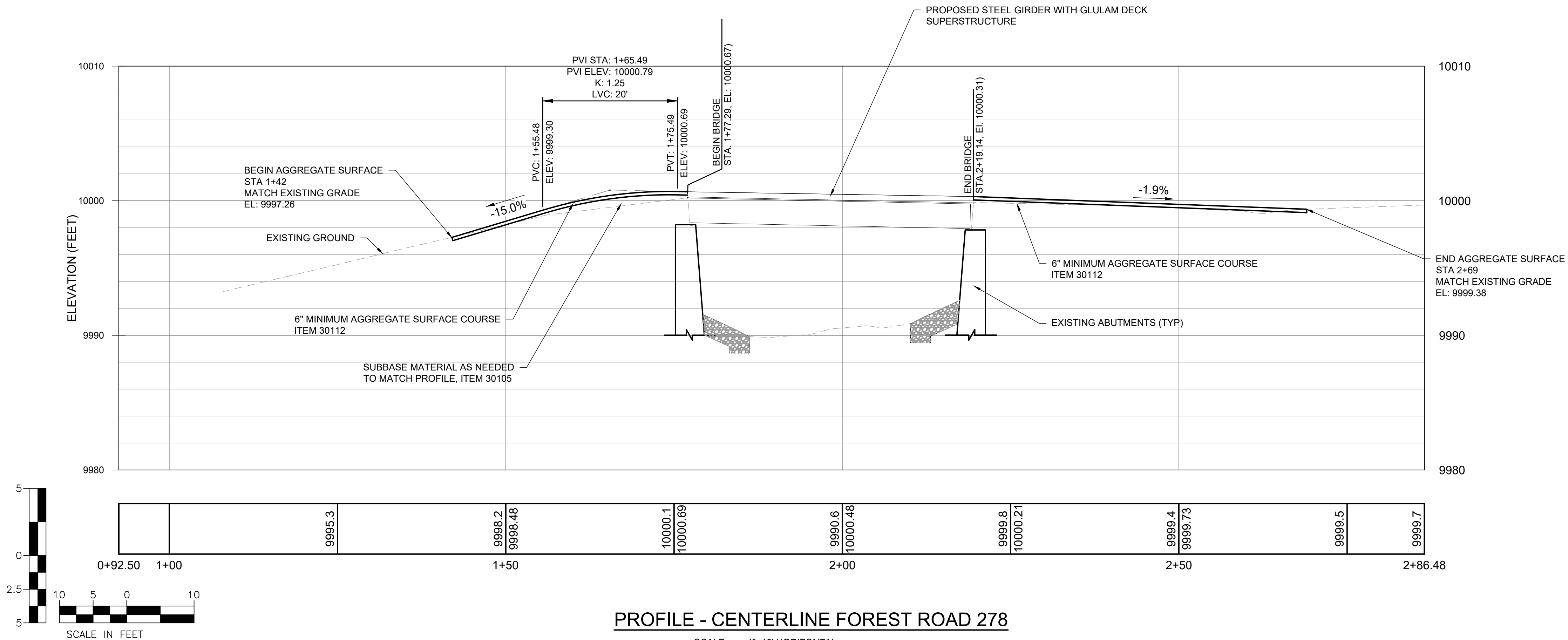
MANCHESTER RANGER DISTRICT

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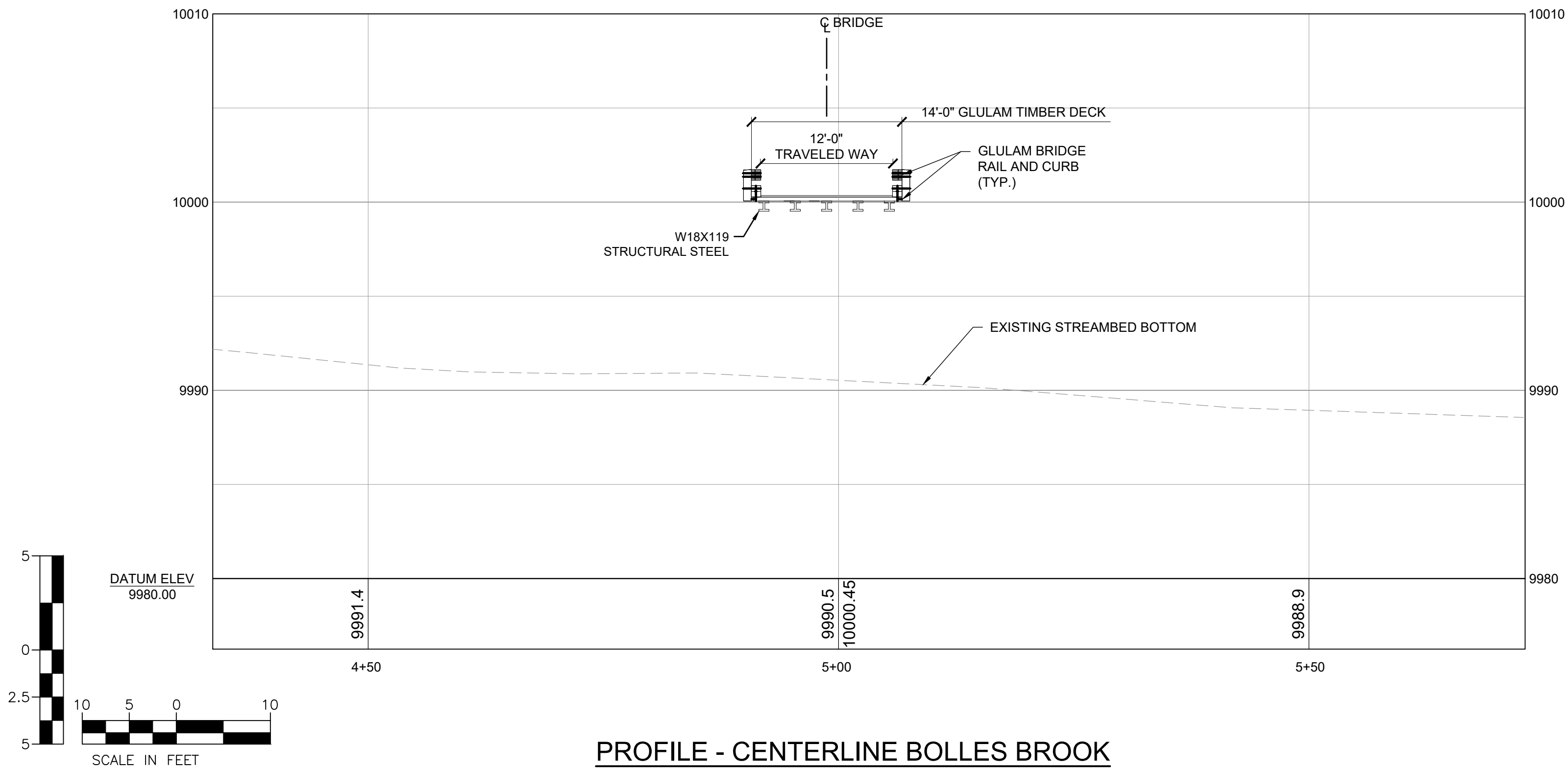
**EXISTING CONDITIONS
PLAN**

DATE 7/12/2024	ARCHIVE NO.
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DRAWN B. GRAVES	
CHECKED D. CONGER	
PROJECT NO. 629065	SHEET 04 OF 18

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PROFILE - CENTERLINE FOREST ROAD 278



PROFILE - CENTERLINE BOLLES BROOK



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PROFILES

DATE 7/12/2024	ARCHIVE NO.
DESIGNER A. HUMMERS	DRAWING SHEET NO. C-05
DRAWN B. GRAVES	
CHECKED D. CONGER	
PROJECT NO. 629065	SHEET 05 OF 18

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inc

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LEGEND:



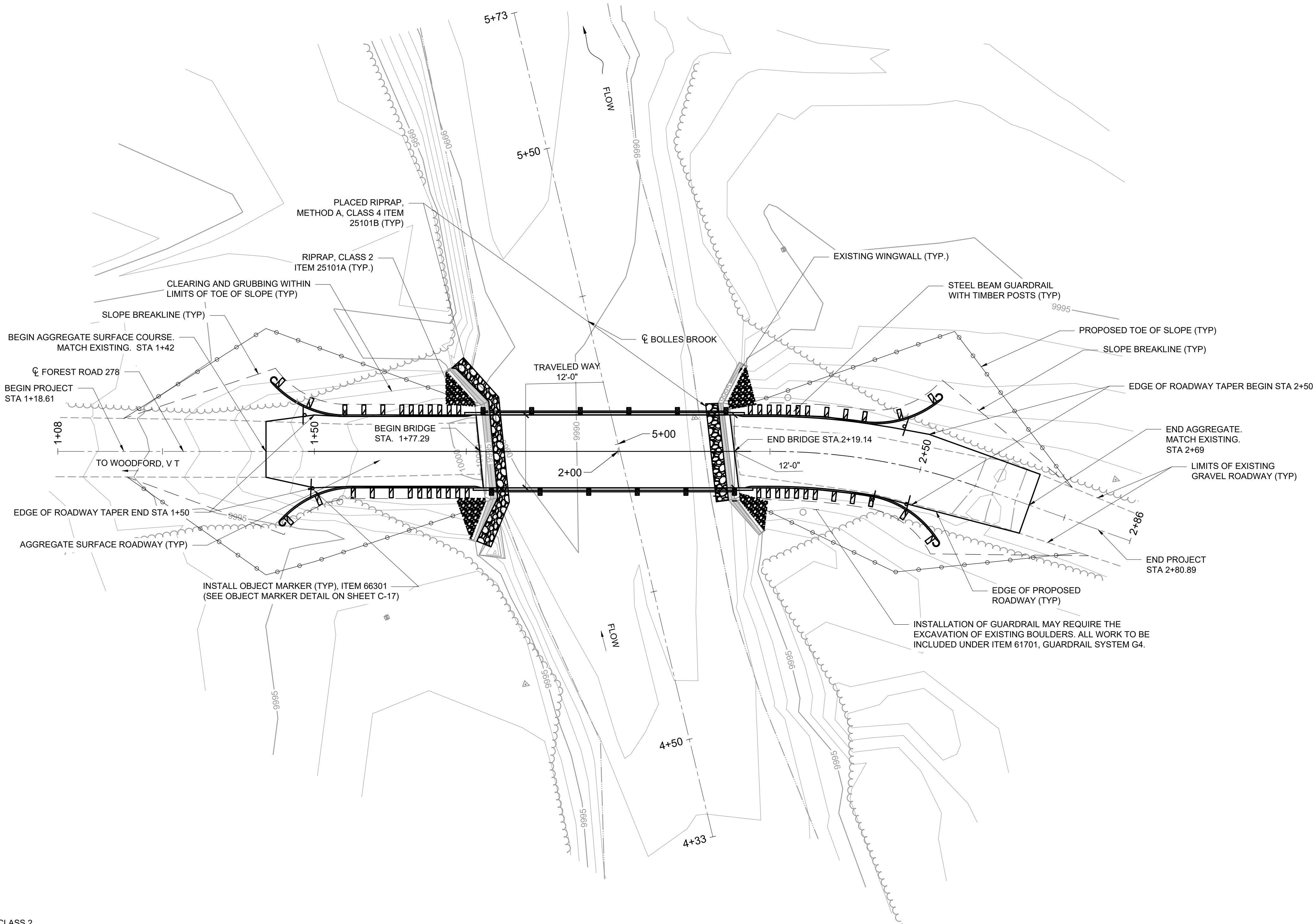
RIPRAP CLASS 2



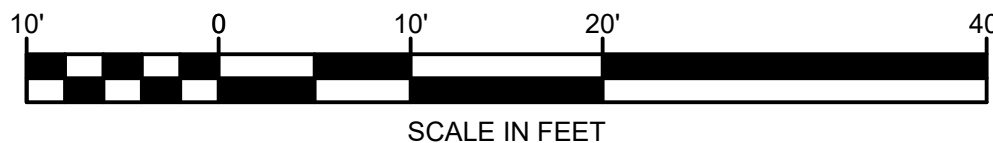
RIPRAP CLASS 4



AGGREGATE SURFACE COURSE



SITE PLAN



SCALE IN FEET



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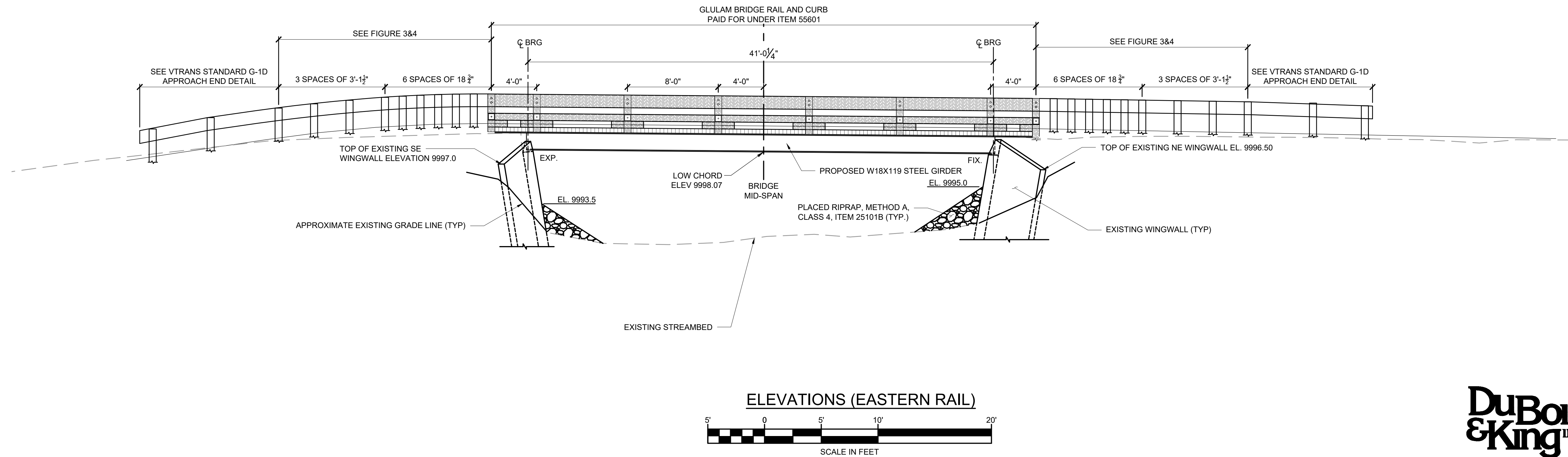
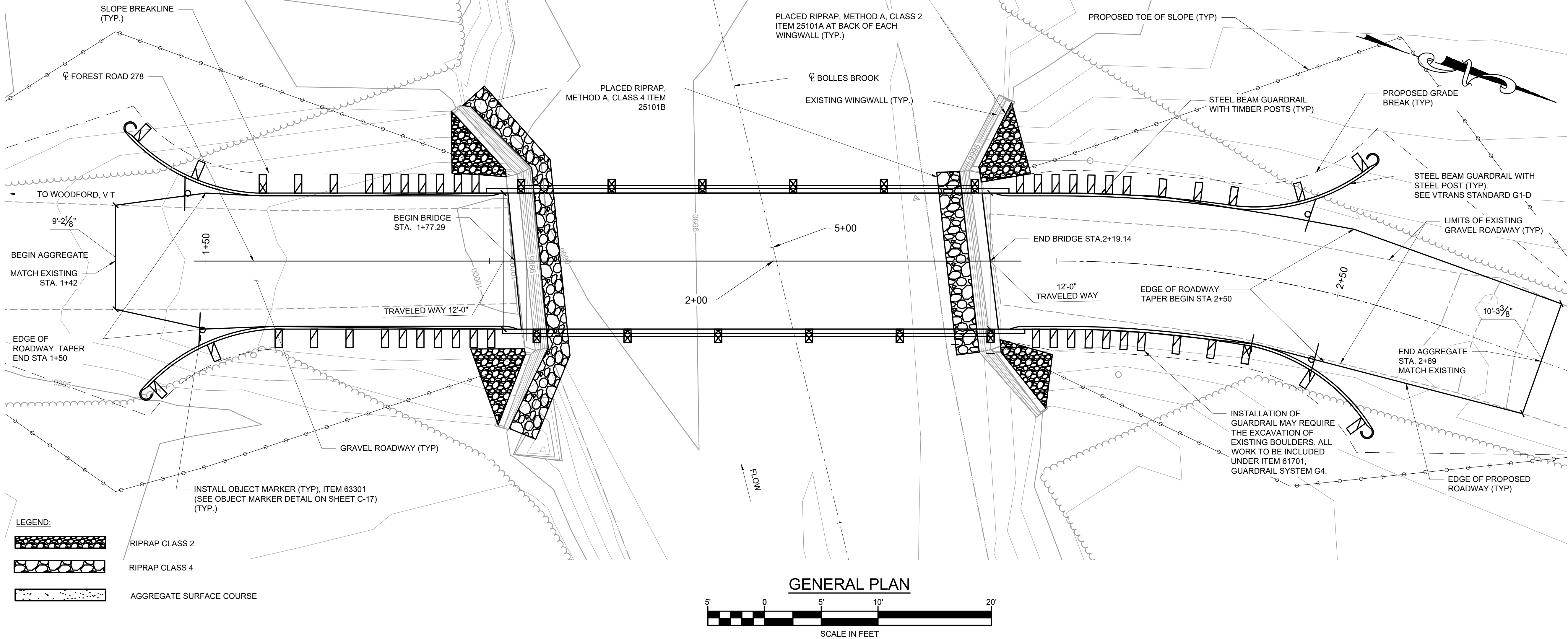
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SITE PLAN

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DESIGNER A. HUMMERS	DRAWING SHEET NO. C-06
DRAWN B. GRAVES	
CHECKED D. CONGER	
PROJECT NO. 629065	SHEET 06 OF 18

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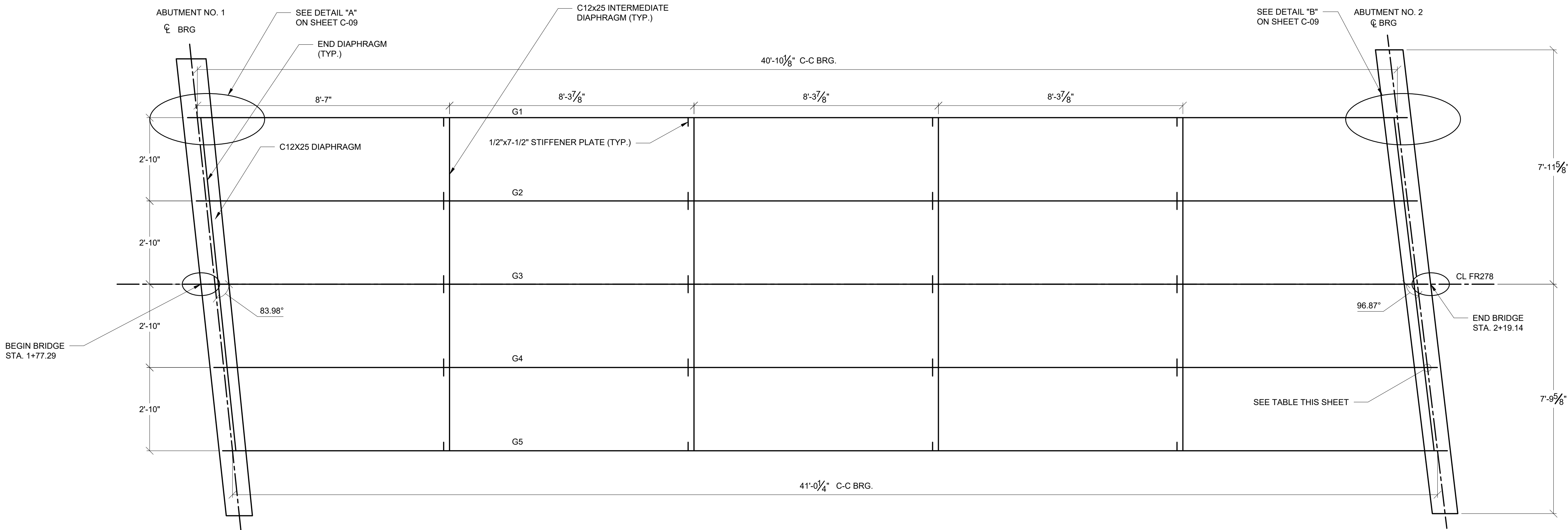
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GENERAL PLAN AND
ELEVATIONS

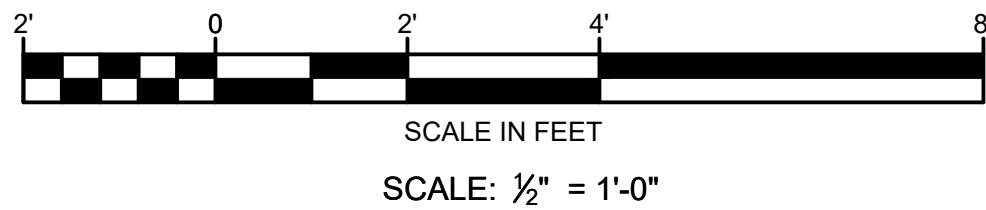
DATE 7/12/2024	ARCHIVE NO.
DESIGNER A. HUMMERS	DRAWING SHEET NO. C-07
DRAWN B. GRAVES	
CHECKED D. CONGER	
PROJECT NO. 629065	SHEET 07 OF 18

DuBois
& King inc

7/10/24 16:46 BGR/AVES I:\629065 GMMF FR278 BRIDGE SUPERSTRUCTURE REPLACEMENT\DRAWINGS\CAD\629065_07 FRAMING.DWG;

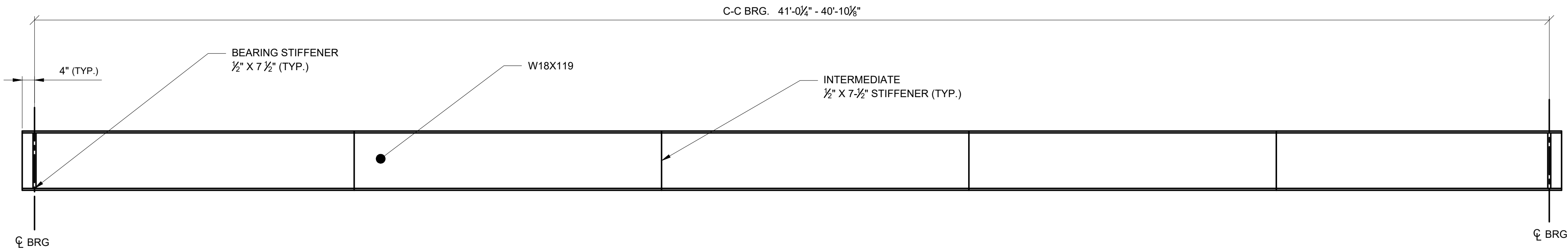


BRIDGE FRAMING PLAN

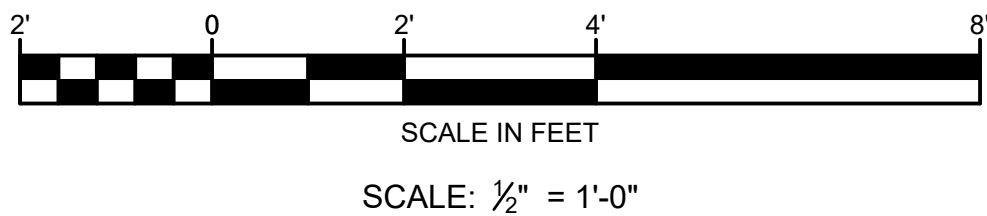


TOP OF STEEL EL @ BRG CL (ft)		
	ABUT. 1	ABUT. 2
G1	10000.08	9999.74
G2	10000.08	9999.74
G3	10000.08	9999.74
G4	10000.08	9999.73
G5	10000.07	9999.73

TOP OF STEEL



GIRDER ELEVATION



NOTE:
CONTRACTOR TO FIELD VERIFY ALL DIMENSIONS
PRIOR TO SUBMITTING FABRICATION DRAWINGS.



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EASTERN REGION

STAMPS, LOGOS, AND SEALS



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PROJECT NAME

FR 278-0.24
BRIDGE
SUPERSTRUCTURE
REPLACEMENT

GREEN MOUNTAIN
NATIONAL FOREST

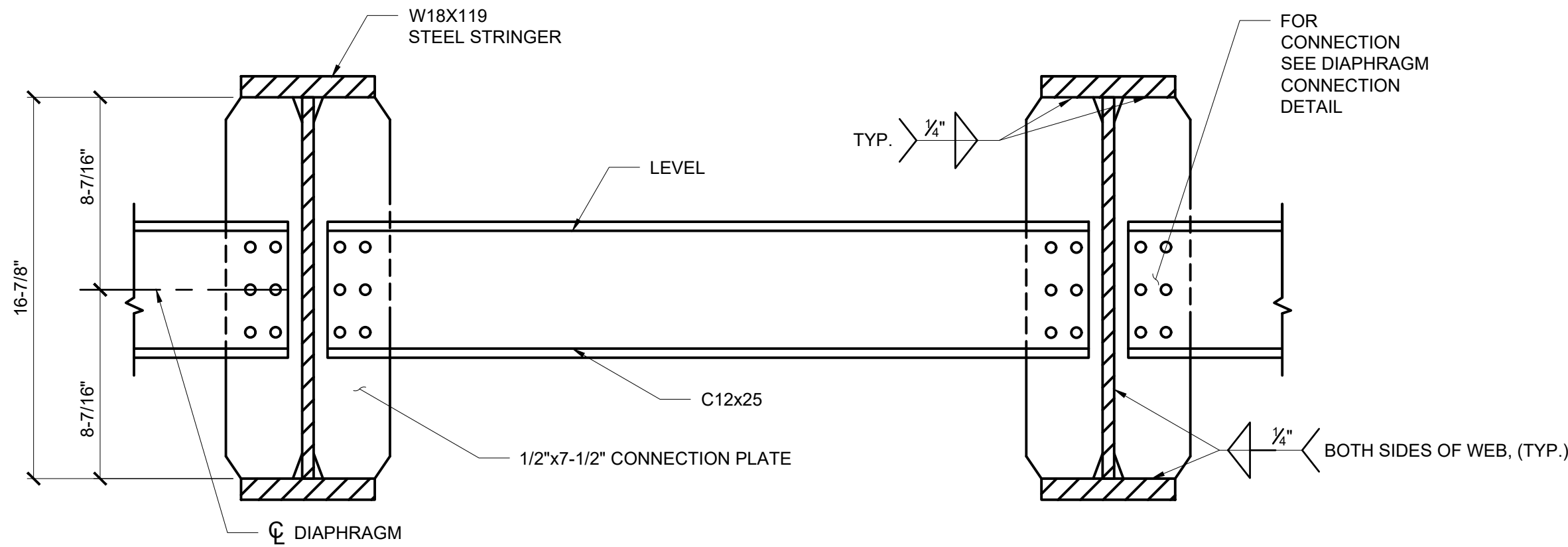
MANCHESTER RANGER DISTRICT

DRAWING TITLE

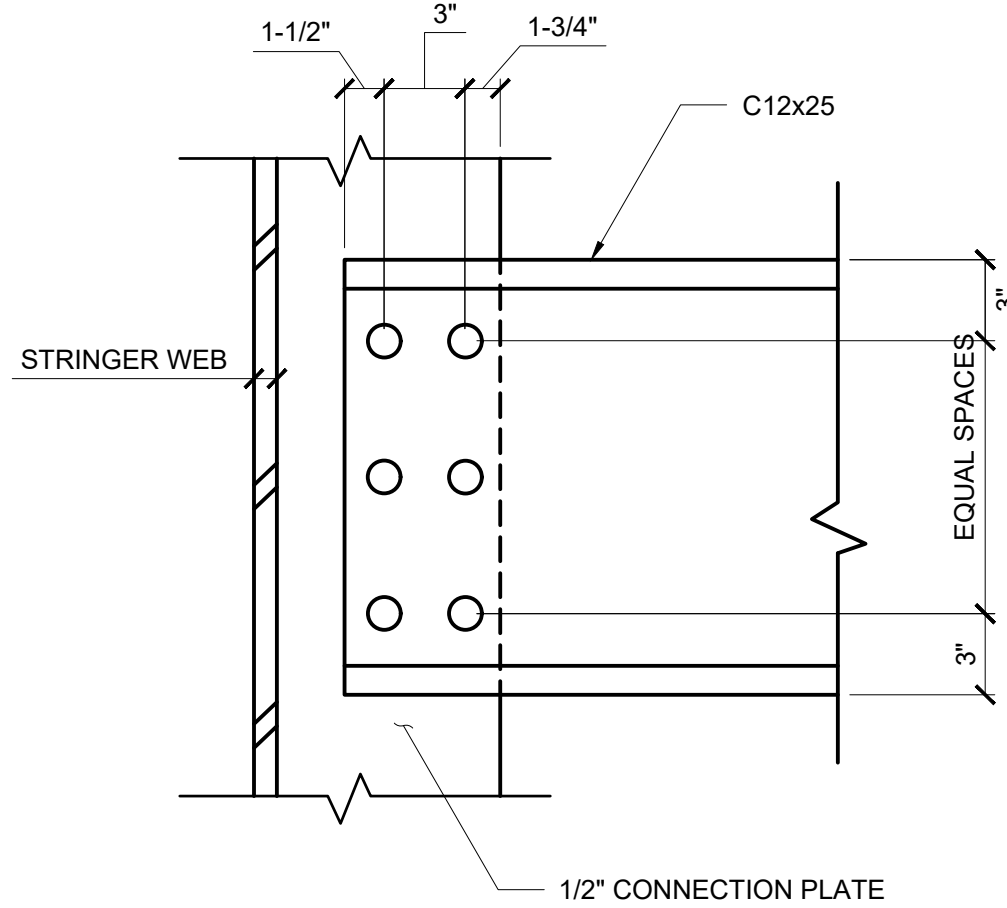
FRAMING PLAN AND
GIRDER ELEVATION

DATE 7/12/2024	ARCHIVE NO.
DESIGNER A. HUMMERS	DRAWING SHEET NO. C-08
DRAWN B. GRAVES	
CHECKED D. CONGER	
PROJECT NO. 629065	SHEET 08 OF 18

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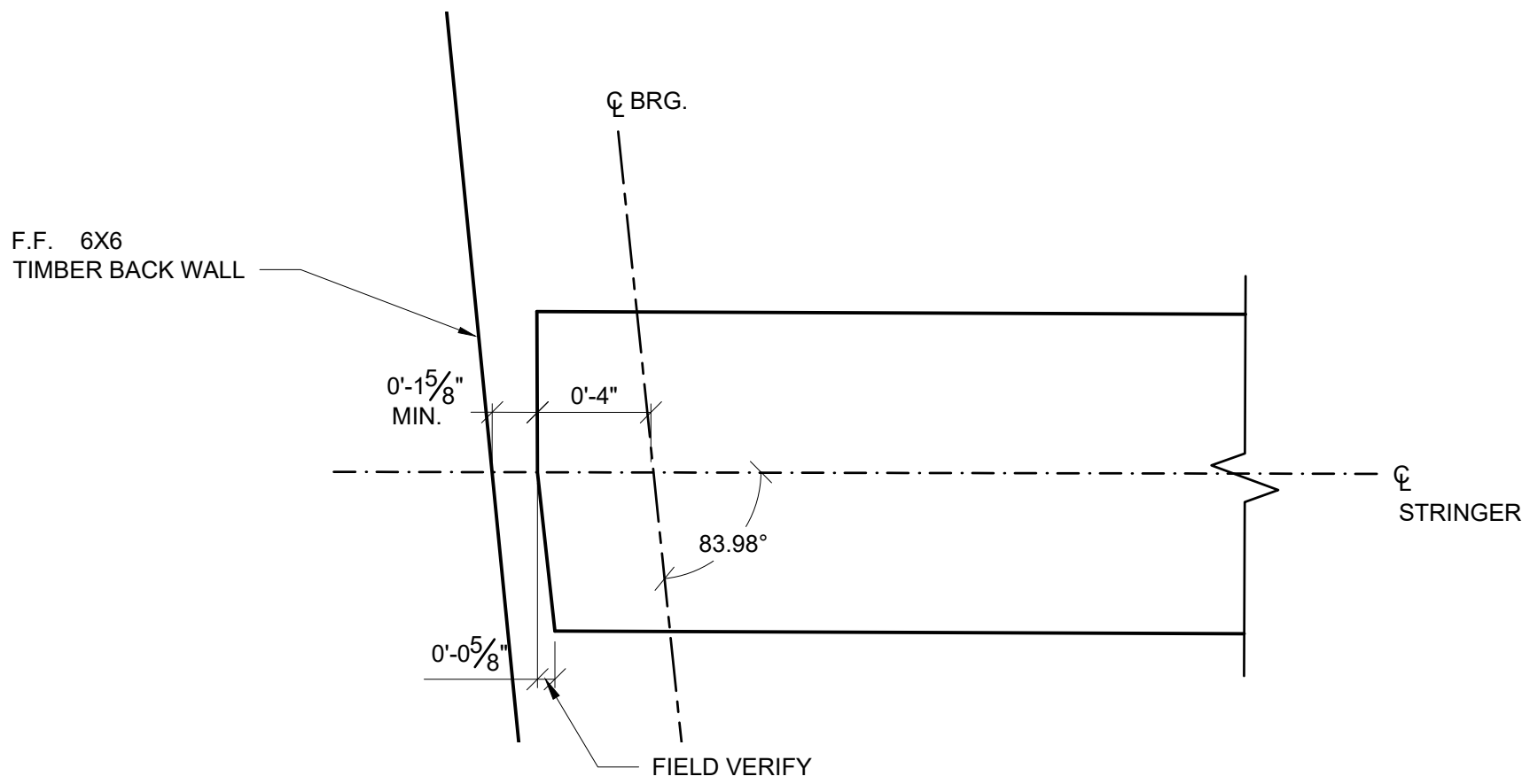


DIAPHRAGM DETAIL
SCALE: NOT TO SCALE

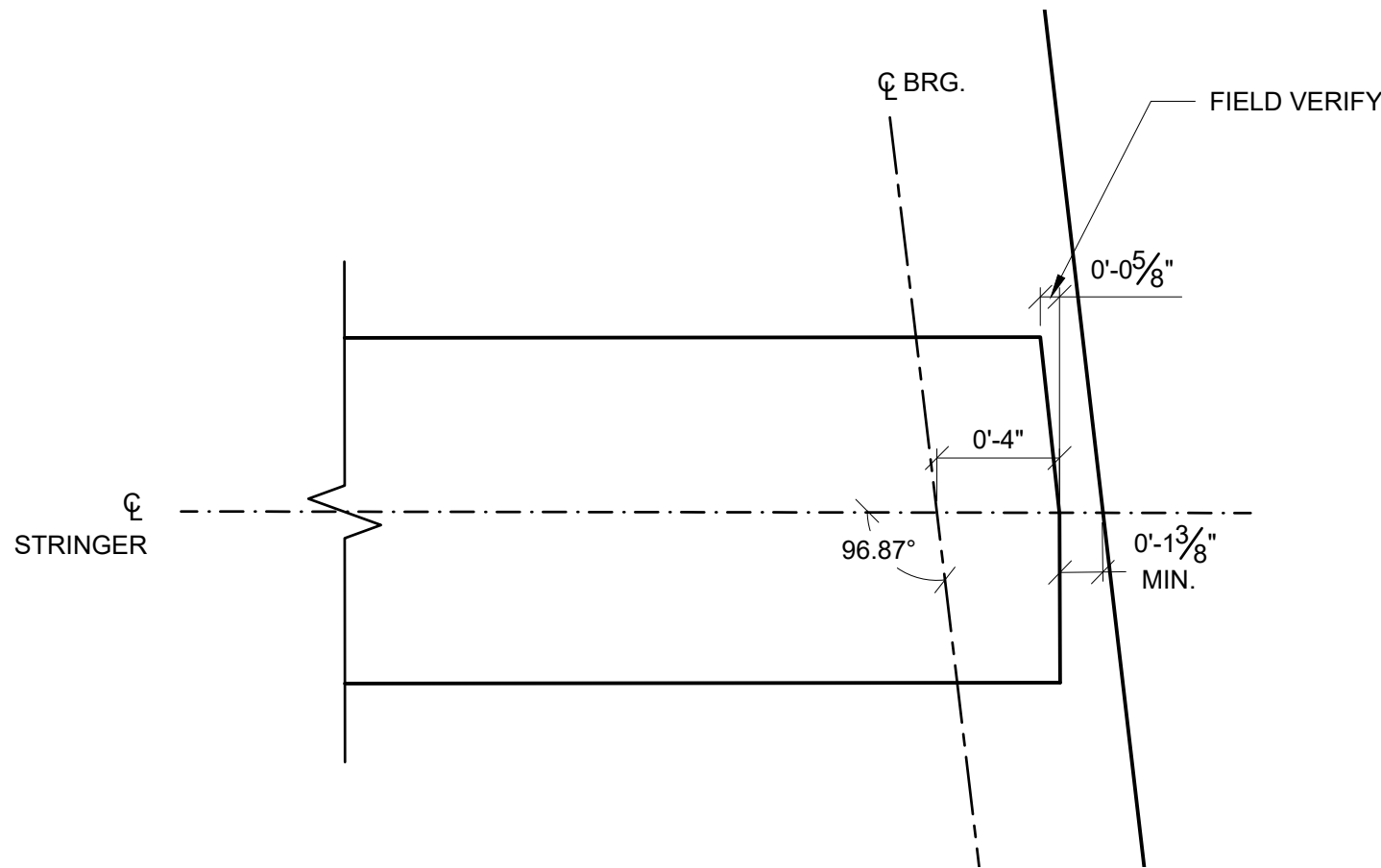


DIAPHRAGM CONNECTION DETAIL
SCALE: NOT TO SCALE

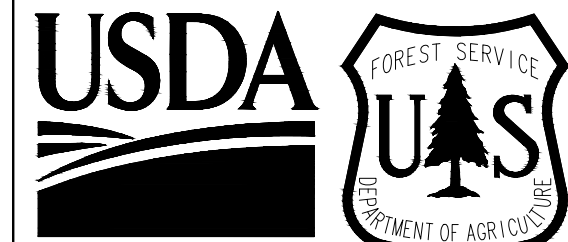
- NOTES:
- 1-1/16" DIAMETER HOLE IN CONNECTION PLATE; 15/16" DIAMETER HOLE IN CONNECTING MEMBER FOR 7/8" DIAMETER ASTM A325 BOLTS STD. SIZE HOLES ARE PERMITTED
 - USE 7/8" DIAMETER ASTM A325 BOLTS HAVING AN UNTHREADED SHANK OF SUFFICIENT LENGTH TO NOT ALLOW ANY THREADS TO EXIST IN THE PLANE BETWEEN THE TWO CONNECTED PARTS (SHEAR PLANE)



A - TYP. CLIPPING DETAIL @ ABUT. 1
SCALE: 2" = 1'

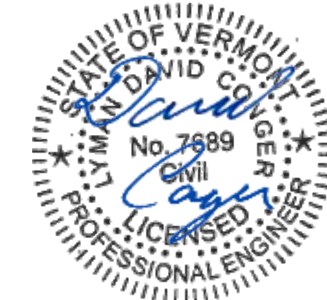


B - TYP. CLIPPING DETAIL @ ABUT. 2
SCALE: 2" = 1'



(R09)
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BRIDGE
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NATIONAL FOREST

MANCHESTER RANGER DISTRICT

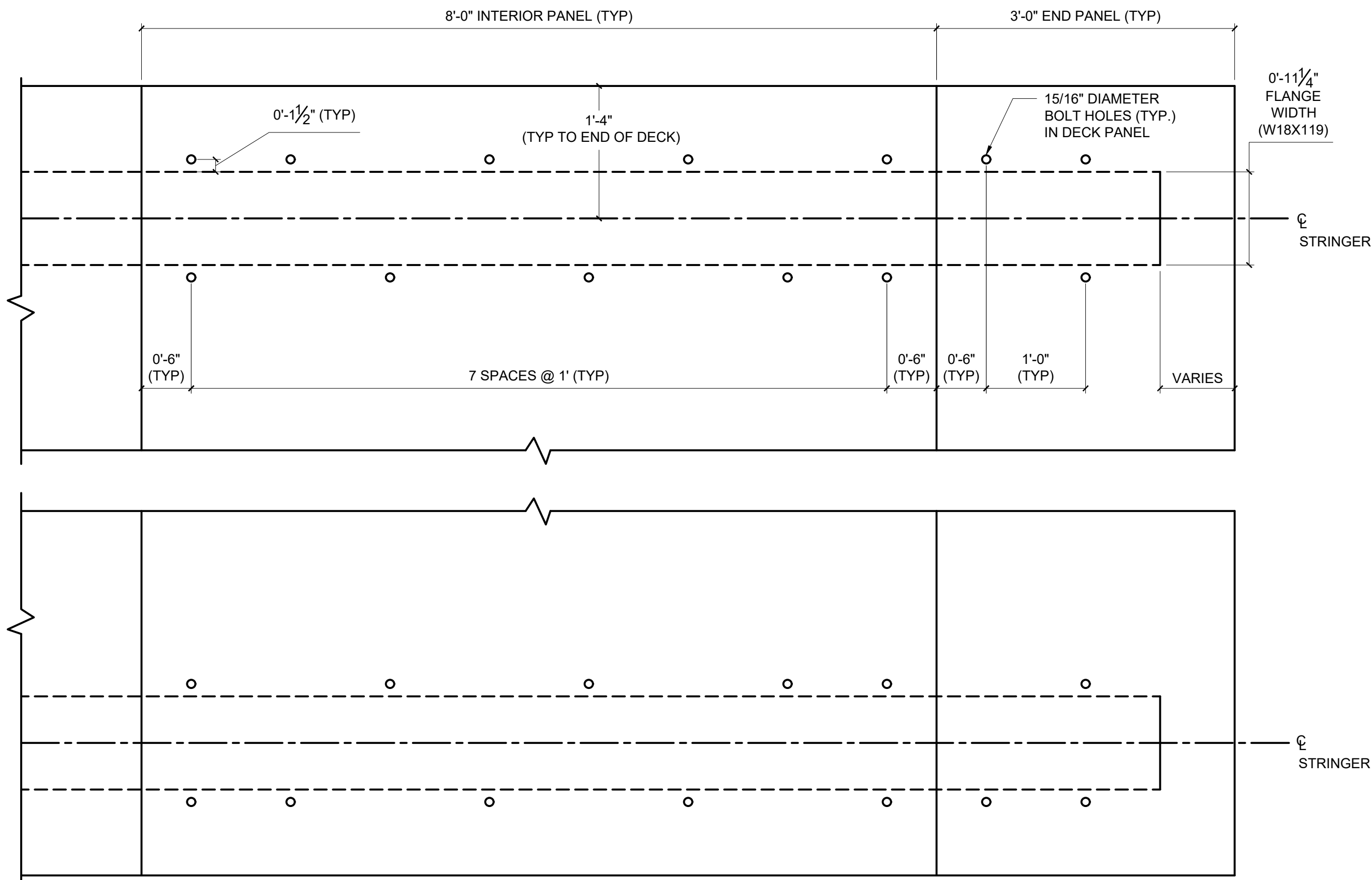
DRAWING TITLE

DIAPHRAGM AND
CLIPPING DETAILS

DATE 7/12/2024	ARCHIVE NO.
DESIGNER A. HUMMERS	DRAWING SHEET NO. C-09
DRAWN B. GRAVES	
CHECKED D. CONGER	
PROJECT NO. 629065	SHEET 09 OF 18

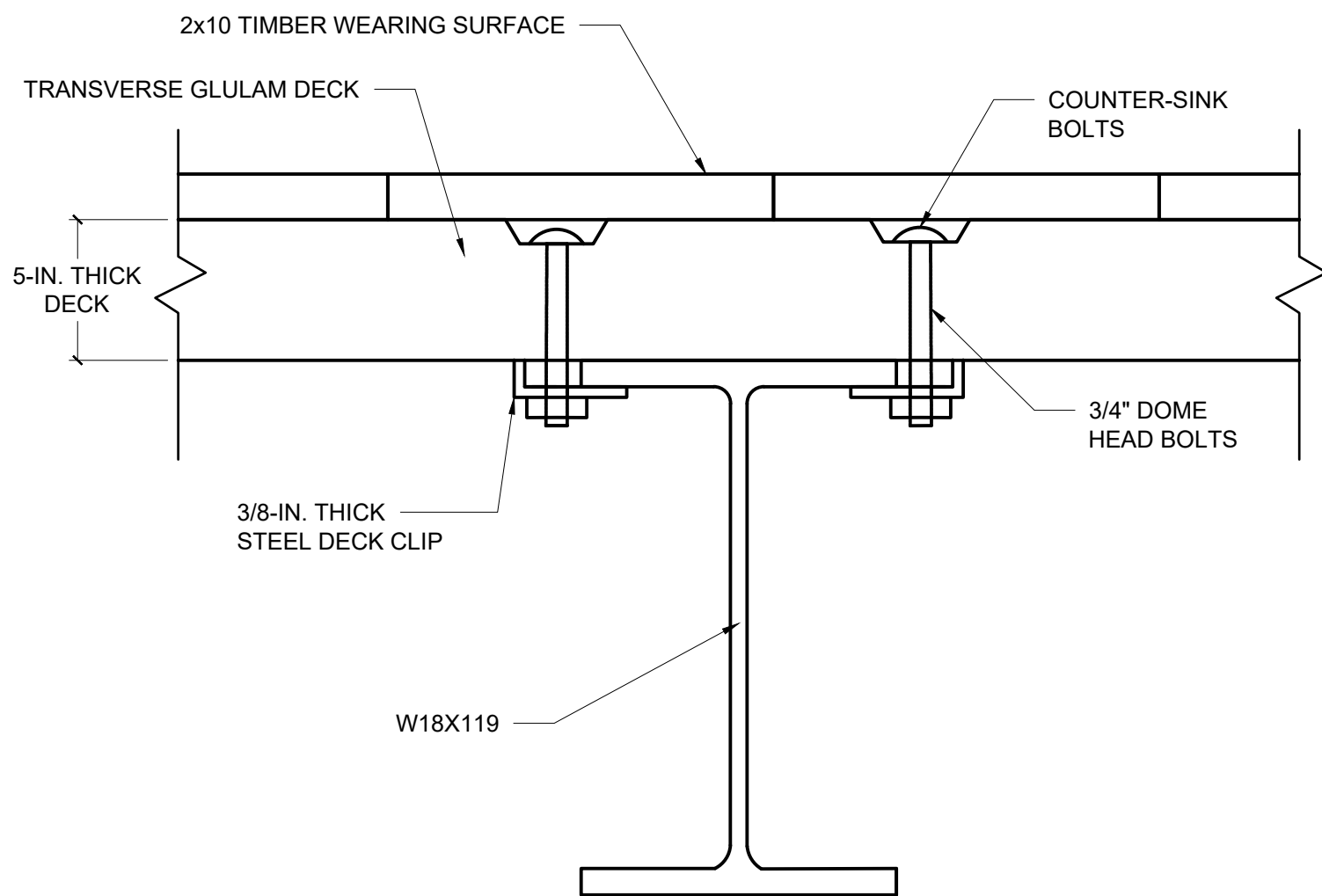
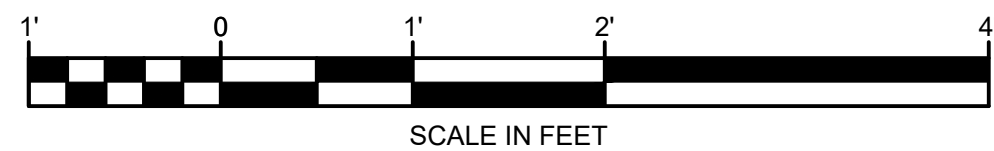
DuBois
& King
INC

5/3/24 10:54 ODAIMER I:\629065 04MF FR278 BRIDGE SUPERSTRUCTURE REPLACEMENT\DRAWINGS\629065_06 DECK DETAILS.DWG

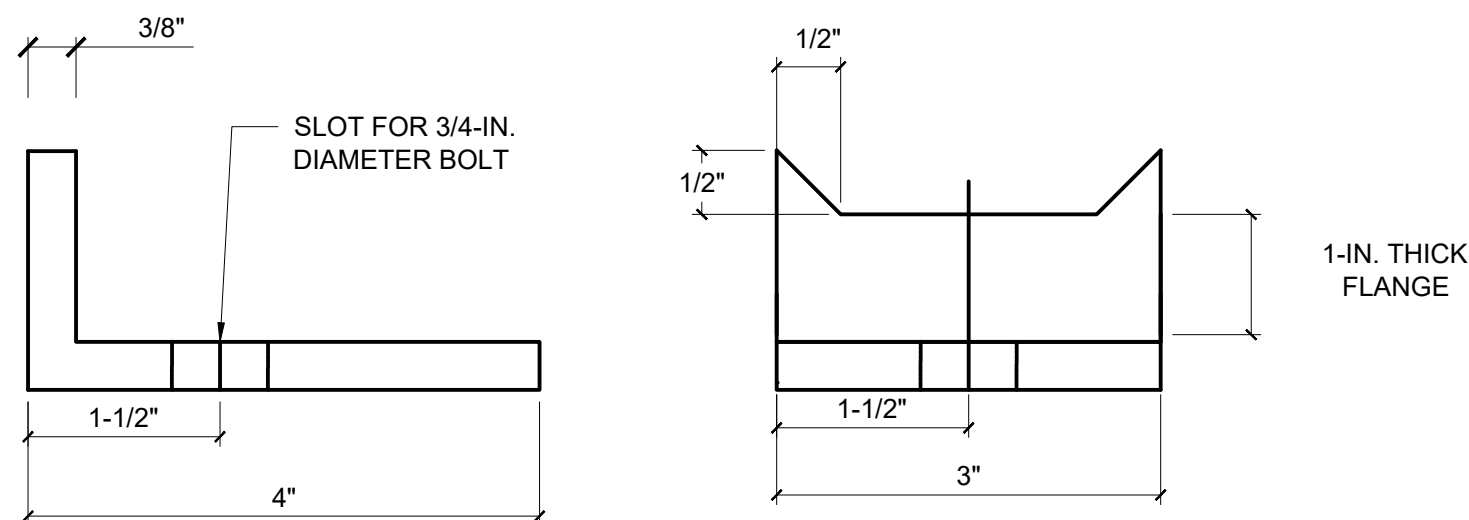


NOTE:
SYMMETRIC ABOUT CENTERLINE OF BRIDGE.
CONTRACTOR TO FIELD VERIFY END PANEL DIMENSION.

DECK TO GIRDER CONNECTION PLAN



DECK TO GIRDER CONNECTION DETAIL
NOT TO SCALE



CLIP DETAIL
NOT TO SCALE

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STAMPS, LOGOS, AND SEALS



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PROJECT NAME

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BRIDGE
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MANCHESTER RANGER DISTRICT

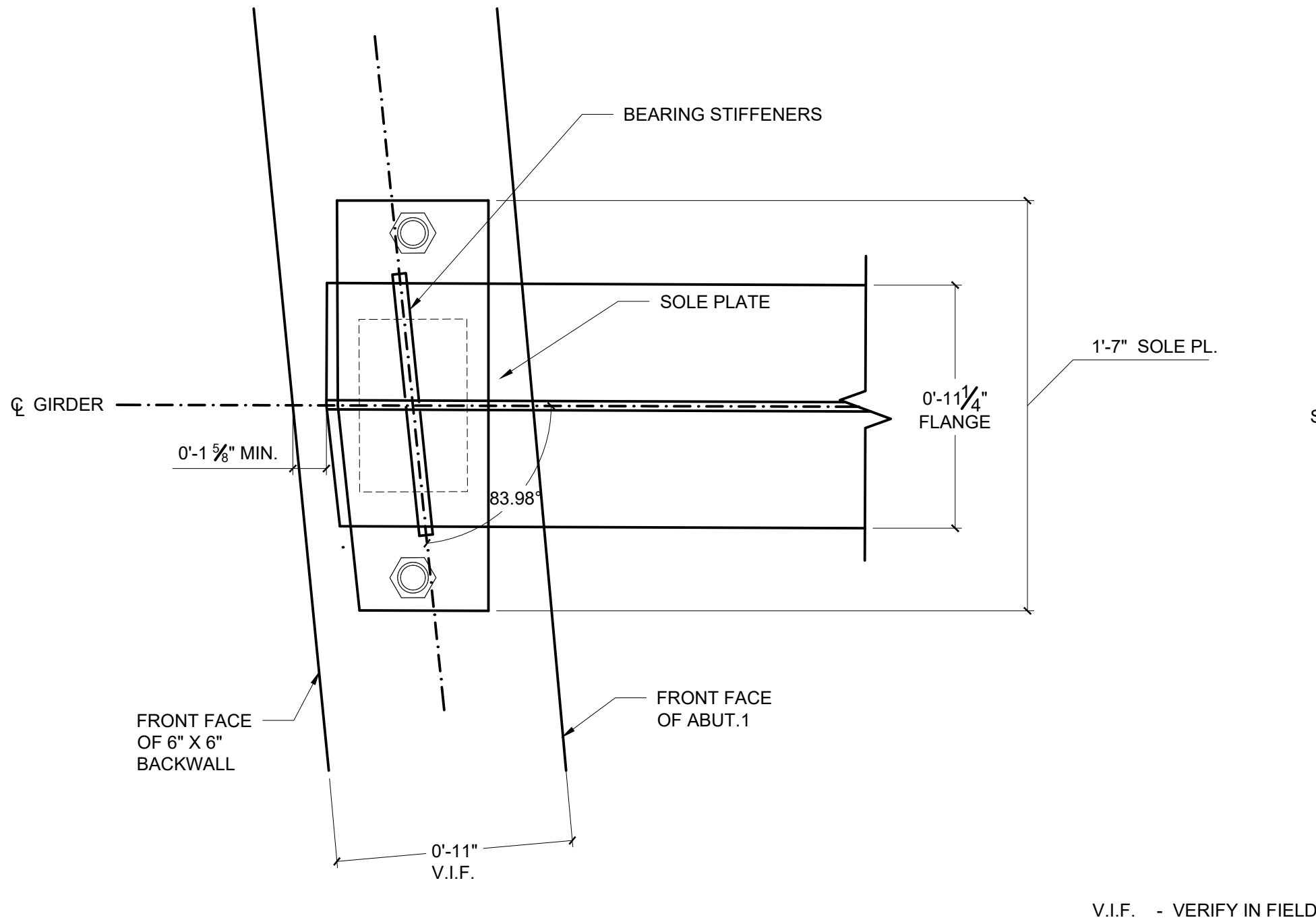
DRAWING TITLE

DECK PLAN AND
DETAILS

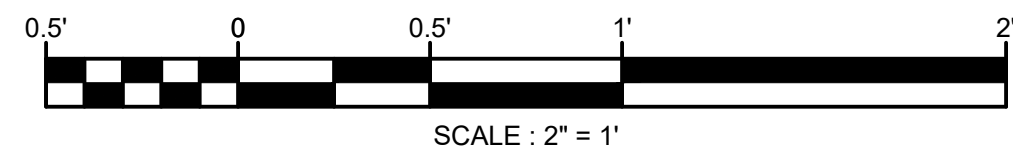
DATE 7/12/2024	ARCHIVE NO.
DESIGNER A. HUMMERS	DRAWING SHEET NO. C-10
DRAWN B. GRAVES	
CHECKED D. CONGER	
PROJECT NO. 629065	SHEET 10 OF 18

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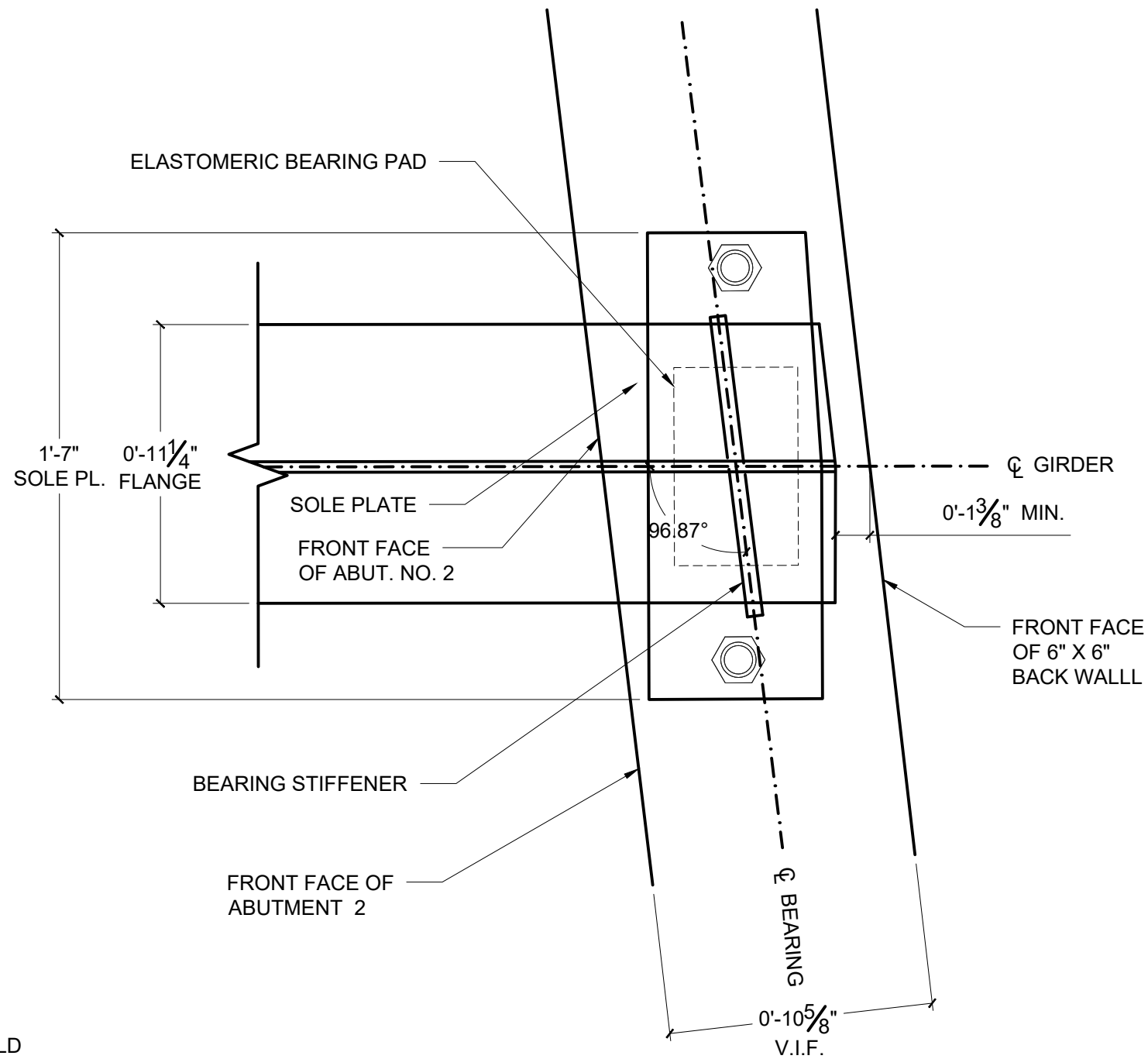
7/12/24 14:48 BGRAVES I:\0629065 QMNF FR278 BRIDGE SUPERSTRUCTURE REPLACEMENT\DRAWINGS\CAD\629065_06 BEARING DEVICES.DWG;



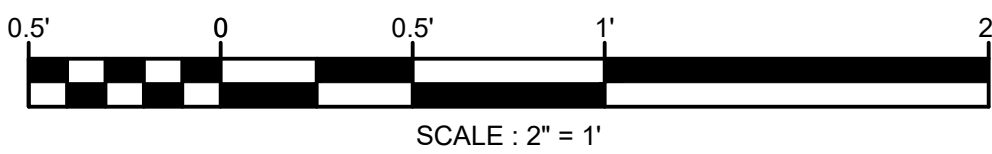
TYPICAL EXPANSION BEARING PLAN



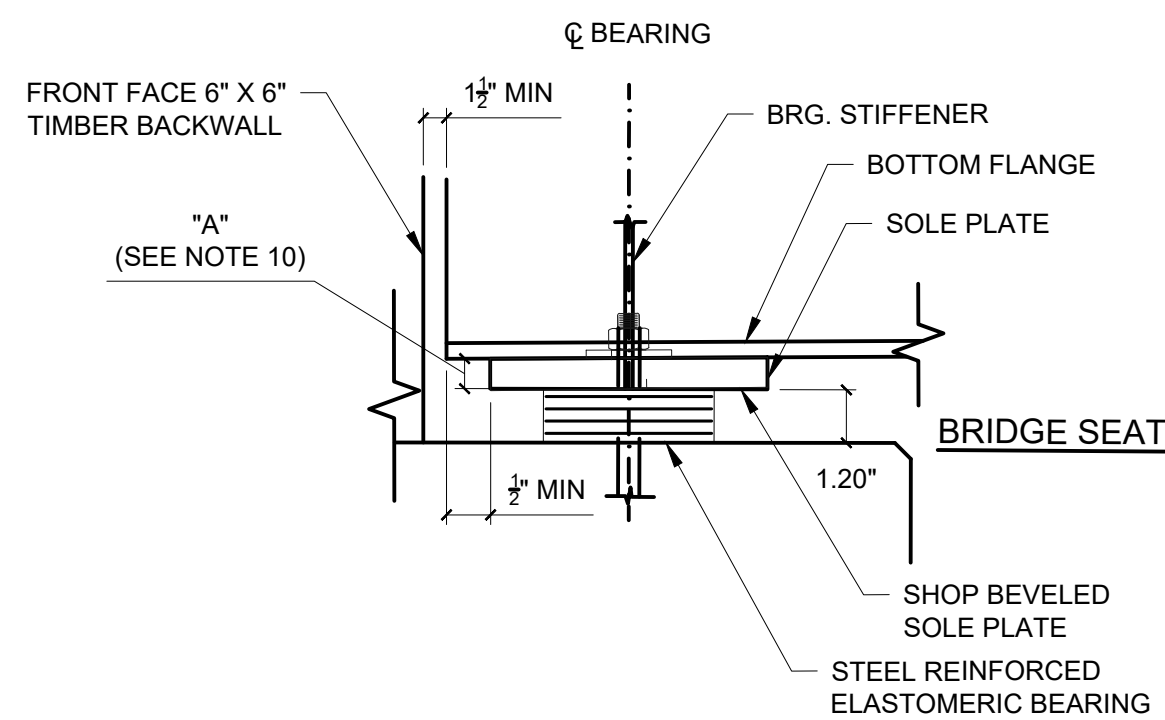
SCALE : 2" = 1'



TYPICAL FIXED BEARING PLAN

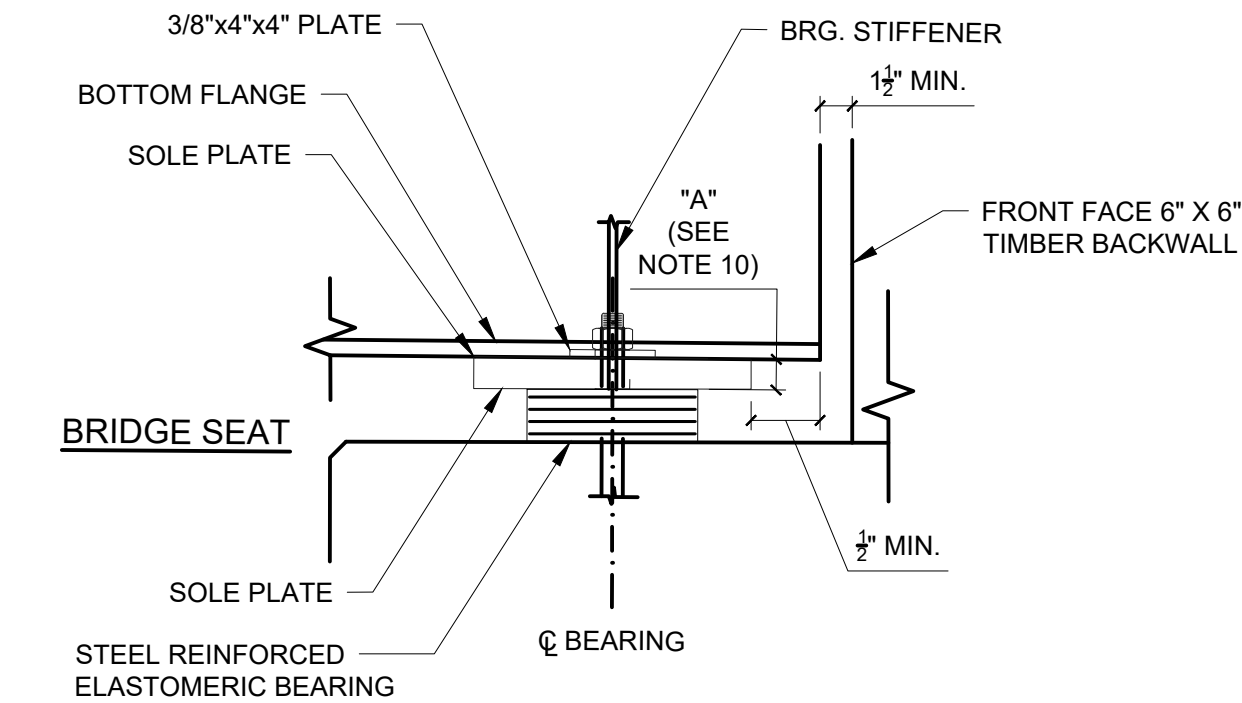


SCALE : 2" = 1'



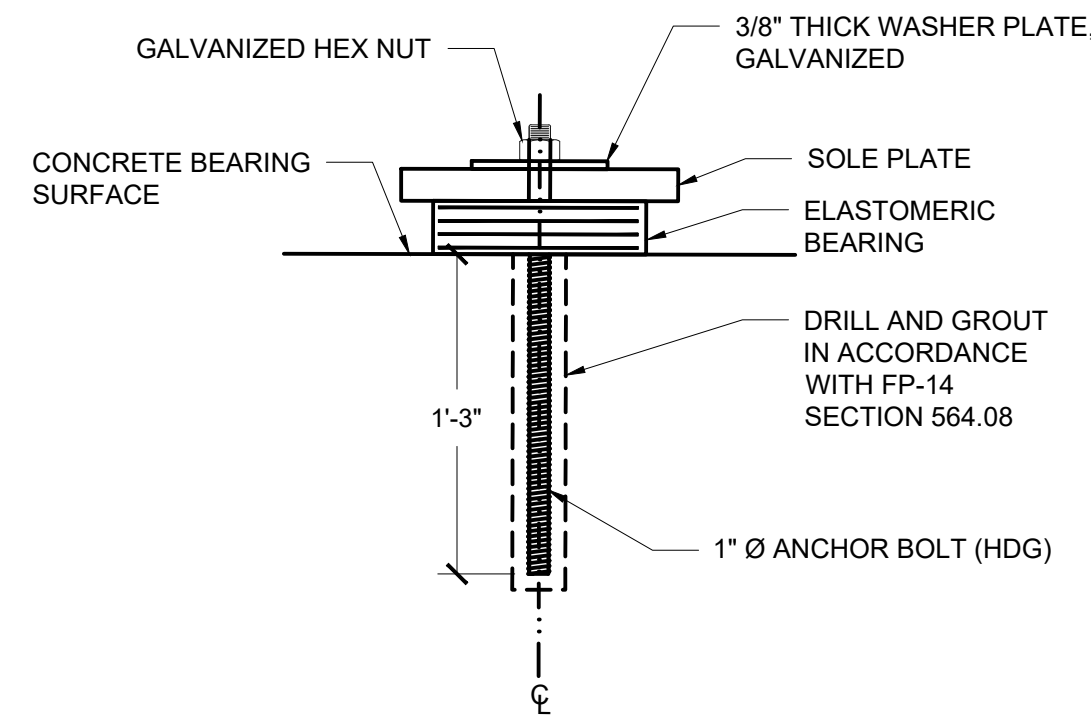
TYPICAL EXPANSION BEARING ELEVATION

NOT TO SCALE



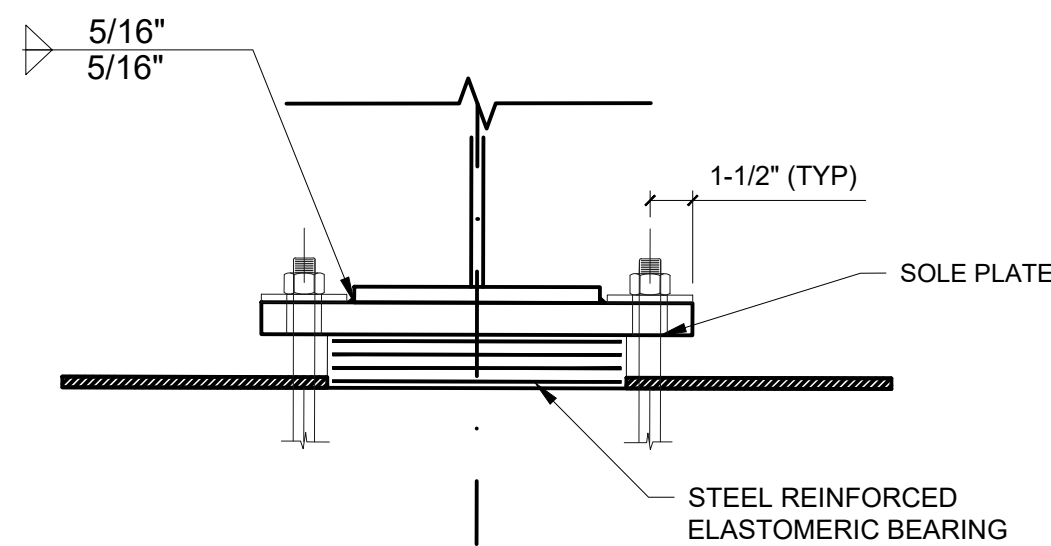
TYPICAL FIXED BEARING ELEVATION

NOT TO SCALE



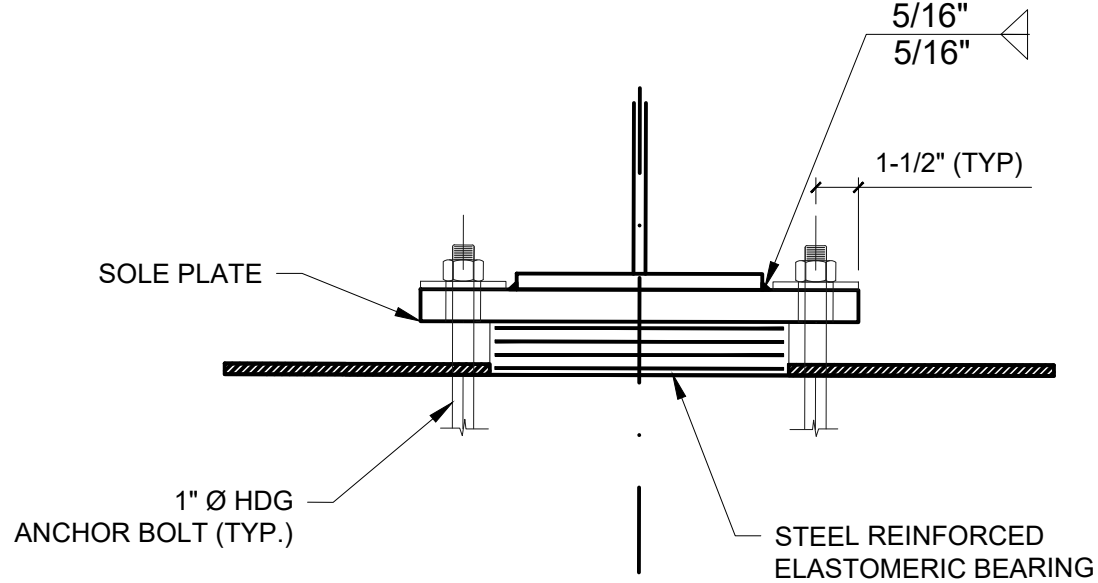
ANCHOR BOLT DETAIL

NOT TO SCALE



TYPICAL EXPANSION BEARING SECTION

NOT TO SCALE



TYPICAL FIXED BEARING SECTION

NOT TO SCALE

BEARING NOTES:

- BEARINGS WILL BE PAID FOR UNDER ITEM 56401, "BEARING DEVICE ELASTOMERIC", AND SHALL CONFORM TO APPLICABLE SUBSECTIONS OF SECTION 564.
- FABRICATION DRAWINGS, INCLUDING ASSOCIATED WELDING PROCEDURES, SHALL BE SUBMITTED CONFORMING TO SUBSECTION 564.
- DESIGN CRITERIA :
 - DEAD LOAD UNFACTORED: 38.1 KIPS
 - LIVE LOAD UNFACTORED: 71.9 KIPS
 - DEAD LOAD ROTATION: .007 RADIANS
 - ALL STEEL IN BEARINGS DEVICES SHALL BE AASHTO M 270/M 270 GRADE 36, UNLESS OTHERWISE NOTED.
- ALL INTERNAL STEEL SHIMS, IN THE ELASTOMERIC BEARINGS SHALL BE SAND BLASTED AND FREE OF COATINGS, RUST AND MILL SCALE. THE SHEETS SHALL BE FREE OF SHARP EDGES AND BURRS.
- STEEL REINFORCED ELASTOMERIC BEARINGS SHALL HAVE MINIMUM 1/8" EDGE SEAL OF ELASTOMER INTEGRAL WITH BEARING OVER ALL INTERNAL SHEETS.
- CONTRACTOR TO PROVIDE LEVEL CONCRETE BRIDGE SEATS AT BEARINGS. GROUT LEVELING PADS SHALL BE PERMITTED. SEE SHEET C-12.
- ANCHOR BOLTS SHALL HAVE A MINIMUM 15 INCHES EMBEDMENT INTO THE CONCRETE AND SHALL CONFORM TO SUBSECTION 717.01. ANCHOR BOLTS SHALL BE GRADE 36.
- THE ELASTOMERIC MATERIAL SHALL CONFORM TO SECTION 717 WITH THE FOLLOWING EXCEPTIONS:
HARDNESS = 60 (SHORE A)
SHEAR MODULUS = 165 PSI
- ALL BEARINGS SHALL BE MARKED PRIOR TO SHIPPING. MARKS SHALL INCLUDE THE BEARING LOCATION ON THE BRIDGE AND A DIRECTION ARROW THAT POINTS UP-STATION. ALL MARKS SHALL BE PERMANENT AND VISIBLE AFTER THE BEARING IS INSTALLED.
- VULCANIZED SOLE PLATE THICKNESS SHALL BE AS PER THE SOLE PL. DETAIL, ON SHEET C-12. CONTRACTOR SHALL FIELD VERIFY ABUTMENT SEAT ELEVATIONS AS PART OF BEARING SUBMITTAL.
- ELASTOMERIC BEARING SHALL BE FACTORY VULCANIZED TO THE SOLE PLATES.
- SOLE PLATES SHALL BE PAINTED WITH A SHOP APPLIED PAINT SYSTEM IN ACCORDANCE WITH SECTION 563 OF THE SPECIFICATIONS. ANY AREAS OF DAMAGE DURING THE SHIPPING OR PLACEMENT OF THE STEEL SHALL BE REPAIRED IN ACCORDANCE WITH THIS SECTION. THE COLOR SHALL CONFORM TO FEDERAL STANDARD NO. 595, COLOR CHIP NUMBER 20059 AND SHALL BE APPROVED BY THE C.O.R

NOTES:

- WELDS SHALL TERMINATE 1/4" FROM EDGE OF PLATE.
- ALL COSTS ASSOCIATED WITH THE FABRICATION, TRANSPORTATION AND INSTALLATION OF THE SOLE PLATES, INCLUDING ALL HARDWARE AND TOOLS NECESSARY SHALL BE INCIDENTAL TO ITEM 55502, STRUCTURAL STEEL, ROLLED BEAM.
- ALL COSTS ASSOCIATED WITH THE INSTALLATION OF THE ANCHOR BOLTS, INCLUDING HARDWARE, SHALL BE INCIDENTAL TO ITEM 56401.



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PROJECT NAME

FR 278-0.24
BRIDGE
SUPERSTRUCTURE
REPLACEMENT

GREEN MOUNTAIN
NATIONAL FOREST

MANCHESTER RANGER DISTRICT

DRAWING TITLE

BEARING DEVICES

DATE 7/12/2024	ARCHIVE NO.
DESIGNER A. HUMMERS	DRAWING SHEET NO.
DRAWN B. GRAVES	C-11
CHECKED D. CONGER	
PROJECT NO. 629065	SHEET 11 OF 18

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GROUT PAD ELEVATION
NOT TO SCALE



FIX. SOLE PL. PLAN
SCALE: 3" = 1'

- GROUT PAD / BRIDGE SEAT NOTES:**
1. CONTRACTOR SHALL VERIFY EXISTING ABUTMENT BRIDGE SEAT ELEVATIONS AFTER REMOVING EXISTING STRINGERS.
 2. PRIOR TO ERECTION OF PROPOSED STRINGERS, BRIDGE SEATS ARE TO BE THOROUGHLY CLEANED.
 3. IF BRIDGE SEAT ADJUSTMENT IS REQUIRED TO ATTAIN PROPOSED ROADWAY ELEVATIONS, CONTRACTOR SHALL REMOVE EXISTING ABUTMENT MATERIAL OR PROVIDE GROUT LEVELING PAD(S). GROUT PAD SHALL BE MINIMUM 1.5 INCHES THICK.
 4. GROUT FOR ANY LEVELING PAD SHALL CONFORM TO VTRANS 2024 STANDARD SPECIFICATIONS FOR CONSTRUCTION 707.01(E)(1).
 5. BEFORE PLACING GROUT, THE CONCRETE ON WHICH IT IS TO BE PLACED SHALL BE THOROUGHLY CLEANED, ROUGHENED, AND WETTED WITH WATER TO ENSURE PROPER BONDING.
 6. THE GROUT SHALL BE CURED AS RECOMMENDED BY THE MANUFACTURER OR KEPT CONTINUOUSLY WET WITH WATER FOR 3 DAYS. THE GROUT PAD MAY BE LOADED WHEN A MINIMUM 4,000 psi COMPRESSIVE STRENGTH IS ATTAINED.
 7. ALL WORK ASSOCIATED WITH ABUTMENT SEAT CLEANING, PREPARATION AND ASSOCIATED GROUT LEVELING PADS SHALL BE INCIDENTAL TO ITEM NO. 56401



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**FR 278-0.24
BRIDGE
SUPERSTRUCTURE
REPLACEMENT**

**GREEN MOUNTAIN
NATIONAL FOREST**

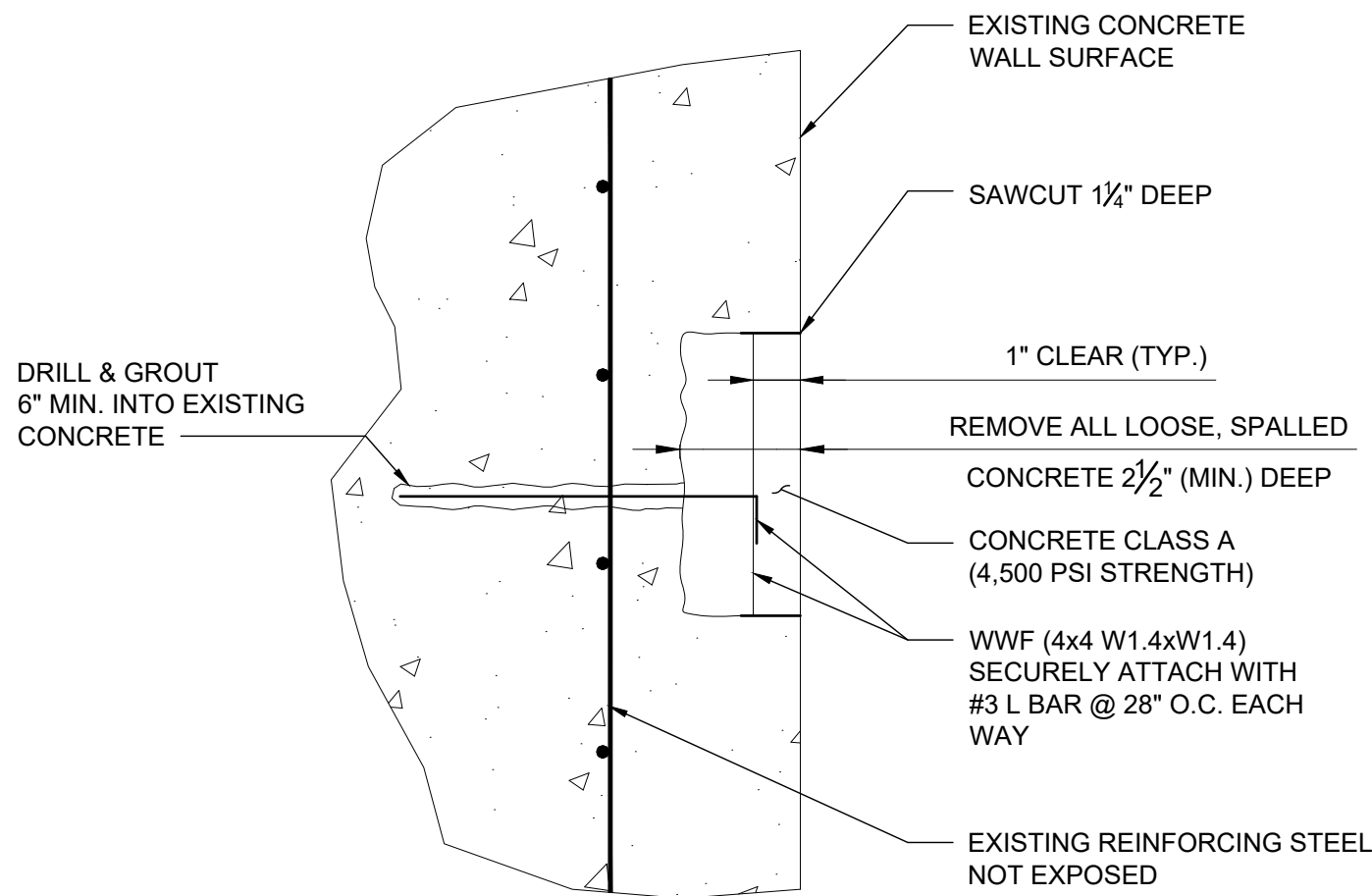
MANCHESTER RANGER DISTRICT

BEARING DEVICES

DATE 7/12/2024	ARCHIVE NO.	
DESIGNER A. HUMMERS	DRAWING SHEET NO.	C-12
DRAWN B. GRAVES		
CHECKED D. CONGER		
PROJECT NO. 629065	SHEET 12 OF 18	

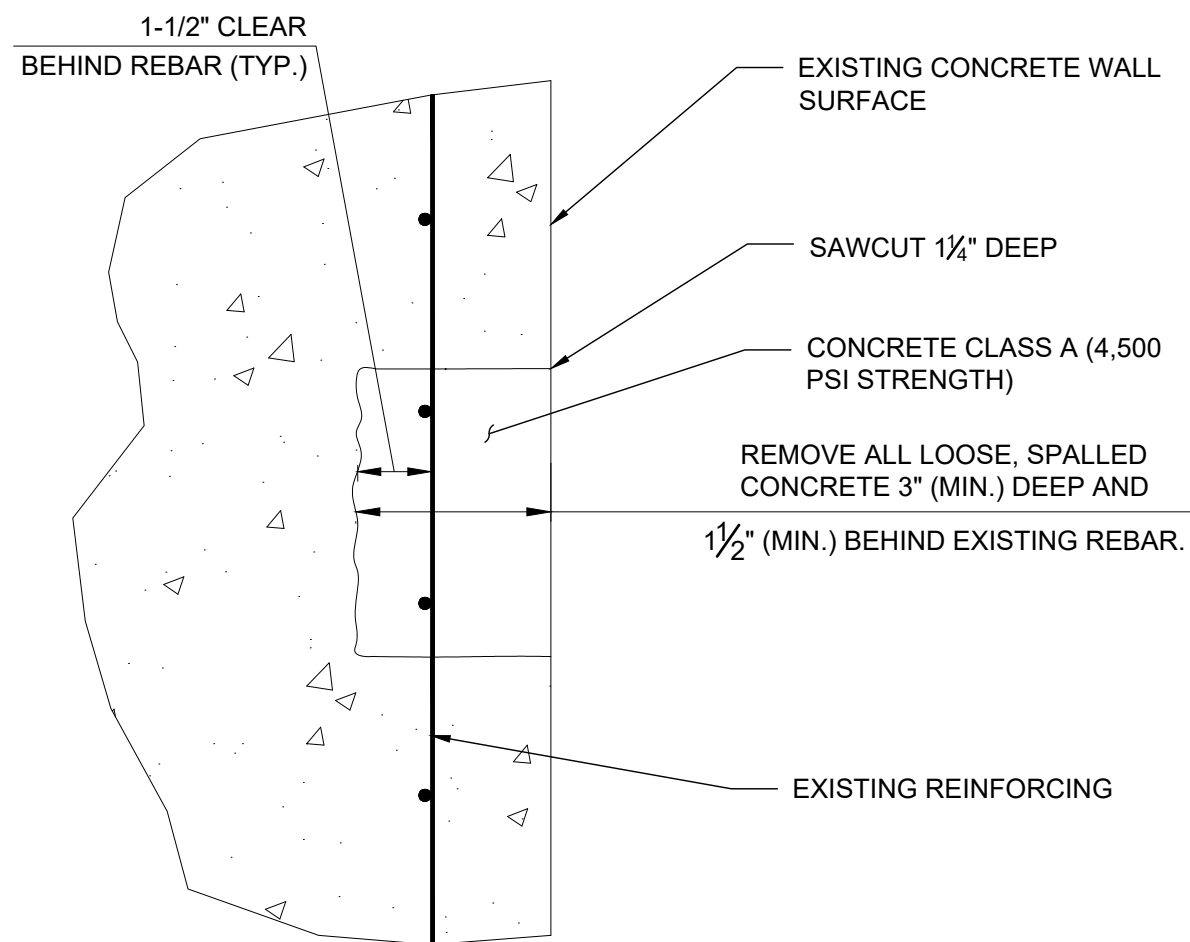
**DuBois
& King inc**

7/10/24 14:39 BGRAVES I:\0629065 QMNF FR278 BRIDGE SUPERSTRUCTURE REPLACEMENT\DRAWINGS\CAD\629065_07 ABUTMENT DETAILS.DWG



CLASS I CONCRETE REPAIR DETAIL

NOT TO SCALE



CLASS II CONCRETE REPAIR DETAIL

NOT TO SCALE

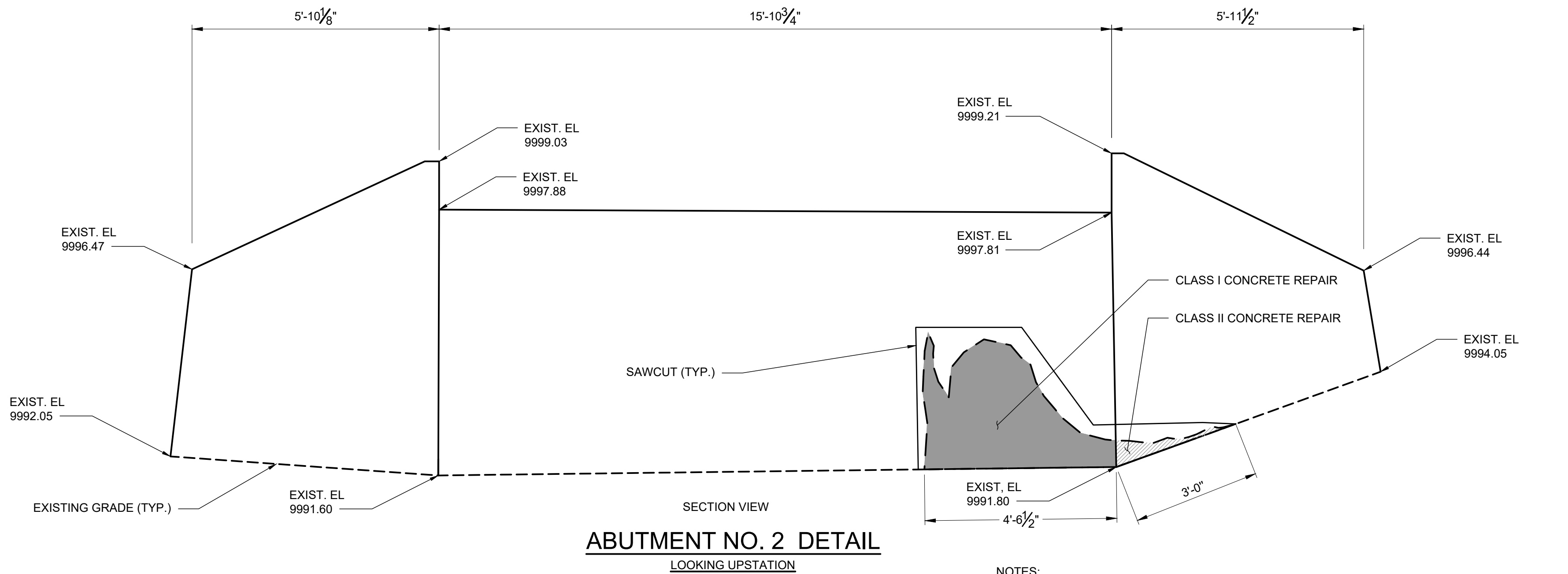
LEGEND:



CLASS I CONCRETE REPAIR AREA



CLASS II CONCRETE REPAIR AREA

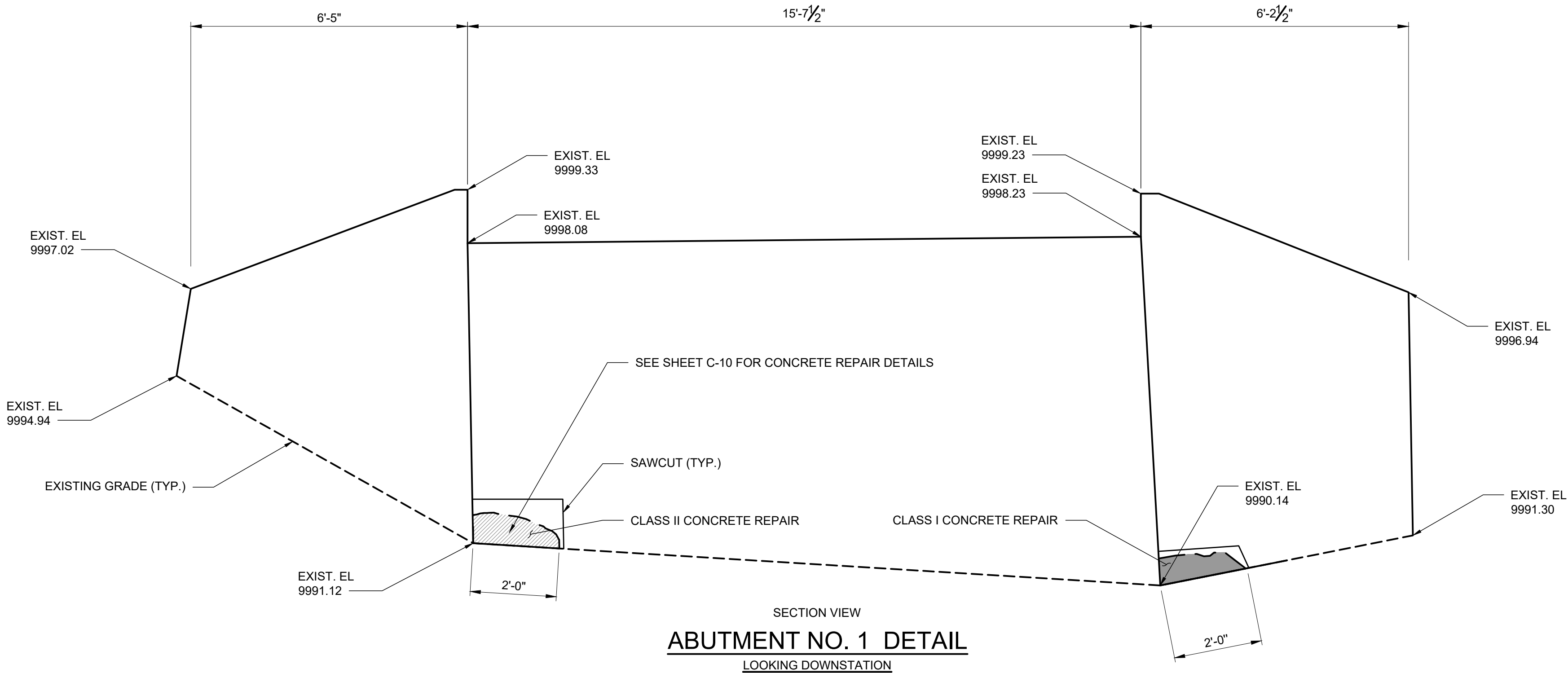


ABUTMENT NO. 2 DETAIL

LOOKING UPSTATION

NOTES:

1. DIMENSIONS OF SPALLS ARE APPROXIMATE
2. CONTRACTOR SHALL FIELD VERIFY ALL DIMENSIONS AND ELEVATIONS PRIOR TO COMMENCEMENT OF WORK
3. ALL COSTS ASSOCIATED WITH THE WWF REQUIRED FOR PATCHING TYPE I SPALLS SHALL BE INCIDENTAL TO ITEM 55221.



ABUTMENT NO. 1 DETAIL

LOOKING DOWNSTATION

NOTES:

1. DIMENSIONS OF SPALLS ARE APPROXIMATE
2. CONTRACTOR SHALL FIELD VERIFY ALL DIMENSIONS AND ELEVATIONS PRIOR TO COMMENCEMENT OF WORK
3. ALL COSTS ASSOCIATED WITH THE WWF REQUIRED FOR PATCHING TYPE I SPALLS SHALL BE INCIDENTAL TO ITEM 55221.



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EASTERN REGION

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PROJECT NAME

FR 278-0.24
BRIDGE
SUPERSTRUCTURE
REPLACEMENT

GREEN MOUNTAIN
NATIONAL FOREST

MANCHESTER RANGER DISTRICT

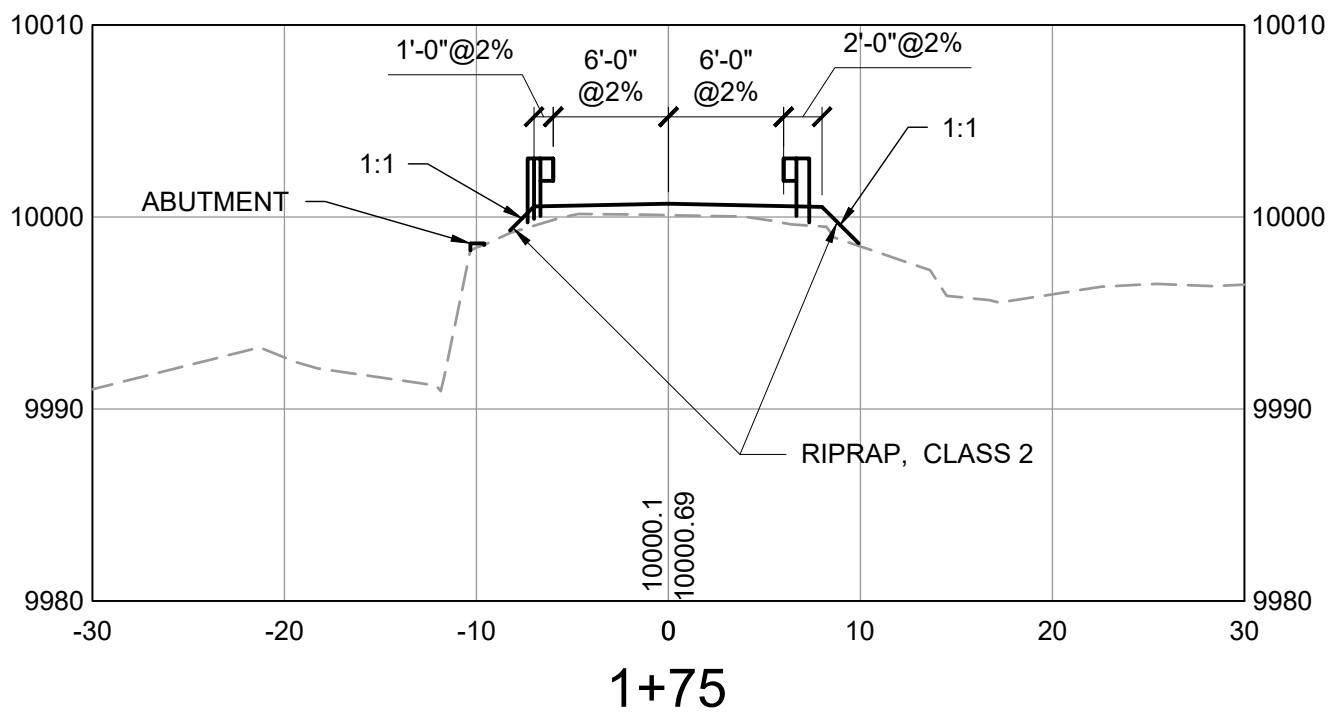
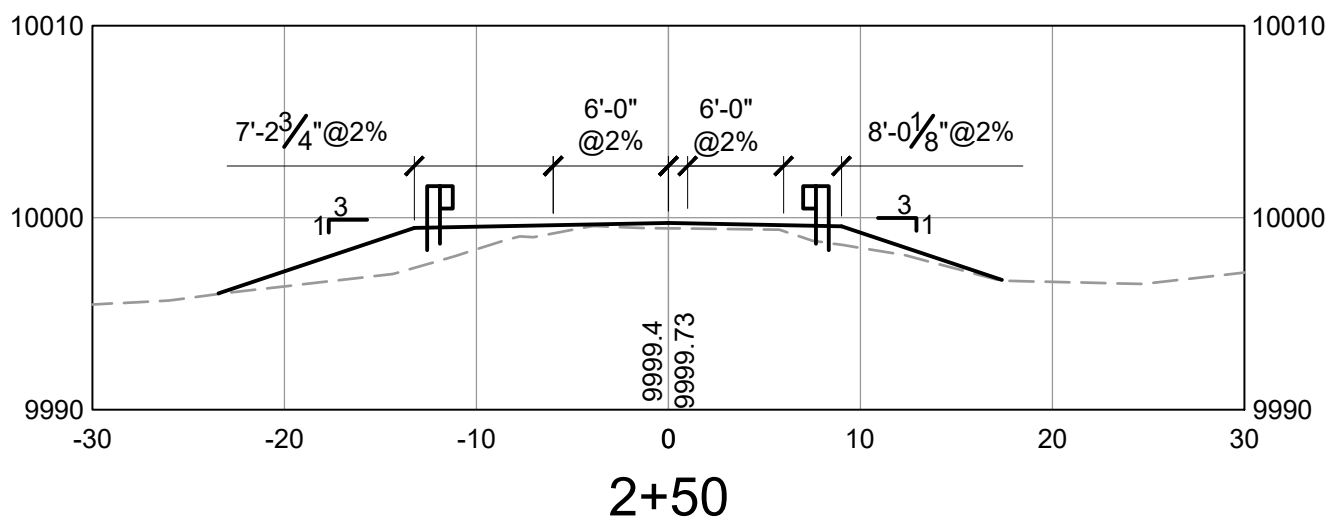
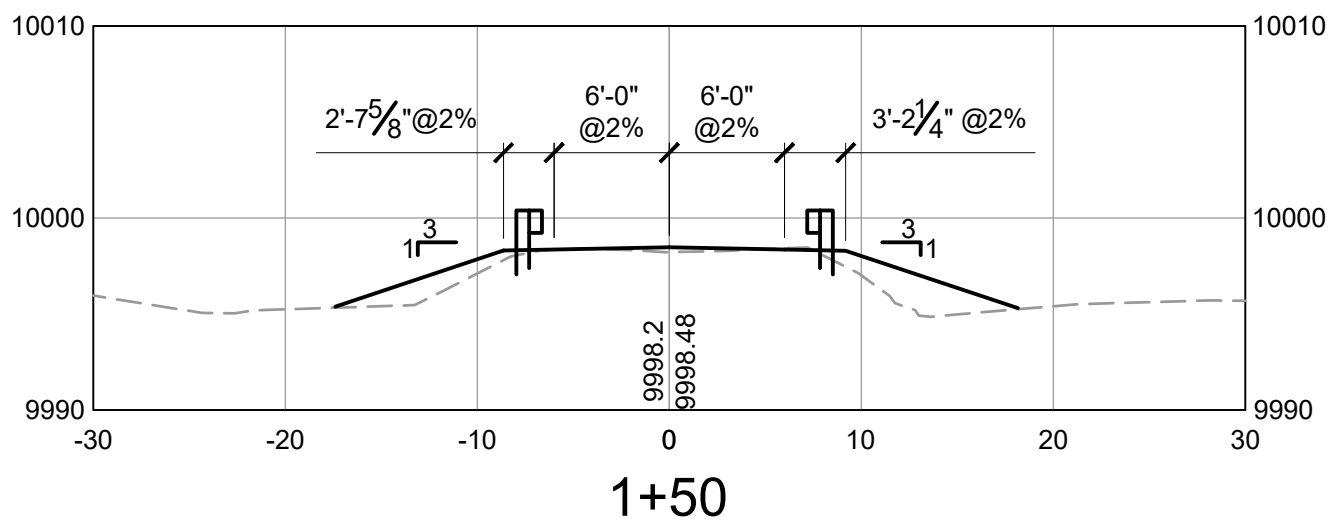
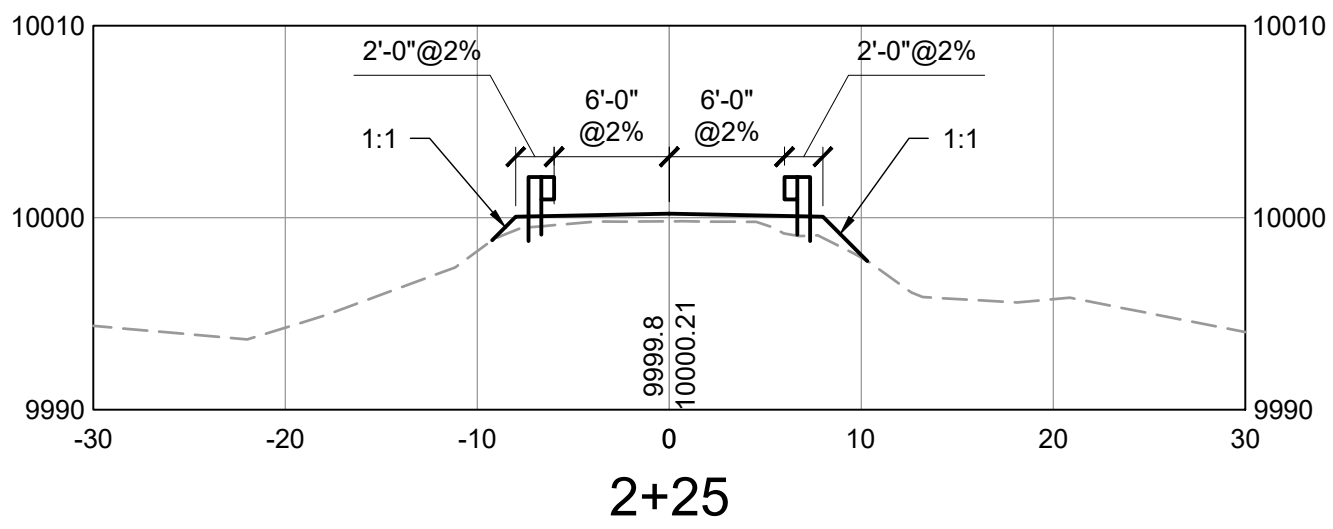
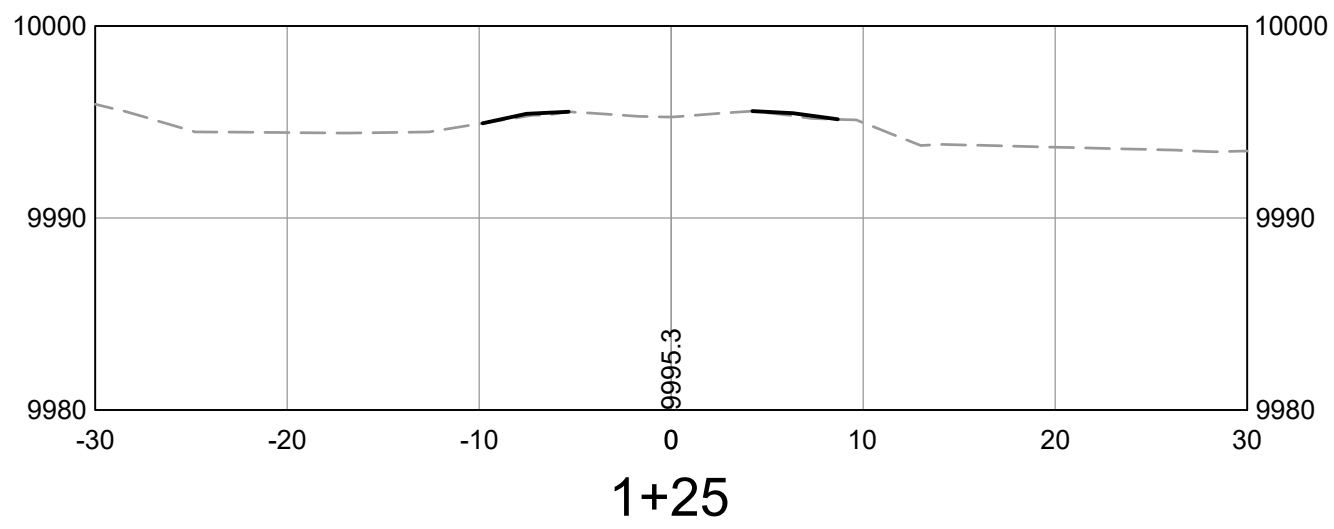
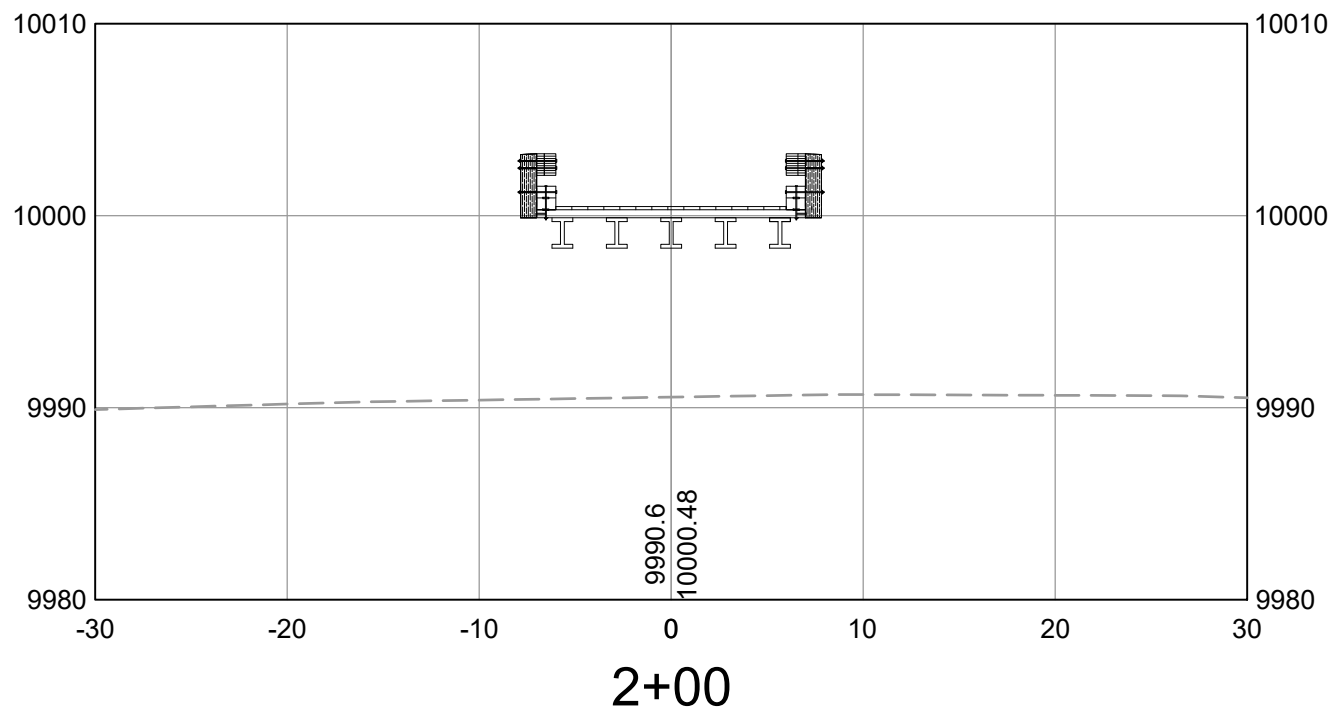
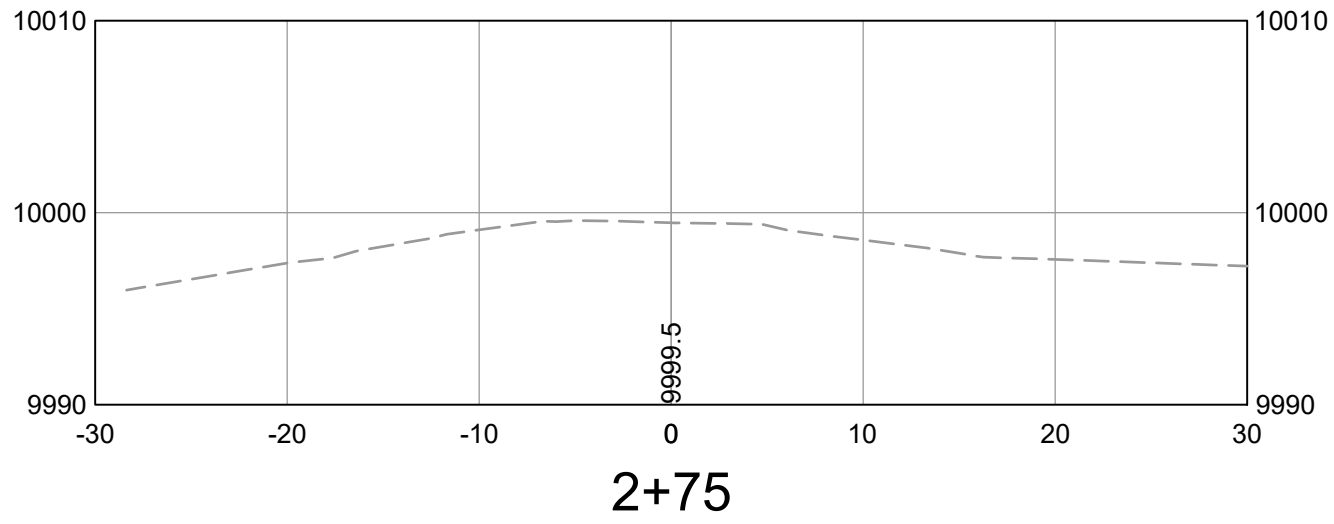
DRAWING TITLE

ABUTMENT 1 & 2
DETAILS

DATE 7/12/2024	ARCHIVE NO.
DESIGNER A. HUMMERS	DRAWING SHEET NO. C-13
DRAWN B. GRAVES	
CHECKED D. CONGER	
PROJECT NO. 629065	SHEET 13 OF 18

DuBois
& King
INC

7/12/24 11:12 BGRAVES I:\629065 GMMF FR278 BRIDGE SUPERSTRUCTURE REPLACEMENT\DRAWINGS\CAD\629065_04 ROAD SECTIONS.DWG;



SECTIONS - CENTERLINE EXISTING FOREST ROAD 278

SCALE: 1"=10' HORIZONTAL
1"=10' VERTICAL



United States Department of Agriculture
Forest Service

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EASTERN REGION

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PROJECT NAME

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BRIDGE
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REPLACEMENT

GREEN MOUNTAIN
NATIONAL FOREST

MANCHESTER RANGER DISTRICT

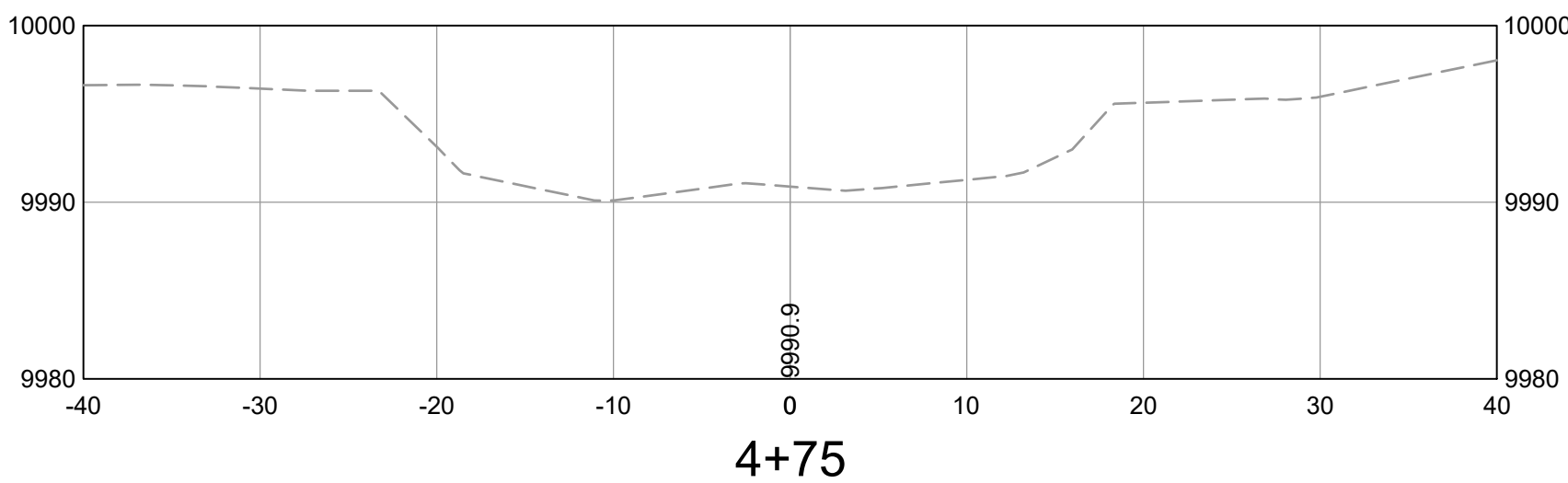
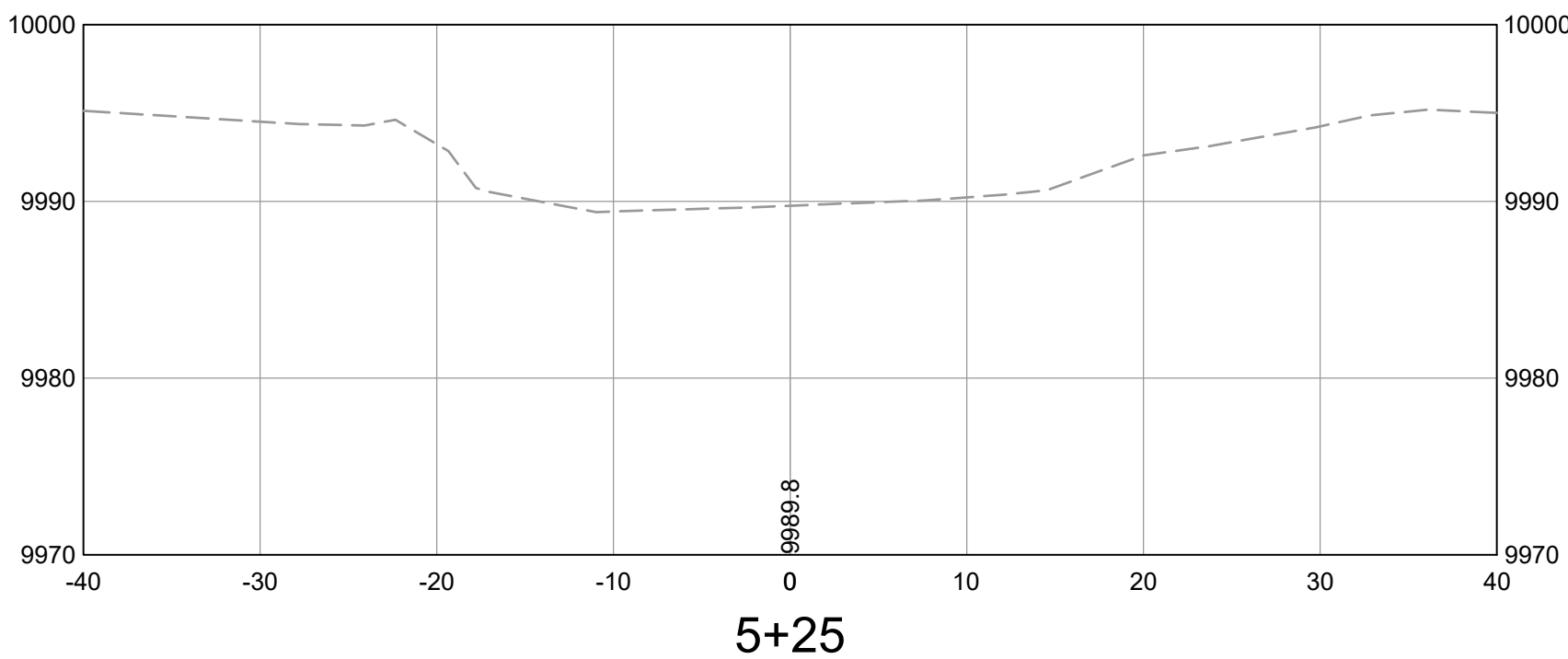
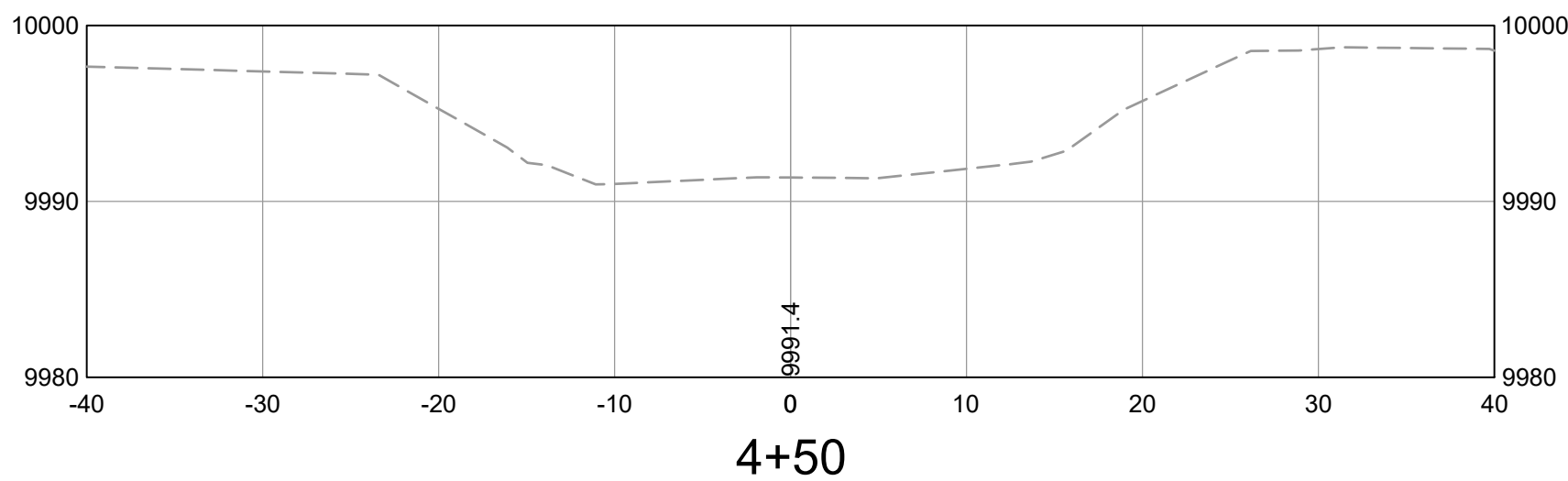
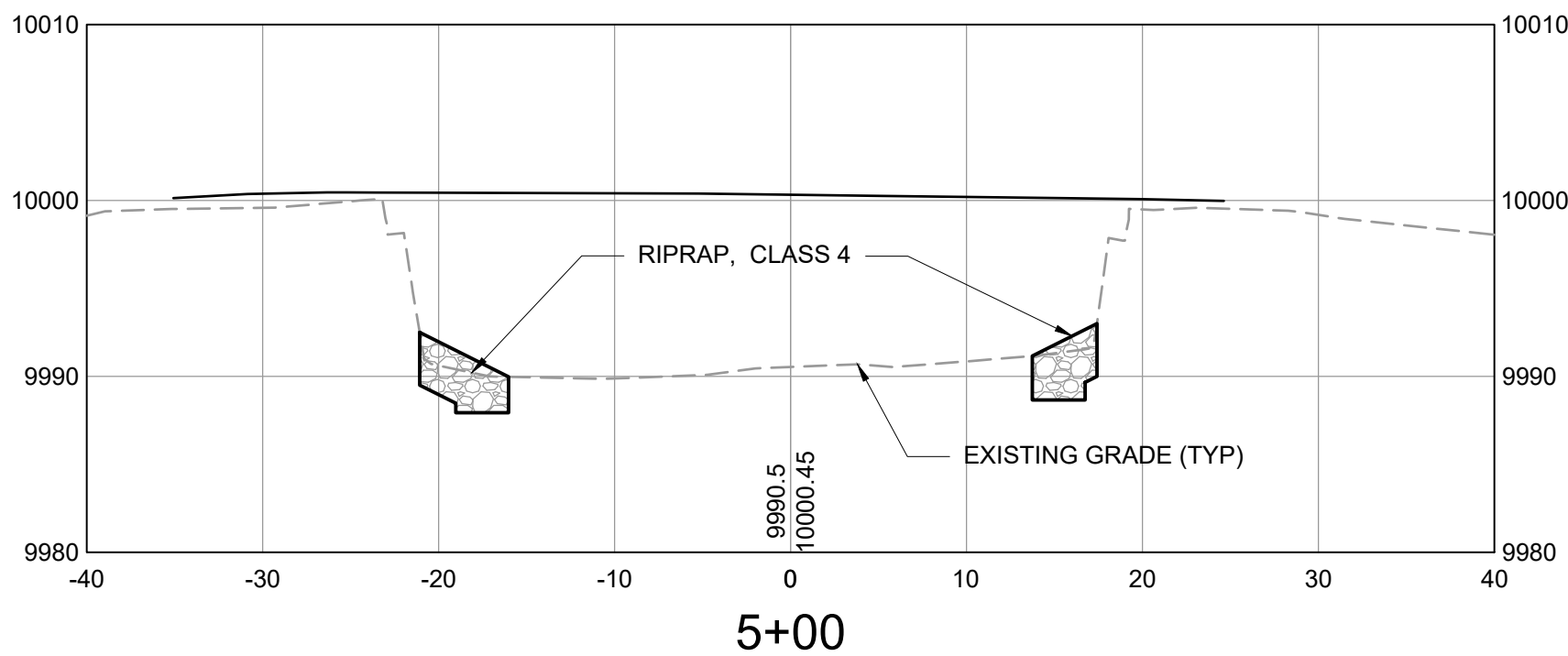
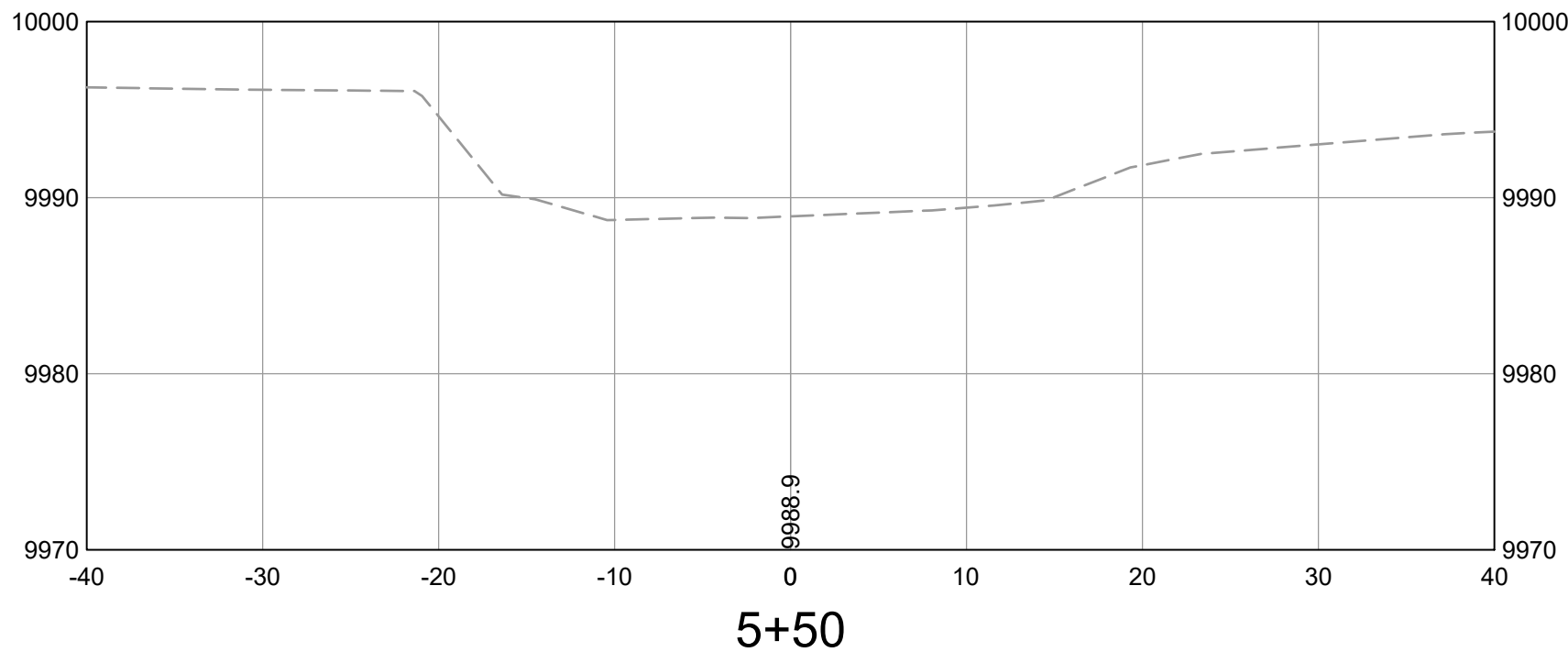
DRAWING TITLE

ROAD CENTERLINE
SECTIONS

DATE 7/12/2024	ARCHIVE NO.
DESIGNER A. HUMMERS	DRAWING SHEET NO. C-14
DRAWN B. GRAVES	
CHECKED D. CONGER	
PROJECT NO. 629065	SHEET 14 OF 18

DuBois
& King
INC

5/3/24 10:57 ODAIMER I:\629065 6MNF FR278 BRIDGE SUPERSTRUCTURE REPLACEMENT\DRAWINGS\CAD\629065_06 CHANNEL SECTIONS.DWG.



SECTIONS - CENTERLINE EXISTING CHANNEL

SCALE: 1"=10' HORIZONTAL
1"=10' VERTICAL

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United States Department of Agriculture

Forest Service

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EASTERN REGION

STAMPS, LOGOS, AND SEALS

STATE OF VERMONT

DAVID COHEN

No. 7689

Civil

PROFESSIONAL ENGINEER

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PROJECT NAME

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DRAWING TITLE

CHANNEL CENTERLINE
SECTIONS

DATE 7/12/2024	ARCHIVE NO.
DESIGNER A. HUMMERS	DRAWING SHEET NO. C-15
DRAWN B. GRAVES	
CHECKED D. CONGER	
PROJECT NO. 629065	SHEET 15 OF 18

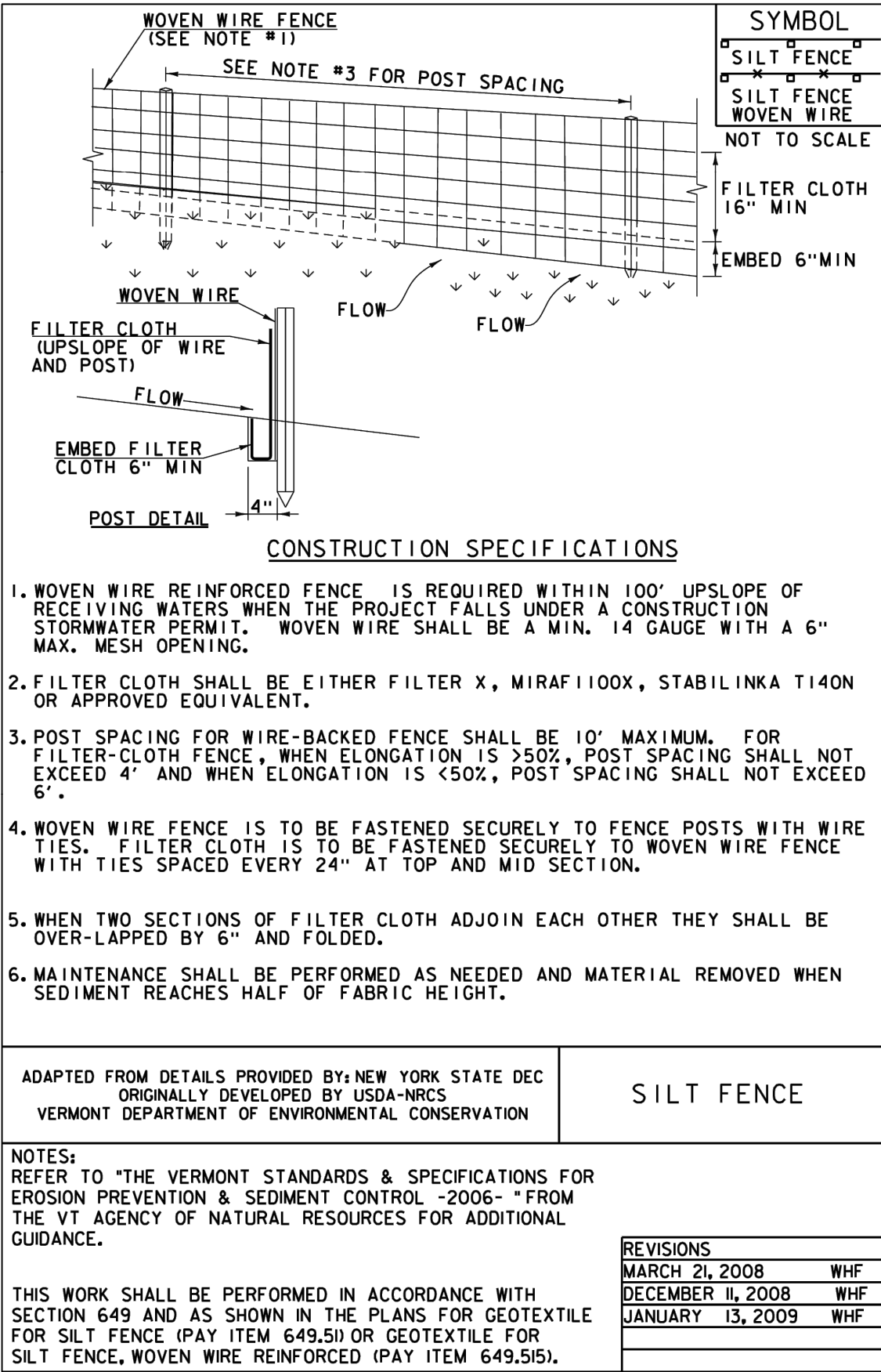
DuBois & King

inc

7/10/24 10:19 BGR/AVES l:\0629065 GMINF FR278 BRIDGE SUPERSTRUCTURE REPLACEMENT\DRAWINGS\CAD\629065_XX.EPSD NOTES AND EXISTING.DWG;

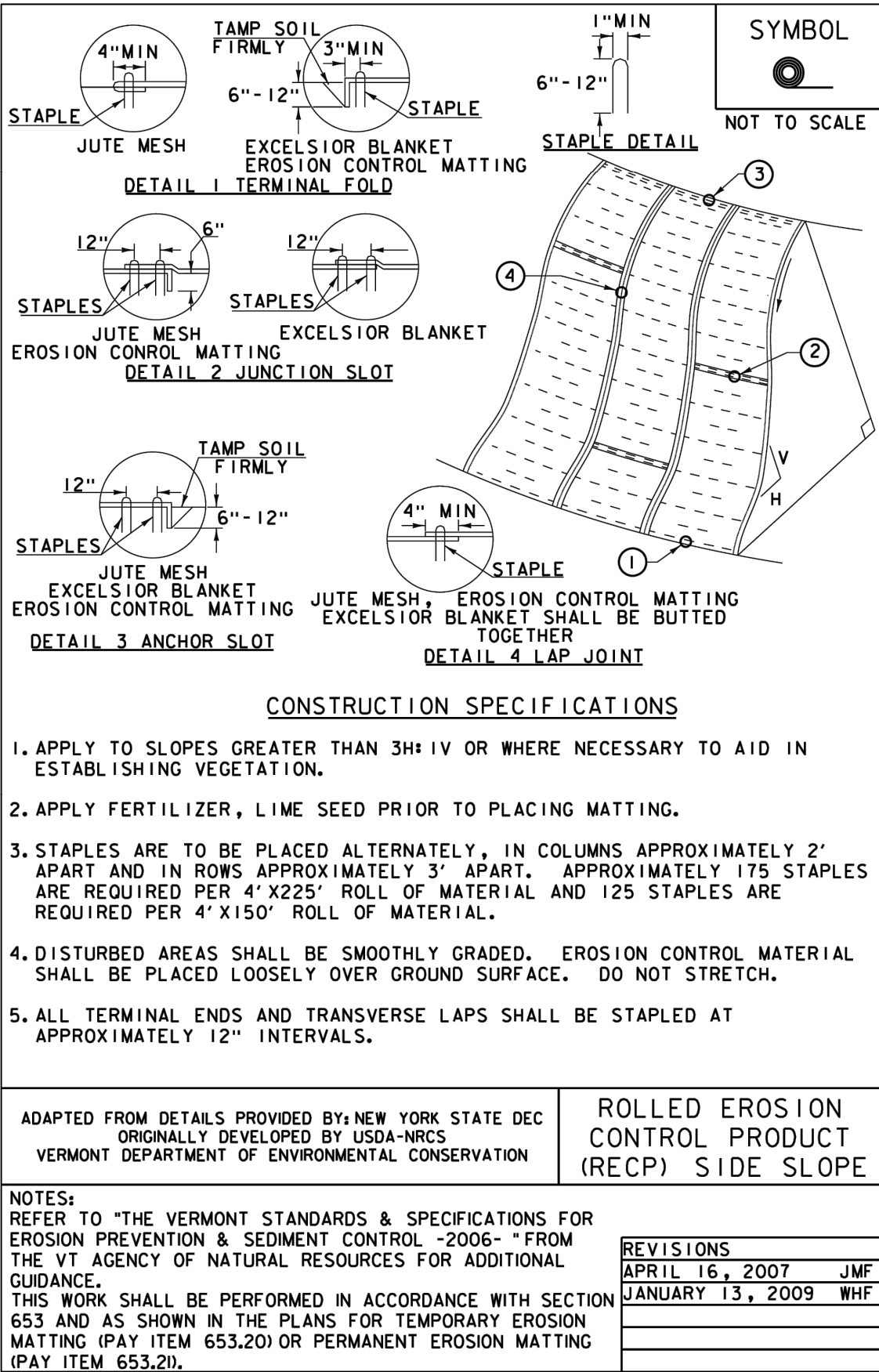
EROSION CONTROL NOTES:

1. PRIOR TO STARTING CONSTRUCTION THE CONTRACTOR SHALL SUBMIT AN EROSION AND SEDIMENT CONTROL PLAN WITH A SCHEDULE OF CONSTRUCTION ACTIVITIES. THE PLAN SHALL BE SITE SPECIFIC AND SHALL BE PAID FOR UNDER ITEM 15701, SOIL EROSION CONTROL.
2. THE CONTRACTOR SHALL MINIMIZE THE AMOUNT OF DISTURBED SOIL EXPOSED TO EROSION FROM STORM WATER AND WIND AT ANY TIME BY USING VEGETATIVE AND STRUCTURAL CONTROLS AND PROPER TIMING AND SEQUENCING OF CONSTRUCTION ACTIVITIES.
3. ONLY DISTURB, CLEAR, OR GRADE AREAS NECESSARY FOR IMMEDIATE CONSTRUCTION ACTIVITY. FLAG OR OTHERWISE DELINEATE AREAS NOT TO BE DISTURBED. EXCLUDE VEHICLES AND CONSTRUCTION EQUIPMENT FROM THESE AREAS TO PRESERVE NATURAL VEGETATION.
4. IF PROJECT IS GREATER THAN 5-ACRES, THE AREA OF UNSTABILIZED SOIL SHALL NOT EXCEED 5 ACRES AT ANY TIME UNLESS PROJECT PERMITS SPECIFICALLY PROVIDE FOR A GREATER AREA OF DISTURBANCE.
5. ALL EROSION AND SEDIMENT CONTROL PRACTICES AND MEASURES TO BE CONSTRUCTED, APPLIED AND MAINTAINED IN ACCORDANCE WITH THESE DRAWINGS, PER THESE NOTES AND DETAILS, AND ACCORDING TO MANUFACTURER'S RECOMMENDATIONS.
6. THE CONTRACTOR SHALL USE PROPER EROSION AND SEDIMENT CONTROL MEASURES TO ENSURE NO SEDIMENT IS TRACKED OFF-SITE (E.G. STABILIZED CONSTRUCTION ENTRANCE).
7. DISTURBED AREAS WITH POTENTIAL TO DISCHARGE SEDIMENT-LADEN WATER INTO SURFACE WATERS OR OFF THE SITE MUST BE PROTECTED WITH TEMPORARY EROSION CONTROL MEASURES (E.G., SILT FENCE).
8. THE CONTRACTOR SHALL DIVERT OFFSITE STORMWATER RUNOFF FROM HIGHLY ERODIBLE AREAS AND STEEP SLOPES AND CONVEY OFFSITE STORMWATER RUNOFF TO STABLE AREAS. INSTALL PERIMETER SILT FENCE PRIOR TO CLEARING AND GRUBBING AND INSTALL NECESSARY EROSION AND SEDIMENT CONTROL PRACTICES AS WORK TAKES PLACE.
9. AREAS TO BE FILLED SHALL BE CLEARED, GRUBBED, AND STRIPPED OF TOPSOIL TO REMOVE TREES, VEGETATION, ROOTS OR OTHER OBJECTIONABLE MATERIALS.
10. ALL STOCKPILES, BORROW AREAS, AND SPOILS SHALL BE PROTECTED FROM EROSION AND SURROUNDED BY SILT FENCE AND COVERED WITH A BLANKET OF STRAW MULCH.
11. FROZEN MATERIAL OR SOFT, MUCKY OR HIGHLY COMPRESSIBLE MATERIALS ARE SUSCEPTIBLE TO ACCELERATED SETTLEMENT AND POTENTIAL ACCELERATED EROSION. WORK IN THESE MATERIALS TO BE PERFORMED UNDER THE DIRECTION OF A PROFESSIONAL ENGINEER.
12. REPAIR OR REPLACEMENT OF EROSION CONTROL MEASURES SHALL BE MADE PROMPTLY AS NEEDED, AND AS DESCRIBED IN THE INSPECTION, MONITORING AND MAINTENANCE NOTES ON THESE DRAWINGS.
13. THE OUTER FACE OF THE FILL SLOPE SHALL BE ALLOWED TO STAY LOOSE, NOT ROLLED, COMPACTED, OR BLADED SMOOTH. ROUGHEN THE SURFACE OF ALL SLOPES DURING THE CONSTRUCTION OPERATION TO RETAIN WATER, INCREASE INFILTRATION, AND FACILITATE VEGETATION ESTABLISHMENT.
14. USE SLOPE BREAKS, SUCH AS DIVERSIONS, BENCHES, OR CONTOUR FURROWS AS APPROPRIATE, TO REDUCE THE LENGTH OF CUT-AND-FILL SLOPES TO LIMIT SHEET AND RILL EROSION AND PREVENT GULLY EROSION. ALL BENCHES TO BE KEPT FREE OF SEDIMENT DURING ALL PHASES OF DEVELOPMENT.
15. AREAS SHALL BE SCARIFIED TO A MINIMUM DEPTH OF 3 INCHES PRIOR TO PLACEMENT OF TOPSOIL. TOPSOIL TO BE PLACED WITHOUT SIGNIFICANT COMPACTION TO PROVIDE A LOOSE BEDDING FOR PLACEMENT OF SEED.
16. STABILIZE ALL GRADED AREAS WITH VEGETATION, CRUSHED STONE, COMPOST BLANKET, OR OTHER GROUND COVER AS SOON AS GRADING IS COMPLETED OR IF WORK IS INTERRUPTED FOR 21 WORKING DAYS OR MORE. USE MULCH OR OTHER APPROVED METHODS TO STABILIZE AREAS TEMPORARILY WHERE FINAL GRADING MUST BE DELAYED.
17. ALL GRADED AREAS TO BE PERMANENTLY STABILIZED IMMEDIATELY FOLLOWING FINISHED GRADING.
18. SEED MIXTURE FOR THE PURPOSES OF TURF ESTABLISHMENT SHALL BE THE NEW ENGLAND EROSION CONTROL/ RESTORATION MIX (FOR DETENTION BASINS AND MOIST SITES) BY NEW ENGLAND WETLANDS, INC. OR AN APPROVED EQUAL. THE MIX SHALL BE APPLIED PER THE SEEDING FORMULA AND MANUFACTURERS RECOMMENDATIONS. ALL SEEDED AREAS SHALL RECEIVE EROSION CONTROL MATTING IMMEDIATELY FOLLOWING SEEDING. THIS WORK SHALL BE PAID UNDER ITEM 15701.
19. PERMANENT OR TEMPORARY COVER MUST BE IN PLACE BEFORE THE GROWING SEASON ENDS. WHEN SEEDED AREAS ARE MULCHED, PLANTINGS MAY BE MADE FROM EARLY SPRING TO EARLY OCTOBER. WHEN SEEDED AREAS ARE NOT MULCHED, SEED SHOULD BE PLACED FROM EARLY SPRING TO MAY 20 OR FROM AUGUST 15 TO SEPTEMBER 15. NO DISTURBED AREA SHALL BE LEFT EXPOSED DURING WINTER MONTHS (NOVEMBER THROUGH MARCH).
20. ONCE DISTURBED AREAS HAVE BEEN STABILIZED AND VEGETATION IS ESTABLISHED, ALL TEMPORARY EROSION CONTROL MEASURES SUCH AS SILT FENCE SHALL BE REMOVED. AREAS DISTURBED BY REMOVAL OF THESE MEASURES SHALL BE IMMEDIATELY SEEDED ACCORDING TO SEEDING SPECIFICATIONS ON THESE DRAWINGS.
21. AN AREA IS CONSIDERED "STABLE" IF ONE OF THE FOLLOWING HAS OCCURRED: A MINIMUM OF 85% VEGETATED GROWTH HAS BEEN ESTABLISHED; A MINIMUM OF 3" OF NON-EROSIVE MATERIAL (SUCH AS STONE RIP RAP OR A CERTIFIED COMPOST BLANKET) HAS BEEN INSTALLED. EROSION CONTROL BLANKETS HAVE BEEN PROPERLY INSTALLED.
22. ALL AREAS OF EXPOSED OR DISTURBED SOIL TO BE TEMPORARILY STABILIZED AS SOON AS PRACTICABLE BUT NO LATER THAN 45 DAYS FROM THE TIME OF INITIAL DISTURBANCE, UNLESS A SHORTER TIME IS SPECIFIED BY LOCAL AUTHORITIES, THE CONSTRUCTION SEQUENCE APPROVED AS PART OF THE ISSUED PERMIT, OR AN INDEPENDENT MONITOR.
23. ALL AREAS OF EXPOSED OR DISTURBED SOIL TO BE PERMANENTLY STABILIZED AS SOON AS PRACTICABLE BUT NO LATER THAN 3 DAYS FOLLOWING FINAL GRADING.
24. ALL ITEMS SPECIFIED IN THE EROSION CONTROL PLAN BUT NOT INCLUDED AS A SEPARATE PAY ITEM SHALL BE PAID FOR UNDER ITEM 15701, SOIL EROSION CONTROL.
25. THE CONTRACTOR MAY USE SILT FENCE OR BIODEGRADABLE FIBER ROLLS FOR SOIL EROSION AND SEDIMENTATION CONTROL. SILT FENCE MUST BE REMOVED ONCE VEGETATION IS ESTABLISHED. BIODEGRADABLE FIBER ROLLS MAY BE LEFT IN PLACE. THIS WORK SHALL BE INCLUDED IN PAY ITEM 15701.
26. NO MATERIAL SHALL BE CLEANED OR ALLOWED TO ERODE INTO THE WATER COURSE, NOR SHALL MATERIAL BE PLACED INTO THE WATER COURSE EXCEPT AS SHOWN ON THE PLANS.
27. ALL WORK SHALL TAKE PLACE IN THE DRY. THE CONTRACTOR SHALL DEWATER ALL WORK AREAS IN ACCORDANCE WITH THE CONTRACTOR'S APPROVED DEWATERING PLAN, PRIOR TO DISTURBANCE.
28. THE CONTRACTOR IS RESPONSIBLE FOR ALL CONTROL OF WATER MEASURES FROM THE INITIATION OF CONSTRUCTION UNTIL THE CONTROL OF WATER MEASURES ARE REMOVED AT THE CONCLUSION OF THE PROJECT.
29. CONTRACTOR SHALL CONSTRUCT TEMPORARY AND PERMANENT EROSION AND SEDIMENTATION CONTROL FACILITIES IN ACCORDANCE WITH THE CONTRACTOR'S APPROVED SEDIMENT CONTROL PLAN PRIOR TO THE COMMENCEMENT OF EARTHWORK OPERATIONS.
30. THE CONTRACTOR SHALL CONSTRUCT AND MAINTAIN EROSION AND SEDIMENTATION CONTROL DEVICES THROUGHOUT THE PROJECT SITE FOR THE DURATION OF THE CONSTRUCTION. THE CONTRACTOR SHALL INSPECT AND RECORD FINDINGS OF ALL EROSION AND SEDIMENTATION CONTROL DEVICES IN ACCORDANCE WITH THE CONTRACTOR'S SEDIMENT CONTROL PLAN TO ENSURE THAT ALL ITEMS ARE IN STABLE CONDITION. IN THE EVENT THAT SAID ITEMS ARE DETERMINED TO BE IN UNSATISFACTORY CONDITION, THE CONTRACTOR SHALL RECORD THE UNSATISFACTORY ISSUE, AND IMPLEMENT A CORRECTIVE MEASURE IN ACCORDANCE WITH THE CONTRACTOR'S APPROVED SEDIMENT CONTROL PLAN.



NOTES:

1. VTRANS ROLLED EROSION CONTROL PRODUCT (RECP) SIDE SLOPE AND SILT FENCE DETAILS ARE ADDED ONLY FOR REFERENCE OF ACCEPTED PRODUCTS AND PROCEDURES. NO PAYMENT SHALL BE MADE TO THE VTRANS ITEMS REFERENCED ABOVE. ALL PAYMENT SHALL BE MADE UNDER ITEM 15701 SOIL EROSION CONTROL OR ITEM 15705 SOIL EROSION CONTROL, SILT FENCE.



STAMPS, LOGOS, AND SEALS



4		
3		
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1		
NO.	REVISION / ISSUE	DATE

PROJECT NAME

**FR 278-0.24
BRIDGE
SUPERSTRUCTURE
REPLACEMENT**

**GREEN MOUNTAIN
NATIONAL FOREST**

MANCHESTER RANGER DISTRICT

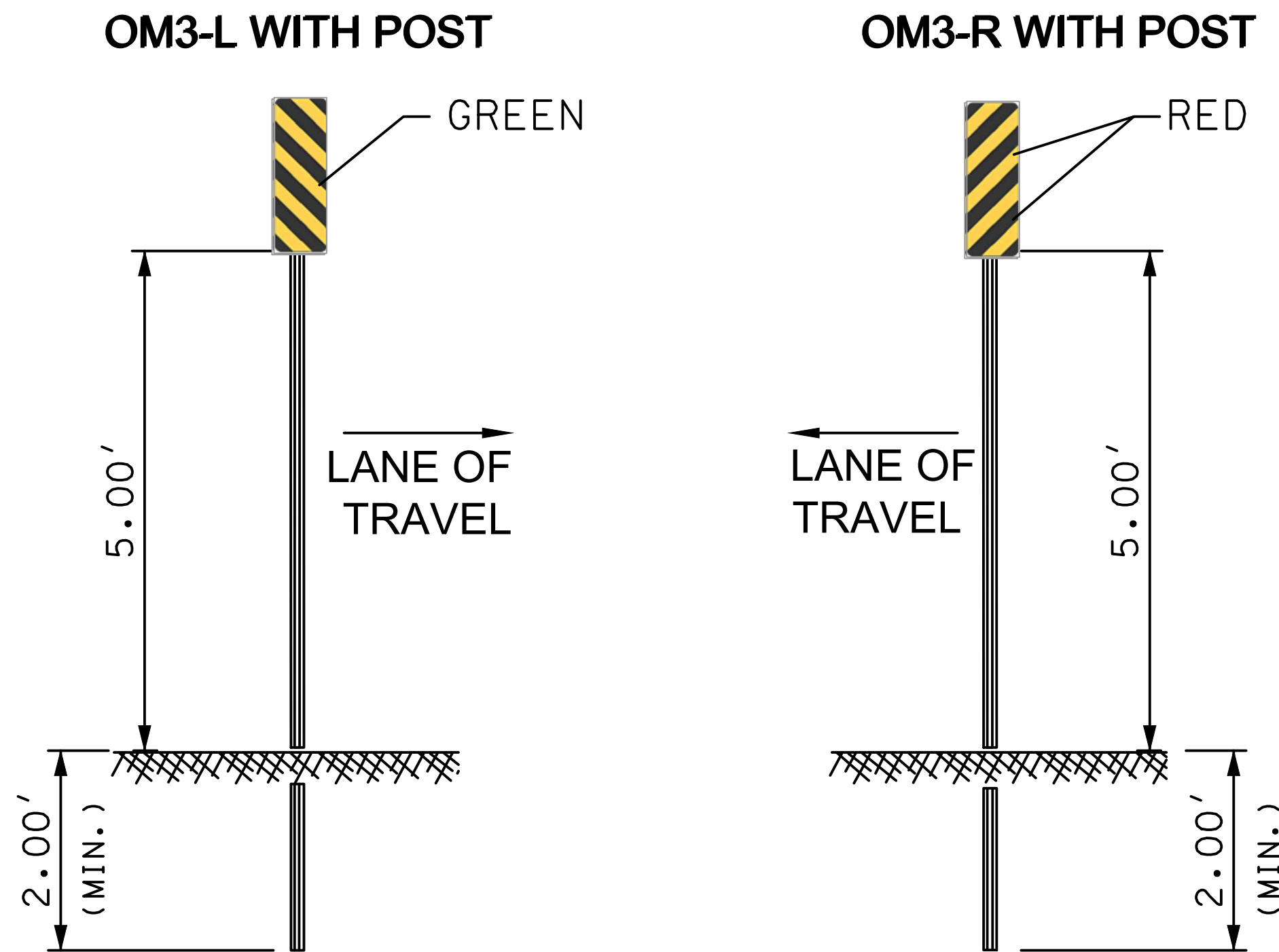
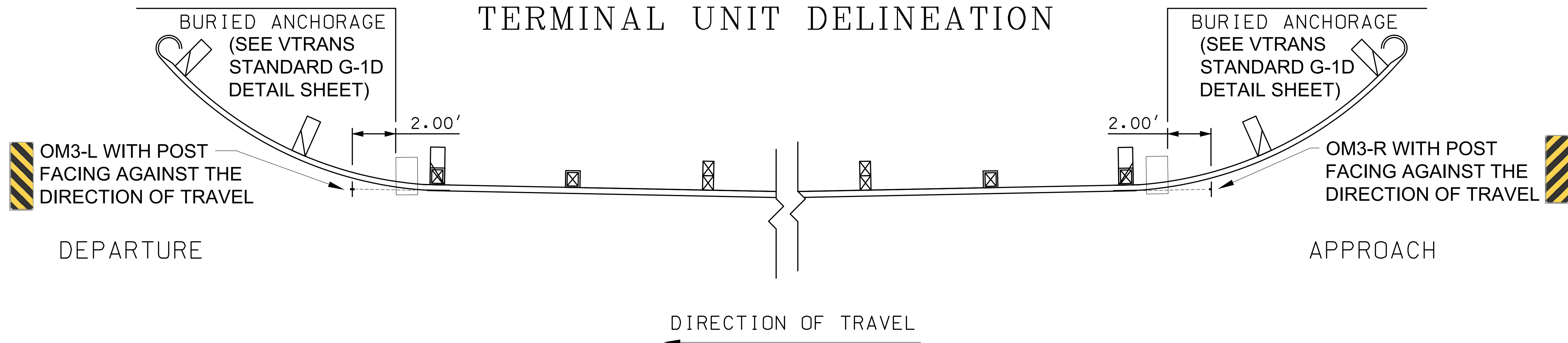
DRAWING TITLE

**EROSION PREVENTION
AND SEDIMENT
CONTROL DETAILS**

DATE 7/12/2024	ARCHIVE NO.
DESIGNER A. HUMMERS	DRAWING SHEET NO. C-16
DRAWN B. GRAVES	
CHECKED D. CONGER	
PROJECT NO. 629065	SHEET 16 OF 18

**DuBois
& King inc**

7/10/24 10:20 BGR/AVES 1:6029065 CMNF FR278 BRIDGE SUPERSTRUCTURE REPLACEMENT\DRAWINGS\CAD\629065_17 RAIL SHEETS.DWG;



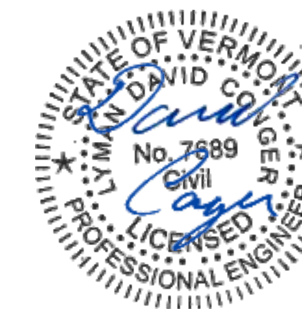
TYPICAL INSTALLATION



United States Department of Agriculture
Forest Service

(R09)
EASTERN REGION

STAMPS, LOGOS, AND SEALS



FOR CONSTRUCTION

4		
3		
2		
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NO.	REVISION / ISSUE	DATE

PROJECT NAME

**FR 278-0.24
BRIDGE
SUPERSTRUCTURE
REPLACEMENT**

**GREEN MOUNTAIN
NATIONAL FOREST**

MANCHESTER RANGER DISTRICT

DRAWING TITLE

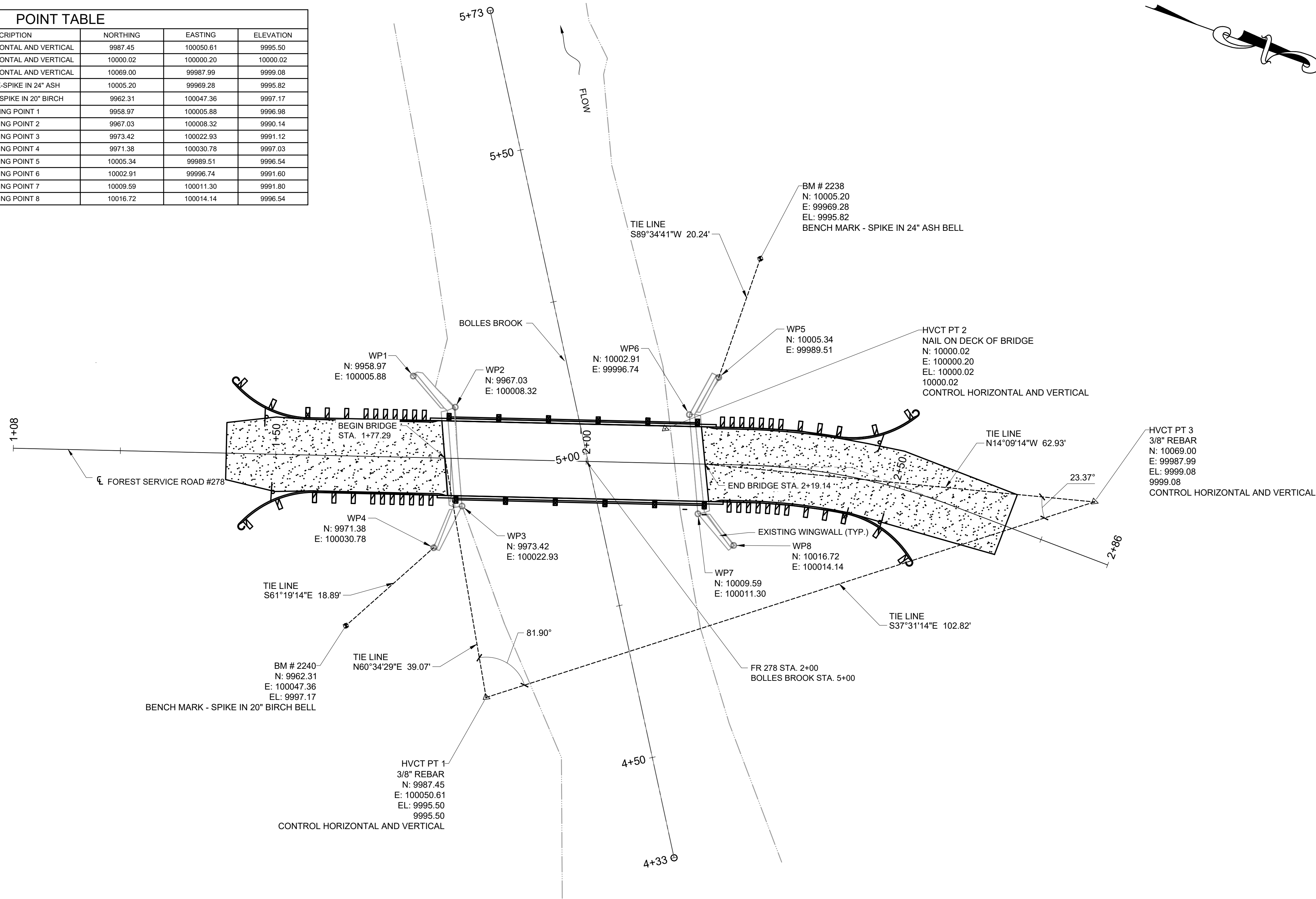
**TERMINAL UNIT
DELINEATION**

DATE 7/12/2024	ARCHIVE NO.
DESIGNER A. HUMMERS	DRAWING SHEET NO. C-17
DRAWN B. GRAVES	
CHECKED D. CONGER	
PROJECT NO. 629065	SHEET 17 OF 18

**DuBois
& King
inc**

7/12/24 17:09 BGRAVES 1:6/629065 GMINF FR278 BRIDGE SUPERSTRUCTURE REPLACEMENTDRAWINGSCAD1629065_06 TIE SHEET.DWG;

POINT TABLE				
POINT NUMBER	DESCRIPTION	NORTHING	EASTING	ELEVATION
HVCT PT 1	CONTROL HORIZONTAL AND VERTICAL	9987.45	100050.61	9995.50
HVCT PT 2	CONTROL HORIZONTAL AND VERTICAL	10000.02	100000.20	10000.02
HVCT PT 3	CONTROL HORIZONTAL AND VERTICAL	10069.00	99987.99	9999.08
BM# 2238	BENCHMARK-SPIKE IN 24" ASH	10005.20	99969.28	9995.82
BM# 2240	BENCHMARK-SPIKE IN 20" BIRCH	9962.31	100047.36	9997.17
WP1	WORKING POINT 1	9958.97	100005.88	9996.98
WP2	WORKING POINT 2	9967.03	100008.32	9990.14
WP3	WORKING POINT 3	9973.42	100022.93	9991.12
WP4	WORKING POINT 4	9971.38	100030.78	9997.03
WP5	WORKING POINT 5	10005.34	99989.51	9996.54
WP6	WORKING POINT 6	10002.91	99996.74	9991.60
WP7	WORKING POINT 7	10009.59	100011.30	9991.80
WP8	WORKING POINT 8	10016.72	100014.14	9996.54



SURVEY TIE PLAN





United States Department of Agriculture
Forest Service

(R09)
EASTERN REGION

STAMPS, LOGOS, AND SEALS



4		
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NO.	REVISION / ISSUE	DATE

PROJECT NAME

FR 278-0.24
BRIDGE
SUPERSTRUCTURE
REPLACEMENT

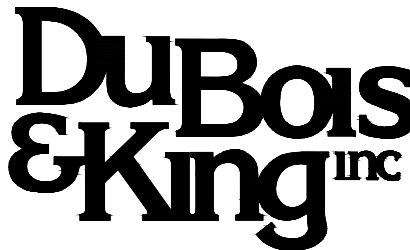
GREEN MOUNTAIN
NATIONAL FOREST

MANCHESTER RANGER DISTRICT

DRAWING TITLE

TIE SHEET

DATE 7/12/2024	ARCHIVE NO.
DESIGNER A. HUMMERS	DRAWING SHEET NO. C-18
DRAWN B. GRAVES	
CHECKED D. CONGER	
PROJECT NO. 629065	SHEET 18 OF 18



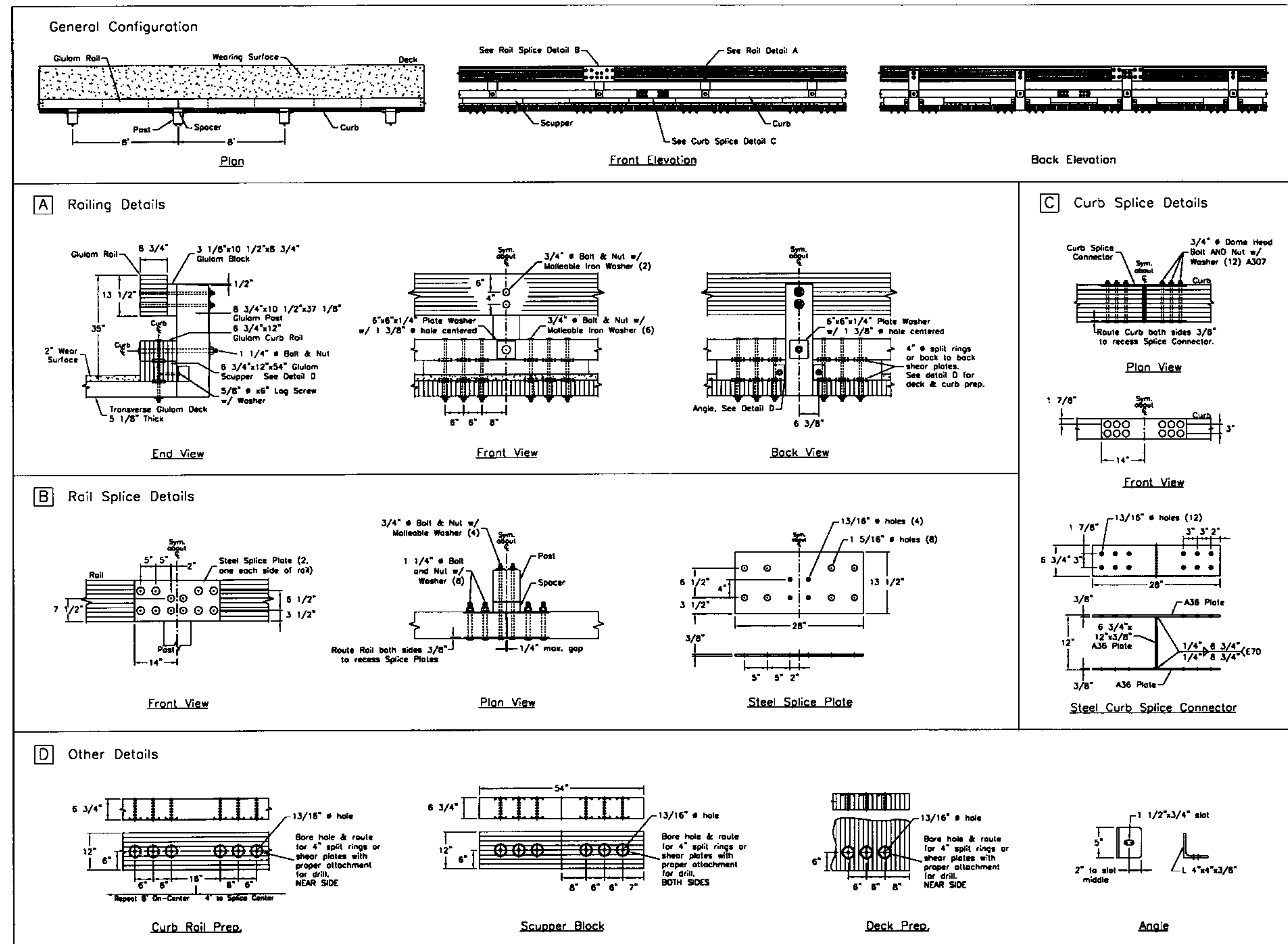


FIGURE 3 System No. 1: wood bridge railing design details (1 in. = 25.4 mm).

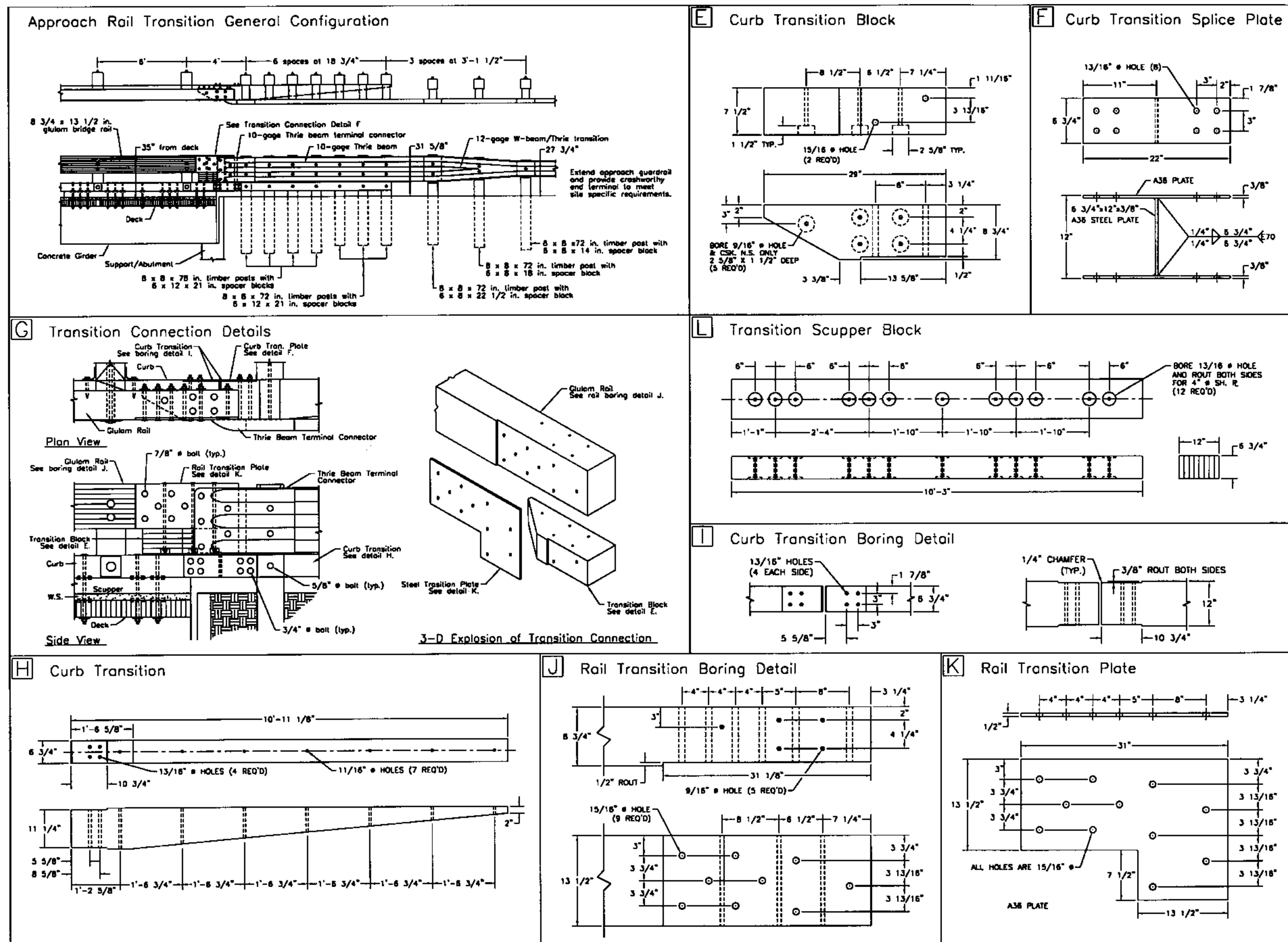
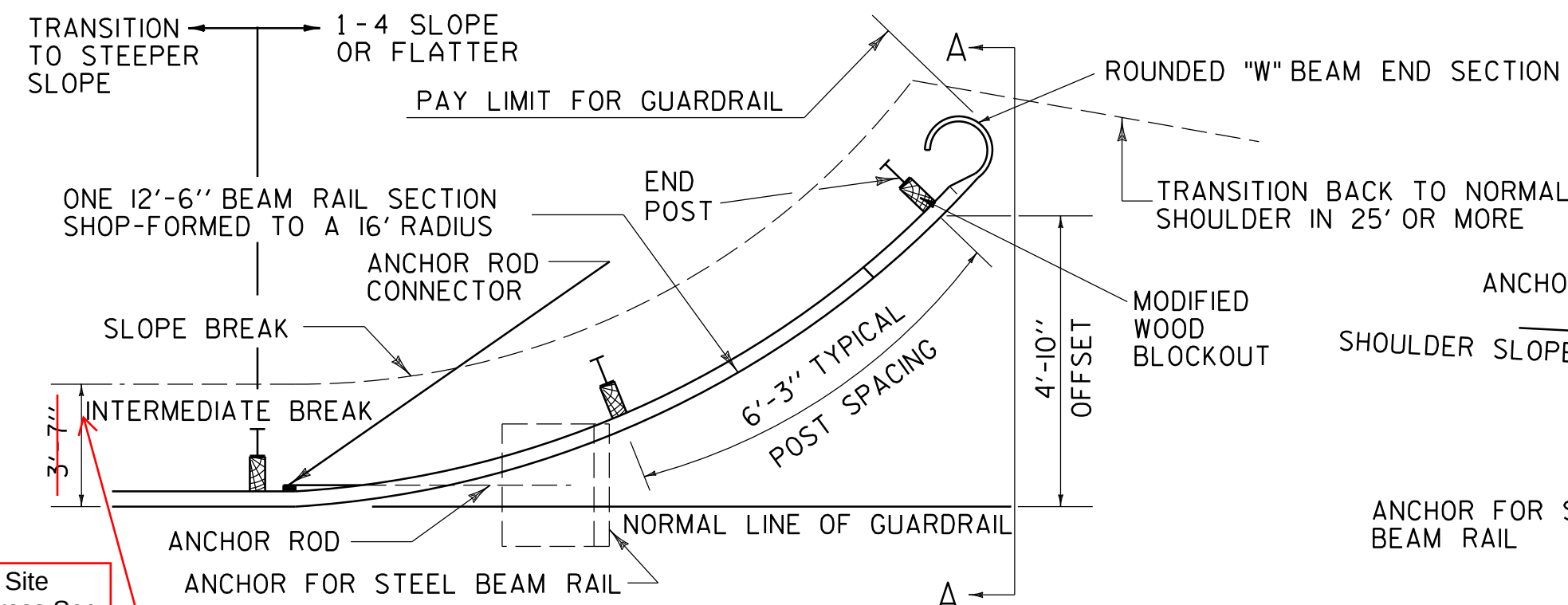
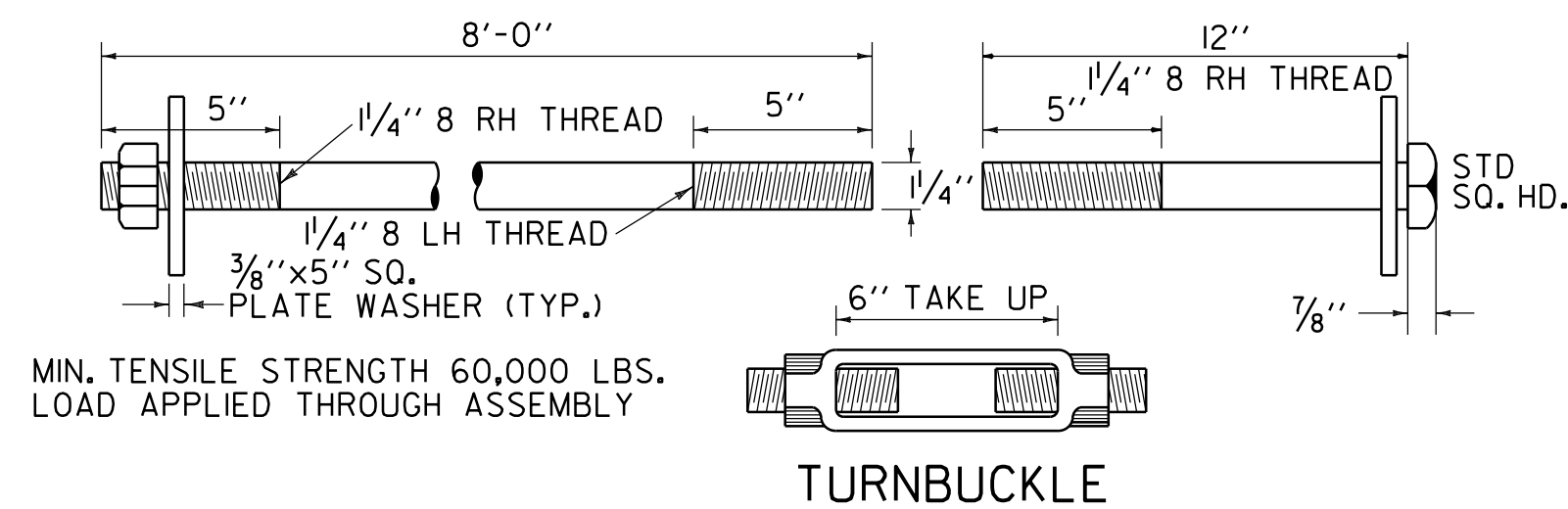
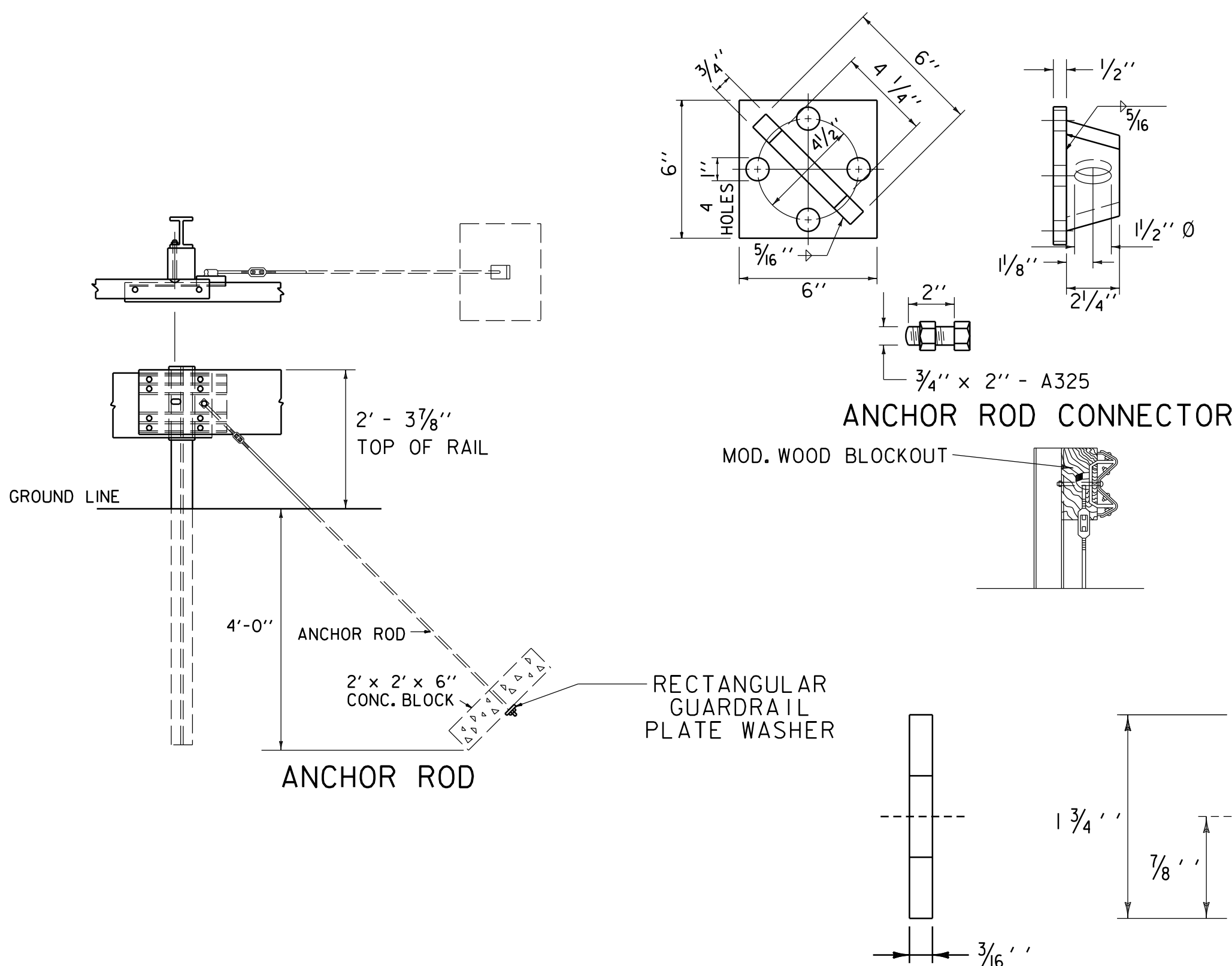
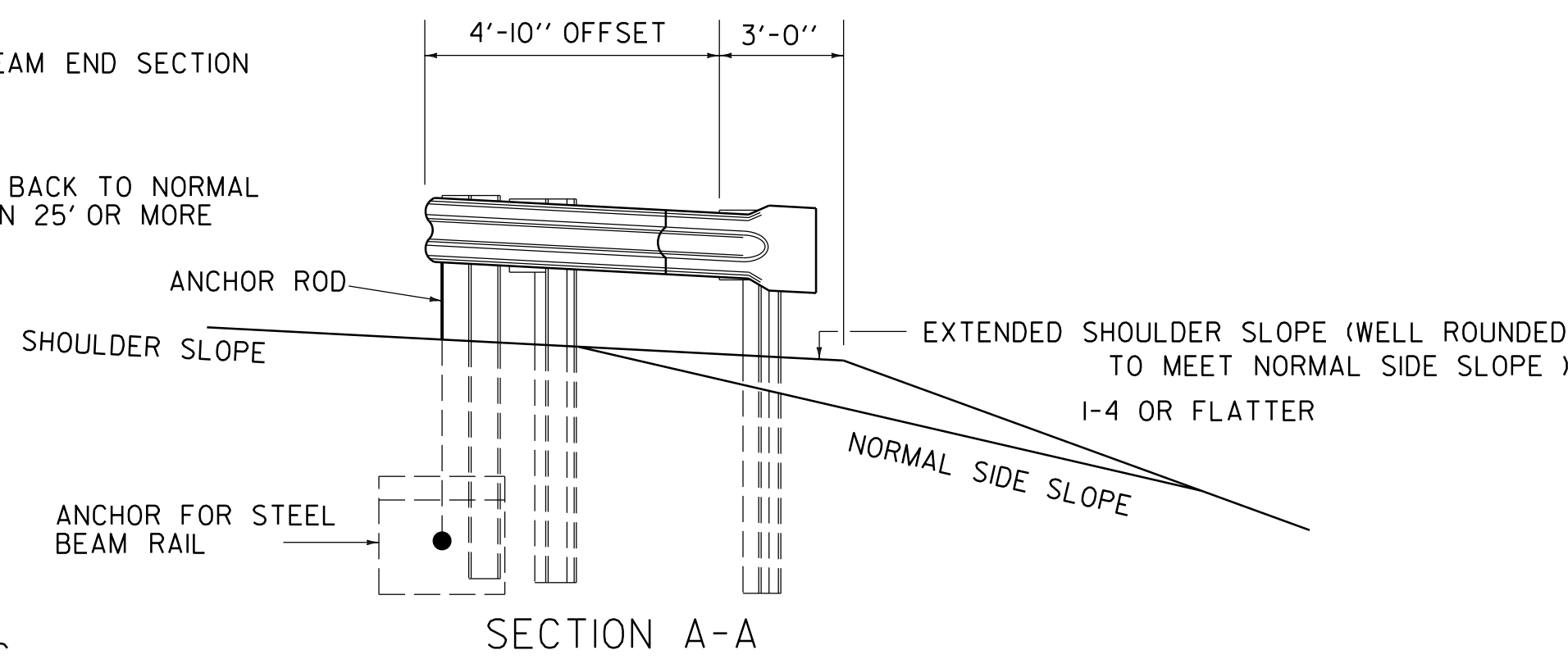


FIGURE 4 System No. 1: approach guardrail transition design details (1 in. = 25.4 mm).

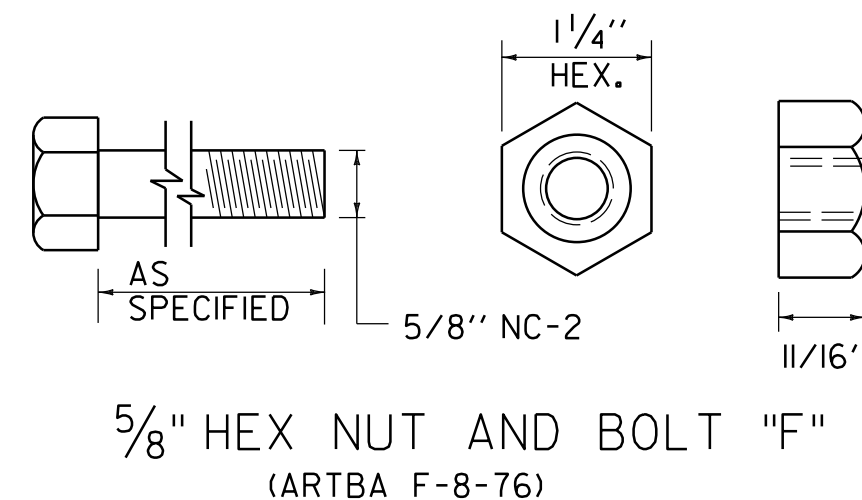
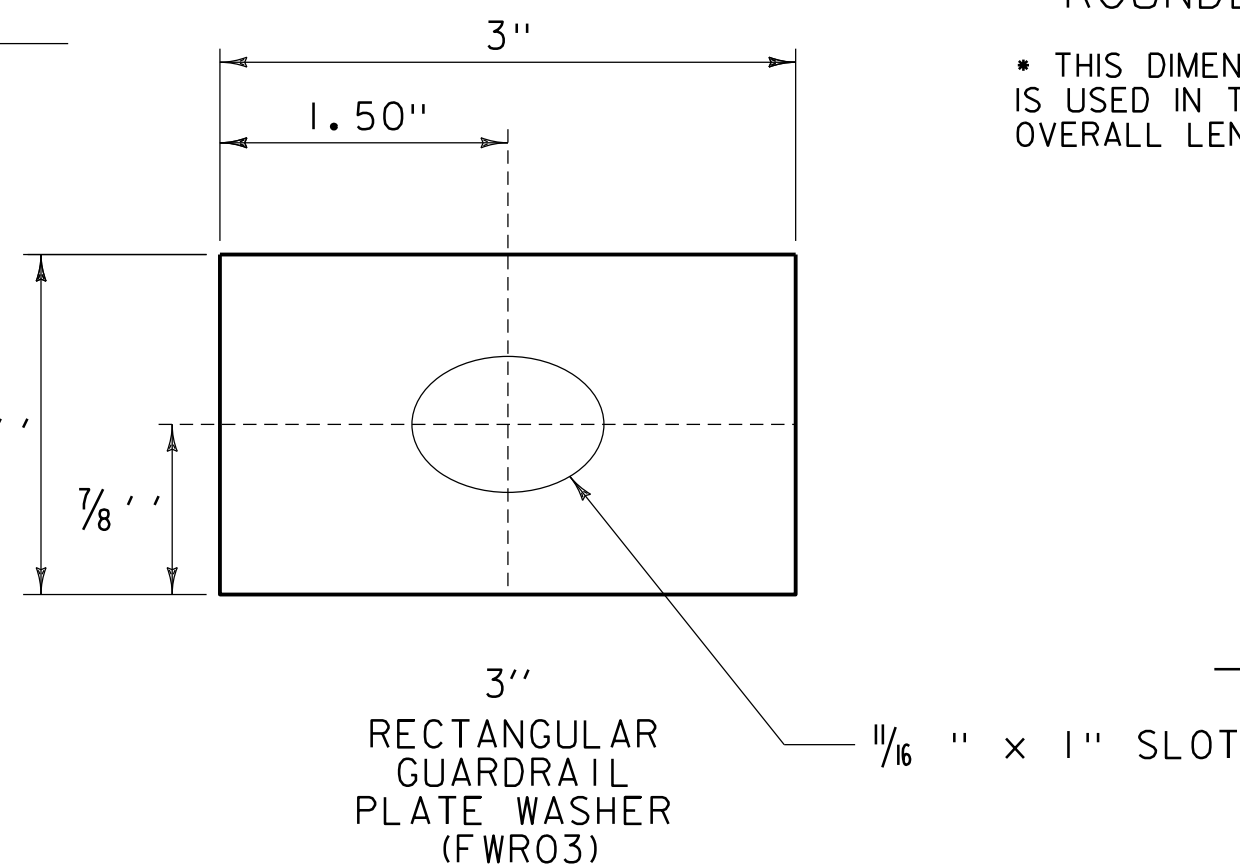
Varies. See Site Plans and Cross Section



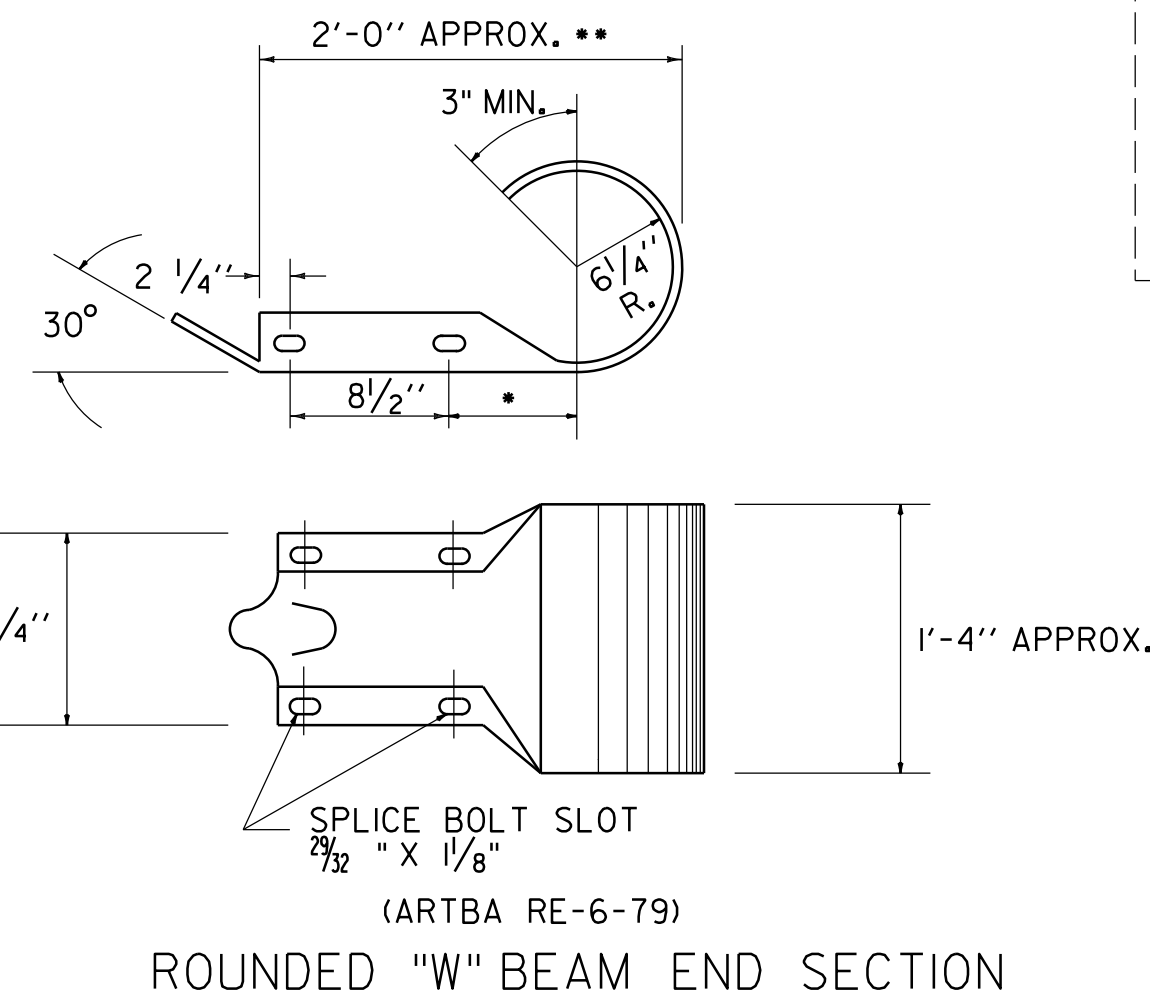
APPROACH END DETAIL
NHS APPROVED FOR USE WHERE DESIGN SPEED IS 40 OR LESS MPH
NON-NHS APPROVED FOR USE WHERE DESIGN SPEED IS 50 OR LESS MPH



MIN. TENSILE STRENGTH 60,000 LBS.
LOAD APPLIED THROUGH ASSEMBLY

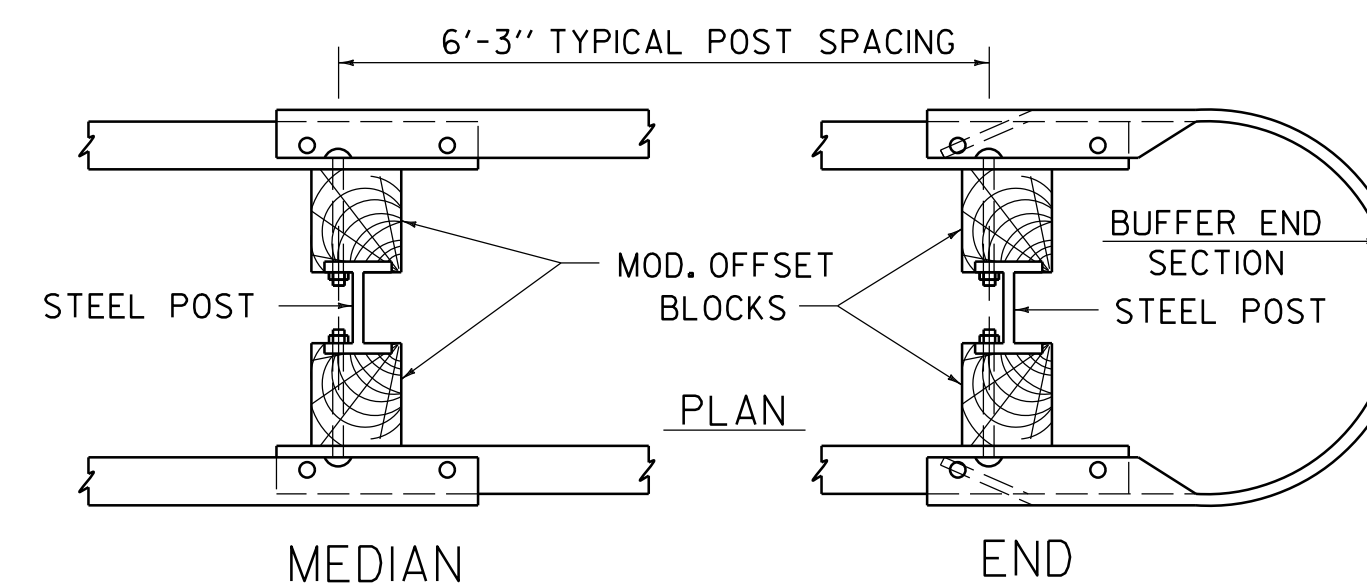
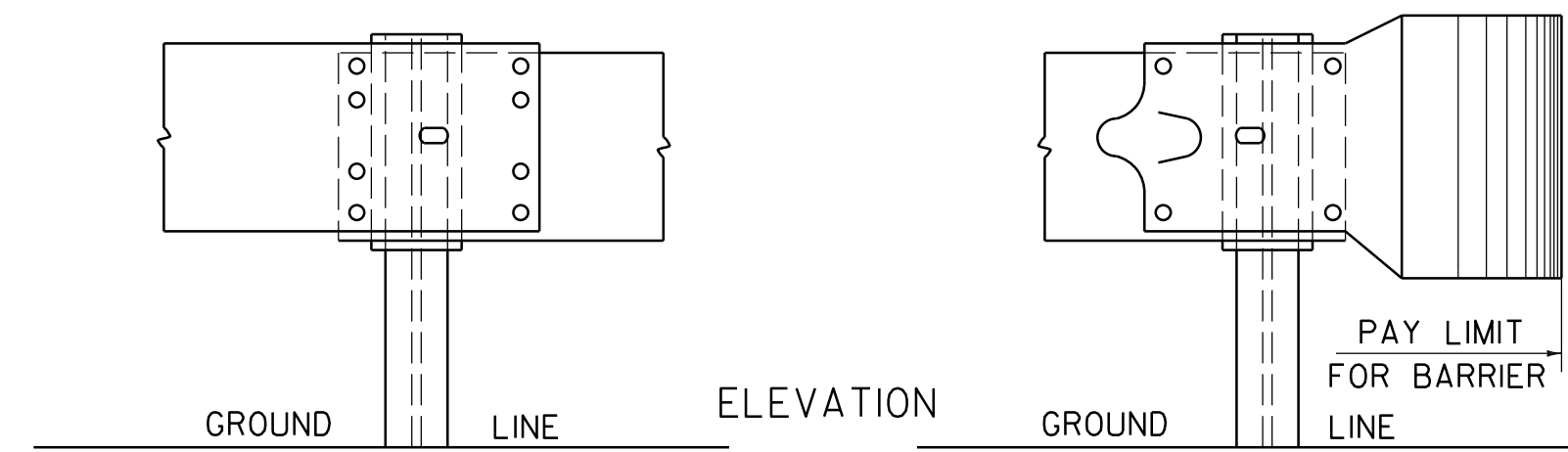


FASTENER DETAILS

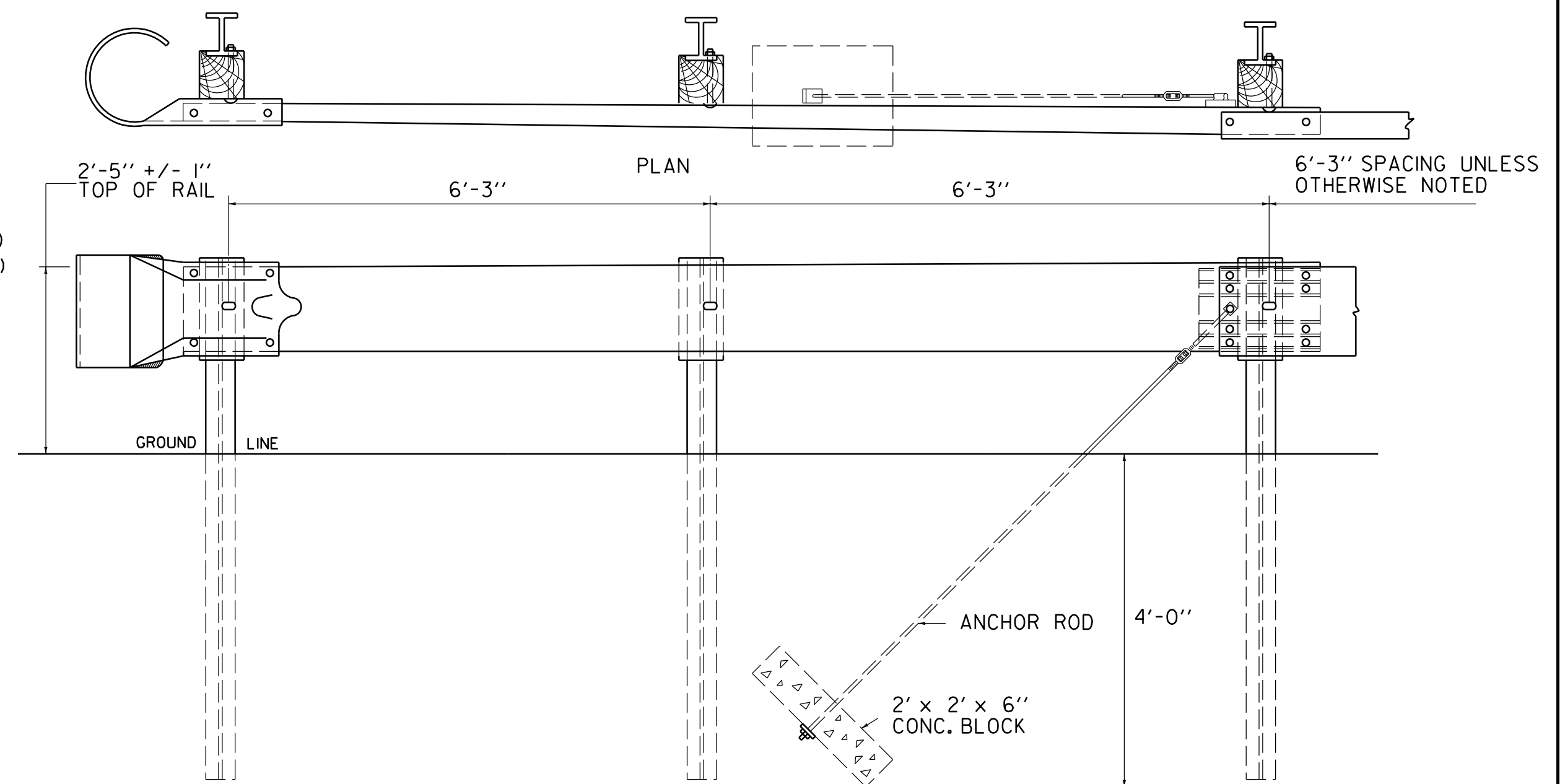


ROUNDED "W" BEAM END SECTION

• THIS DIMENSION IS 7 1/2" IN RE-7-79. IF THE DIMENSION IS USED IN THIS PART, IT WILL GIVE AN ACCEPTABLE OVERALL LENGTH (**) OF APPROXIMATELY 2'-11/2".



STEEL BEAM MEDIAN BARRIER
NOTE: TO BE USED OUTSIDE CLEAR ZONE ONLY.



ASSEMBLY ELEVATION

TRAILING END TERMINAL FOR
USE ON ONE-WAY HIGHWAYS

GENERAL NOTES:

1. ALL METAL PARTS SHALL BE GALVANIZED
2. ALL WOOD POSTS SHALL BE GIVEN A PRESERVATIVE TREATMENT
3. DETAILS PERTINENT TO THE STANDARD INSTALLATION OF "W" BEAM SECTIONS WILL BE FOUND ON STANDARD DRAWING G-1.
4. FOR DESCRIPTION AND SPECIFICATIONS OF PARTS IDENTIFIED BY "(ARTBA...)" AND OTHER DETAILS OF POSTS, POST ACCESSORIES, FASTENERS AND RAIL ELEMENTS, SEE AASHTO-AGC-ARTBA JOINT TASK FORCE NO. 13, TITLED "A GUIDE TO STANDARDIZED HIGHWAY BARRIER HARDWARE", LATEST EDITION.
5. THE TRANSITION FROM THE APPROACH END TO THE STANDARD STEEL BEAM GUARDRAIL SHALL BE 25'-0" UNLESS OTHERWISE SPECIFIED.
6. WHEN STANDARD STEEL BEAM CONNECTS TO BRIDGE APPROACH RAIL OF A DIFFERENT HEIGHT THE LENGTH NEEDED TO TRANSITION THE HEIGHT OF STANDARD STEEL BEAM TO MATCH THE BRIDGE APPROACH RAIL SHALL BE 25'-0" UNLESS OTHERWISE SPECIFIED.
7. WHEN STANDARD STEEL BEAM CONNECTS TO A MANUFACTURED TERMINAL SECTION OF A DIFFERENT HEIGHT THE LENGTH NEEDED TO TRANSITION THE HEIGHT OF STANDARD STEEL BEAM TO MATCH THE MANUFACTURED TERMINAL SECTION SHALL BE 25'-0" UNLESS OTHERWISE SPECIFIED.

REV.	DATE	DESCRIPTION
2	JAN. 3, 2000	UPDATED TO REFLECT METRIC STD. CHANGES
3	FEB. 10, 2014	UPDATED TO REFLECT GUARDRAILS HEIGHT OF 29": AS NOTED IN FHWA LETTER DATED MAY 17, 2000
4	MAR. 10, 2017	UPDATED WASHER DETAILS
OTHER STANDARDS REQUIRED: G-1		
VTRANS AND FHWA APPROVAL ON FILE WITH CONTRACT ADMINISTRATION		

STEEL BEAM GUARDRAIL END TERMINALS

ANCHOR FOR STEEL BEAM GUARDRAIL

STEEL BEAM MEDIAN BARRIER



STANDARD
G-1D